Subspace Identifiability Boundaries for Piecewise Stationary Low Rank Bandits

Anonymous Author(s)
Affiliation
Address
email

Abstract

Low-rank structure is pervasive in high-dimensional sequential decision-making, yet exploiting it under nonstationarity remains open: existing low-rank bandit methods assume a fixed representation, while nonstationary algorithms pay the full ambient cost $\widetilde{O}(d\sqrt{T})$. We study the fundamental question of *when and how* a changing r-dimensional subspace ($r \ll d$) can be identified from single-play scalar feedback $y_t = x_t^\top \theta_t + \varepsilon_t$. Under the standing latent excitation assumptions, we characterize the *probe side identification boundary*: three conditions (sub Gaussian noise with unknown variance estimated online, bounded state noise coupling, and full dimensional probe coverage) are each sufficient and essentially necessary for subspace recovery. Under these conditions, each probe yields a quadratic measurement of the predictable second moment $\widetilde{M}_t$ with controlled bias, enabling recovery via a lifted matrix operator; complementary impossibility results show that removing *any one* condition destroys identifiability, while approximate violations degrade gracefully. Building on this characterization, we propose *Single-Play Subspace-Calibrated Optimism* (SPSC), achieving costed dynamic regret $\widetilde{O}(r\sqrt{T}) + \widetilde{O}(T^{2/3}) + O(V)$. An $\Omega(\sqrt{cT})$ lower bound confirms the probe–subspace tradeoff is fundamental for explore-then-commit policies; an adaptive variant removes the need for oracle segment boundaries. Experiments confirm the predicted phase transition: SPSC achieves regret reductions over ambient-space baselines when $d - r \gg T^{1/6}$, and incurs higher cost when $d - r \lesssim T^{1/6}$, matching theory on synthetic and real-data benchmarks.

1 Introduction

Low-rank structure is a powerful lever for high-dimensional sequential decision-making: when the reward parameter $\theta_t \in \mathbb{R}^d$ lies in an r-dimensional subspace with $r \ll d$, estimation can scale with the intrinsic rank r rather than the ambient dimension d. Existing low-rank bandit methods [Jun et al., 2019, Jedra et al., 2024, Lu et al., 2021] exploit this but require a *fixed* representation. Nonstationary methods [Russac et al., 2019, Cheung et al., 2019] handle changing parameters but pay the full ambient cost $\widetilde{O}(d\sqrt{T})$, where T is the time horizon. No existing method achieves both.

Consider a learner who, at each round $t = 1, \ldots, T$, selects an action x_t from a given action set $\mathcal{A}_t \subset \mathbb{R}^d$ and observes a scalar reward

$$y_t = x_t^\top \theta_t + \varepsilon_t, \qquad \theta_t = B_k^\star w_t \in \mathbb{R}^d, \tag{1}$$

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

where $B_k^\star \in \mathbb{R}^{d\times r}$ ($r \ll d$) is an unknown orthonormal representation matrix that is piecewise constant over K segments $\mathcal{I}_k$ ($k = 1,\dots,K$), $w_t \in \mathbb{R}^r$ is a latent state evolving via a stable linear dynamical system $w_t = A_k w_{t-1} + \eta_{t-1}$ with segment-specific dynamics matrix $A_k \in \mathbb{R}^{r\times r}$ and zero-mean innovation $\eta_{t-1} \in \mathbb{R}^r$, and ε_t is observation noise. The learner's goal is to minimize the *costed dynamic regret* $\mathrm{DynReg}_T^{(c)} := \sum_{t=1}^T \big(\max_{x\in\mathcal{A}_t} x^\top\theta_t - x_t^\top\theta_t\big) + c\,|\mathcal{T}_{\mathrm{probe}}|$, which jointly penalizes suboptimal actions and the number of dedicated *probe rounds* $\mathcal{T}_{\mathrm{probe}} \subseteq [T]$ (rounds used for active sensing rather than exploitation), each costing a known constant $c > 0$.

However, when $B_k^\star$ changes across segments and the learner receives only a single scalar y_t per round, two mathematical difficulties arise. Simply resetting a d-dimensional estimator at each change point does not help: each segment still requires $\Omega(d)$ rounds to build a well-conditioned design matrix, so regret scales with d, not r.

1. **Identification from scalar feedback.** The observation $y_t = x_t^\top B_k^\star w_t + \varepsilon_t$ is a scalar projection of a rank-r signal. Recovering the r-dimensional column space $\mathrm{range}(B_k^\star)$ from such projections requires extracting second-order information: the predictable second moment $\widetilde{M}_t := \mathbb{E}[\theta_t\theta_t^\top \mid \mathcal{H}_{t-1}]$ (the conditional covariance of the reward parameter given the learner's history $\mathcal{H}_{t-1}$) satisfies $\mathrm{range}(\widetilde{M}_t) \subseteq \mathrm{range}(B_k^\star)$, but a single observation y_t does not directly reveal $\widetilde{M}_t$.
2. **Probe–subspace cost tradeoff.** Probing with random isotropic actions u_t drawn from a known distribution Q (satisfying $\mathbb{E}[u_t u_t^\top] \succeq \rho I_d$ for a constant $\rho > 0$, ensuring coverage in every direction) enables subspace recovery, but each probe round forfeits exploitation and incurs cost c. With m probes per segment, the subspace estimation error is $\varepsilon = O(1/\sqrt{m})$ and the exploitation regret from mismatch is $O(\varepsilon T)$, while the probe cost is $O(cm)$. Optimizing gives $m^\star = \Theta(T^{2/3})$ and a tradeoff term of $\widetilde{O}(T^{2/3})$.

We resolve both difficulties simultaneously. Under three structural conditions—sub-Gaussian noise, bounded state–noise coupling ($\|\mathbb{E}[\varepsilon_t\theta_t \mid \mathcal{H}_{t-1}, u_t]\|_2 \le \epsilon_\times$ for a known bound $\epsilon_\times \ge 0$), and full-dimensional probe coverage—the centered statistic $s_t := y_t^2 - \hat\sigma^2$ (where $\hat\sigma^2$ is an online estimate of the noise variance) yields a quadratic measurement of $\widetilde{M}_t$, and inverting the probe moment operator $\mathcal{K}$ lifts these scalar measurements into matrix-valued estimates from which the top-r eigenspace recovers $\mathrm{range}(B_k^\star)$. These conditions constitute an *identification boundary*: each is individually necessary, no two jointly sufficient (Propositions 7–9), and approximate violations degrade gracefully (Theorem 3).

Building on this identification result, we propose *Single-Play Subspace-Calibrated Optimism* (SPSC), which interleaves probing with projected UCB exploitation in the learned r-dimensional subspace. For K stationary segments with total within-segment drift $V := \sum_k \sum_{t\in\mathcal{I}_k} \|\theta_{t+1} - \theta_t\|_2$, the costed dynamic regret satisfies

$$\mathrm{DynReg}_T^{(c)} = \widetilde{O}(r\sqrt{T}) + \widetilde{O}(T^{2/3}) + O(V),$$

replacing the ambient $d\sqrt{T}$ dependence with intrinsic rank r. An $\Omega(\sqrt{cT})$ lower bound for explore-then-commit policies (Theorem 17)—a class that includes SPSC—confirms that the probe–subspace tradeoff is inherent to any method that learns a subspace from probes and then exploits within it. The remaining gap to the $\widetilde{O}(c^{1/3}T^{2/3})$ upper bound arises from $O(\varepsilon)$ per-round bias under hard projection versus the $\Omega(\varepsilon^2)$ information-theoretic cost (Remark 15), and closing it via soft-projection schemes is an explicit open problem. An adaptive variant detects change points without oracle boundaries and achieves the same leading rate with $O(K)$ overhead.

Contributions. Our work brings together low-rank bandits, piecewise-stationary LDS dynamics, and priced exploration in a single online decision-making framework.

1. **Formulation.** We introduce an online decision-making model with piecewise-stationary low-rank rewards, stable within-segment LDS dynamics, and explicit single-play probing, evaluated through a *costed dynamic regret* criterion (§3).
2. **Identification and impossibility.** Under three probe-side conditions—sub-Gaussian noise (unknown variance), bounded state–noise coupling, and full-dimensional coverage—

we show that single-play probes yield quadratic measurements of the predictable second moment with controlled bias. Complementary impossibility results show each condition is individually necessary; approximate violations degrade gracefully (Appendix E.2).

3. **Algorithm.** We propose SPSC, which interleaves probing with optimistic exploitation *directly* in the learned r-dimensional subspace via a windowed projected UCB rule. We give both an oracle-boundary version for analysis and an adaptive-reset version for unknown boundaries (§5).
4. **Regret under exact and approximate probe conditions.** For a fixed number of stationary segments, we prove a dynamic regret bound of $\widetilde{O}(r\sqrt{T}) + \widetilde{O}(T^{2/3}) + O(V)$, and show that the leading rate is preserved under $T^{-1/3}$-level probe-side misspecification.
5. **Empirical validation.** Experiments on synthetic and real benchmarks show that SPSC consistently improves over strong baselines across a broad (d, r) regime, with regret reductions of 6–38% (§7).

2 Related Work

Our work lies at the intersection of three lines of research: low-rank and representation-structured bandits, nonstationary or piecewise-stationary bandits, and active information acquisition for sequential decision-making. The closest prior works typically capture at most two of these ingredients at once. Table 3 summarizes the main distinction in the paper body; a broader comparison appears in Appendix C.

Low-rank and representation-structured bandits. In the stationary setting $\theta_t = \theta \in \mathbb{R}^d$ with $\mathrm{rank}(\Theta) = r$, Recent work achieves tighter rates via explore-then-refine [Kang et al., 2022], two-to-infinity recovery [Jedra et al., 2024], optimal experimental design [Jang et al., 2024], and spectral methods [Stojanovic et al., 2023]. All assume a *fixed* representation $B^\star$—none handle changing segment-wise representations that must be re-identified online.

Nonstationary and piecewise-stationary bandits. When the reward parameter θ_t varies over time, dynamic-regret analyses bound $\sum_t (\max_x x^\top \theta_t - x_t^\top \theta_t)$ in terms of the variation budget $V_T := \sum_t \|\theta_{t+1} - \theta_t\|$ or the number of switches K. Sliding-window LinUCB [Russac et al., 2019] and discounted methods [Cheung et al., 2019] achieve $\widetilde{O}(d^{2/3} V_T^{1/3} T^{1/3})$; Abbasi-Yadkori et al. [2023] give $\widetilde{O}(\sqrt{KdT})$ for piecewise-stationary K-armed bandits; and Hou et al. [2024] study best-arm identification under piecewise stationarity. Their regret scales with d, not r—because they do not recover or exploit a latent low-rank representation.

Active information acquisition and probing. In costly-observation bandits, the learner pays c per observed reward: Seldin et al. [2014] study label-efficient prediction, Tucker et al. [2023] prove $\Omega(c^{1/3} T^{2/3})$ lower bounds for K-armed bandits with costly observations, and Elumar et al. [2025] analyze multi-armed bandits with costly probes. Active query models [Sekhari et al., 2023] and system-identification methods [Memmel et al., 2024] also study exploration under cost constraints. Our setting differs structurally: probing must support *identification of a changing low-rank subspace* $\mathrm{range}(B_k^\star)$ from single-play scalar feedback $y_t \in \mathbb{R}$, not merely estimation of a fixed parameter. The interaction between the probe cost $c|\mathcal{T}_{\mathrm{probe}}|$ and the subspace estimation rate $O(1/\sqrt{m})$ creates the $\widetilde{O}(T^{2/3})$ tradeoff term that is absent in stationary low-rank or unstructured nonstationary settings.

3 Problem Formulation

At each round $t = 1, \ldots, T$, the environment reveals an action set $\mathcal{A}_t \subset \mathbb{R}^d$. The learner selects an action $x_t \in \mathcal{A}_t$ and observes a scalar reward

$$y_t = x_t^\top \theta_t + \varepsilon_t, \tag{2}$$

where $\theta_t \in \mathbb{R}^d$ is an unobserved reward parameter and ε_t is observation noise. The learner may designate certain rounds as *probe rounds*; on such rounds, the action is drawn from a known probe distribution, $x_t = u_t$ with $u_t \sim Q$, rather than chosen by the control policy. The learner faces two coupled tasks: controlling well under changing rewards, and deliberately acquiring informative observations about the currently active low-dimensional structure. Our formulation makes both aspects explicit.

Piecewise low-rank structure. The reward parameter admits a low-rank factorization $\theta_t = B_k^\star w_t$, where $B_k^\star \in \mathbb{R}^{d\times r}$ has orthonormal columns, $w_t \in \mathbb{R}^r$ is the latent state vector, and $B_k^\star$ is piecewise constant over $K \geq 1$ segments $\mathcal{I}_k$ $(k = 1, \ldots, K)$ with change points $1 = \tau_0 < \tau_1 < \cdots < \tau_K \leq T+1$, where $\mathcal{I}_k := \{\tau_{k-1}, \ldots, \tau_k - 1\}$. Within segment k, the latent state evolves according to a stable linear dynamical system:

$$w_t = A_k w_{t-1} + \eta_{t-1}, \qquad t \in \mathcal{I}_k, \tag{3}$$

where $A_k \in \mathbb{R}^{r\times r}$ is an unknown segment-specific transition matrix $(k = 1, \ldots, K)$ and $\eta_{t-1} \in \mathbb{R}^r$ is a zero-mean innovation term.

Assumption 1 (Latent-state stability and excitation)**.** *For each segment k, $\rho(A_k) \leq 1-\alpha$ for some $\alpha \in (0,1)$ for which a known lower bound $\alpha_0 \in (0, \alpha]$ is available, and the innovations satisfy $\eta_{t-1} \perp\!\!\!\perp \mathcal{H}_{t-1}$, $\mathbb{E}[\eta_{t-1} \mid \mathcal{H}_{t-1}] = 0$, $\mathbb{E}[\eta_{t-1}\eta_{t-1}^\top \mid \mathcal{H}_{t-1}] = \Sigma_{\eta,k}$, with $\lambda_r(\Sigma_{\eta,k}) \geq \underline{\lambda} > 0$.*

Assumption 2 (Uniform action boundedness)**.** *There exists a known constant $R_{\mathcal{A}} < \infty$ such that $\sup_{x\in\mathcal{A}_t} \|x\|_2 \leq R_{\mathcal{A}}$ for all t.*

Probe design. The learner may designate selected rounds as probe rounds. On such rounds it plays an action drawn from a known probe distribution, rather than an action chosen purely for exploitation. The next assumptions specify when these probes are both feasible and informative.

Assumption 3 (Probe realizability)**.** *There exists a known distribution Q on $\mathbb{R}^d$ such that $\mathbb{E}[uu^\top] \succeq \rho I_d$ for some known constant $\rho > 0$, u is sub-Gaussian, $\|u\|_2 \leq L$ almost surely, and $\mathrm{supp}(Q) \subseteq \mathcal{A}_t$ for all t.*

Assumption 4 (Probe-round noise properties)**.** *On non-probe rounds, $\mathbb{E}[\varepsilon_t \mid \sigma(\mathcal{H}_{t-1}, x_t)] = 0$ and ε_t is conditionally σ_ε-sub-Gaussian. On probe rounds $t \in \mathcal{T}_{\mathrm{probe}}$, where $x_t = u_t$:*

(i) ***Conditional mean zero:*** $\mathbb{E}[\varepsilon_t \mid \sigma(\mathcal{H}_{t-1}, u_t)] = 0$.

(ii) ***Sub-Gaussian noise with unknown variance:*** *ε_t is conditionally σ_ε-sub-Gaussian given $\sigma(\mathcal{H}_{t-1}, u_t)$, where $\sigma_\varepsilon > 0$ is* unknown *to the learner. The algorithm estimates σ_ε^2 online from probe-round residuals (Remark 1).*

(iii) ***Bounded state–noise coupling:*** *$\big\|\mathbb{E}[\varepsilon_t\theta_t \mid \sigma(\mathcal{H}_{t-1}, u_t)]\big\|_2 \leq \epsilon_\times$ almost surely, for a known bound $\epsilon_\times \geq 0$.*

These conditions are probe-side identification requirements, not generic modeling restrictions; their practical scope and formal necessity are discussed in Appendix B.

Remark 1 (Online variance estimation at no asymptotic cost)**.** *Since σ_ε^2 is unknown, the algorithm estimates it from early probe-round residuals. Dedicating the first $T^{2/3}$ probe rounds to variance estimation yields $|\hat{\sigma}^2 - \sigma_\varepsilon^2| \leq \delta_\sigma = O(T^{-1/3})$; this cost is absorbed into the existing $\widetilde{O}(T^{2/3})$ probe–subspace tradeoff term in Theorem 3, so the estimation is* free *in the regret bound. For isotropic Gaussian probes, the resulting bias $\widetilde{B} \approx \frac{\delta_\sigma}{2} I_d$ is a scaled identity that shifts eigenvalues uniformly without corrupting eigenvectors (Proposition 5), so subspace recovery is unaffected.*

The concentration analysis requires bounded noise; the standard truncation argument is in Appendix E.4(Remark 10).

Costed dynamic regret. Let $\mathcal{T}_{\mathrm{probe}} \subseteq \{1, \ldots, T\}$ denote the set of probe rounds (on which $x_t = u_t \sim Q$). Performance is measured by the *costed dynamic regret*

$$\mathrm{DynReg}_T^{(c)} := \sum_{t=1}^{T} \left(\max_{x\in\mathcal{A}_t} x^\top\theta_t - x_t^\top\theta_t \right) + c\,|\mathcal{T}_{\mathrm{probe}}|, \tag{4}$$

where $c > 0$ is a known per-probe cost. The first term measures control loss relative to the instantaneous best action; the second prices information acquisition.

Predictable second moment. The central inferential target is the predictable second moment $\widetilde{M}_t := \mathbb{E}[\theta_t\theta_t^\top \mid \mathcal{H}_{t-1}] = B_k^\star \mathbb{E}[w_t w_t^\top \mid \mathcal{H}_{t-1}](B_k^\star)^\top, \qquad t \in \mathcal{I}_k$. Its importance is that $\widetilde{M}_t$ encodes the active low-dimensional subspace through its range. At the segment level, define the probe-time average $\bar{M}_k^{\text{probe}} := \frac{1}{m_k}\sum_{t\in\mathcal{T}_k}\widetilde{M}_t$. Here $\mathcal{T}_k := \mathcal{T}_{\text{probe}} \cap \mathcal{I}_k$ denotes the probe rounds within segment k and $m_k := |\mathcal{T}_k|$ is their count. When the latent probe-time covariance has full rank r, this quantity identifies the active segment subspace.

Lemma 1 (Predictable probe-time excitation)**.** *Under Assumption 1, for each segment k, $\lambda_r(\bar{S}_k^{\text{probe}}) \geq \lambda_{\min} > 0$ almost surely, where $\bar{S}_k^{\text{probe}} := \frac{1}{m_k}\sum_{t\in\mathcal{T}_k}\mathbb{E}[w_t w_t^\top \mid \mathcal{H}_{t-1}]$. (The proof is given in Appendix F (§F.1.1).)*

Lemma 2 (Invertibility of the probe operator)**.** *Under Assumption 3, the operator $\mathcal{K}(M) := \mathbb{E}[(u^\top M u)\, uu^\top], \qquad u \sim Q$, is invertible on $\mathbb{R}^{d\times d}_{\text{sym}}$, with $\|\mathcal{K}^{-1}\|_{\text{op}} \leq C_{\mathcal{K}}(\rho, L, d)$.*

Here $C_{\mathcal{K}}(\rho, L, d)$ denotes an explicit constant depending on the probe covariance lower bound ρ, the probe norm bound L, and the ambient dimension d; for isotropic Gaussian probes $u \sim \mathcal{N}(0, I_d)$ truncated to $\|u\| \leq L$, $C_{\mathcal{K}} \leq 1$ (see Remark 6 in the appendix). The proof is given in Appendix F (§F.1.2).

4 Identification from Single-Play Probes

This section establishes the identification pipeline underlying the algorithm. We first show that single-play probe rewards reveal quadratic information about the predictable second moment. We then lift these quadratic observations into matrix-valued estimates whose leading eigenspace recovers the active subspace.

Define the centered probe statistic $s_t := y_t^2 - \hat{\sigma}^2$, where $\hat{\sigma}^2$ is the algorithm's estimate of σ_ε^2. Under Assumption 4:

Proposition 1 (Quadratic measurement identity)**.** *For every probe round $t \in \mathcal{T}_{\text{probe}}$, with $\delta_\sigma := \hat{\sigma}^2 - \sigma_\varepsilon^2$, $\mathbb{E}[s_t \mid \sigma(\mathcal{H}_{t-1}, u_t)] = u_t^\top \widetilde{M}_t u_t + 2\, u_t^\top \mathbb{E}[\varepsilon_t \theta_t \mid \mathcal{H}_{t-1}, u_t] - \delta_\sigma$. Under exact conditions ($\delta_\sigma = 0$, $\epsilon_\times = 0$), this reduces to $\mathbb{E}[s_t \mid \sigma(\mathcal{H}_{t-1}, u_t)] = u_t^\top \widetilde{M}_t u_t$. The proof is given in Appendix F (§F.1.3).*

Proposition 2 (Segment-level factorization)**.** *For each segment k, $\bar{M}_k^{\text{probe}} = B_k^\star \bar{S}_k^{\text{probe}} (B_k^\star)^\top$, and $\operatorname{range}(\bar{M}_k^{\text{probe}}) \subseteq \operatorname{range}(B_k^\star)$. The proof is given in Appendix F (§F.1.4).*

Proposition 3 (Identification of the active segment subspace)**.** *Suppose Lemma 1 holds. Then for every segment k, $\operatorname{range}(\bar{M}_k^{\text{probe}}) = \operatorname{range}(B_k^\star)$. Equivalently, the segment projector $P_k^\star := B_k^\star (B_k^\star)^\top$ is identified by $\bar{M}_k^{\text{probe}}$. The proof is given in Appendix F (§F.1.6).*

Lifted operator and subspace recovery. The preceding results show that probe rounds identify the relevant quadratic form. To turn this into a usable matrix estimator, we invert the probe moment operator. Under Lemma 2, define the *lifted probe sample* $G_t := \mathcal{K}^{-1}(s_t u_t u_t^\top)$. When Q is the isotropic Gaussian $\mathcal{N}(0, I_d)$ (or its bounded truncation), the fourth-moment structure gives an explicit closed form: $\mathcal{K}^{-1}(N) = \frac{1}{2}N - \frac{\operatorname{tr}(N)}{2(d+2)} I_d$, so each lifted sample $G_t = \frac{1}{2}s_t u_t u_t^\top - \frac{s_t\|u_t\|^2}{2(d+2)} I_d$ is computable in $O(d^2)$ operations ($N \in \mathbb{R}^{d\times d}_{\text{sym}}$.)

Proposition 4 (Lifted probe sample)**.** *For every probe round $t \in \mathcal{T}_{\text{probe}}$, $\mathbb{E}[G_t \mid \mathcal{H}_{t-1}] = \widetilde{M}_t + \widetilde{B}_t$, where $\|\widetilde{B}_t\|_2 \leq |\delta_\sigma|\, C_{\mathcal{K}}\, L^2 + 2\, C_{\mathcal{K}}\, L^2\, \epsilon_\times$. When $\delta_\sigma = 0$ and $\epsilon_\times = 0$, the estimator is exactly unbiased: $\mathbb{E}[G_t \mid \mathcal{H}_{t-1}] = \widetilde{M}_t$. The proof is given in Appendix F (§F.1.5).*

The empirical segment-level estimator $\widehat{M}_k := \frac{1}{m_k}\sum_{t\in\mathcal{T}_k} G_t$ estimates $\bar{M}_k^{\text{probe}}$ with bias at most $|\delta_\sigma|\, C_{\mathcal{K}}\, L^2 + 2\, C_{\mathcal{K}}\, L^2\, \epsilon_\times$. Let $\widehat{U}_k$ contain the top r eigenvectors of $\widehat{M}_k$, and define $\widehat{P}_k := \widehat{U}_k \widehat{U}_k^\top$. For isotropic Gaussian probes, the variance-misspecification bias is a scaled identity $\widetilde{B} \approx \frac{\delta_\sigma}{2} I_d$ (Proposition 5), which shifts eigenvalues uniformly without corrupting eigenvectors, so subspace recovery is robust to variance estimation error.

Remark 2 (Identification boundary and practical scope)**.** *Each probe-side condition is provably necessary (Appendix E.3); all three are stated in their weakest usable form:* (i) Noise variance *is unknown and estimated online; burn-in cost $O(T^{2/3})$ is absorbed into the existing tradeoff term (Remark 1). Empirically,* $5\times$ *misspecification causes* $< 5\%$ *regret change.* (ii) State–noise coupling *is bounded ($\epsilon_\times$), not required to be zero; the penalty $O(\epsilon_\times T/\lambda_{\min})$ is explicit in Theorem 3. Even $\epsilon_\times$=1 changes regret by $< 1\%$ empirically.* (iii) Probe realizability: $\operatorname{supp}(Q) \subseteq \mathcal{A}_t$ *holds under ϵ-greedy randomization. This is the sole non-relaxable condition: restricted coverage causes $\Omega(T)$ regret (Appendix E.10).*

5 Algorithm

The central design choice is that exploitation is performed *directly in the learned r-dimensional subspace.* After probing has produced an estimated basis $\widehat{U}_{t-1} \in \mathbb{R}^{d\times r}$, each candidate action $x \in \mathcal{A}_t$ is represented through its reduced coordinates $z_t(x) := \widehat{U}_{t-1}^\top x \in \mathbb{R}^r$. The controller then selects actions by solving a projected upper confidence bound (UCB) problem in $\mathbb{R}^r$: it maintains a windowed ridge estimator $\widehat{a}_t = \widetilde{V}_t^{-1}\widetilde{b}_t$ of the reduced parameter $a_t^\star := \widehat{U}_{t-1}^\top \theta_t \in \mathbb{R}^r$, where $\widetilde{V}_t := \lambda I_r + \sum_{s\in\mathcal{W}_t} z_s z_s^\top$ and $\widetilde{b}_t := \sum_{s\in\mathcal{W}_t} z_s y_s$ aggregate data from a sliding window $\mathcal{W}_t$ of recent exploitation rounds. On non-probe rounds, the action maximizes the UCB index

$$x_t = \operatorname*{arg\,max}_{x\in\mathcal{A}_t} \big\{ z_t(x)^\top \widehat{a}_t + \beta_t^{(r,W)} \|z_t(x)\|_{\widetilde{V}_t^{-1}} + \gamma_t \|x\|_2 \big\},$$

where $\beta_t^{(r,W)} = \sigma_\varepsilon \sqrt{r\log(1 + WR_{\mathcal{A}}^2/(\lambda r)) + 2\log(1/\delta)} + \sqrt{\lambda}\, S_w$ is the confidence radius calibrated to the r-dimensional estimation error, and $\gamma_t = \Gamma_k + V_{k,t}(W)$ absorbs subspace mismatch and local drift (precise definitions in §6).

The key benefit is that $\beta_t^{(r,W)}$ scales with $\sqrt{r}$ not $\sqrt{d}$, yielding per-segment regret depending on intrinsic rank. Algorithm 1 summarizes the procedure.

Remark 3 (Known segment boundaries and reset justification)**.** *Algorithm 1 assumes known segment boundaries $\{\tau_k\}_{k=1}^K$. The reset of the probe accumulator at each τ_k is essential: because $B_k^\star$ may change across segments, pre-change probe data would contaminate $\widehat{M}_k$ and thereby destroy the unbiasedness guaranteed by Proposition 4. The data-driven extension is Algorithm 4 in Appendix E.6.*

Computational cost. Each probe round requires forming the lifted sample $G_t = \mathcal{K}^{-1}(s_t u_t u_t^\top)$ in $O(d^2)$ time and updating the running sum $\sum G_t$ (also $O(d^2)$). After m_k probes, the eigendecomposition of $\widehat{M}_k \in \mathbb{R}^{d\times d}$ costs $O(d^2 r)$ via a partial eigenvalue solver (e.g., Lanczos iteration), amortized over the segment. Each exploitation round involves projecting $|\mathcal{A}_t|$ actions via $z_t(x) = \widehat{U}_{t-1}^\top x$ ($O(dr|\mathcal{A}_t|)$), updating the $r\times r$ ridge system ($O(r^2)$), and solving the UCB maximization ($O(r^2|\mathcal{A}_t|)$). The total per-round cost is $O(dr|\mathcal{A}_t| + r^2)$, which is linear in d and quadratic in r—a significant reduction over the $O(d^2|\mathcal{A}_t|)$ cost of ambient LinUCB when $r \ll d$.

Adaptive-reset variant. The known-boundary version isolates the identification-to-exploitation mechanism cleanly; the adaptive-reset version is the practically deployable form and inherits the same leading rate up to lower-order overhead. Algorithm 4 (Appendix E.6) extends Algorithm 1 to the case of unknown segment boundaries through a data-driven change-point detector. The detector compares two non-overlapping rolling-window estimates of the predictable second moment:

$$S_t := \|\widehat{M}_t^{\text{recent}} - \widehat{M}_t^{\text{past}}\|_{\text{op}}. \tag{5}$$

A reset is triggered when S_t exceeds a threshold b calibrated to within-segment variation and concentration noise (the precise form is given in §6, Eq. (12)). After each detected reset, m_{relearn} consecutive probe rounds are used to rebuild the subspace estimate from scratch.

6 Main Regret Results

This section presents the main regret guarantees. Theorem 1 decomposes regret into probe cost, statistical regret, subspace mismatch, and drift (Appendix F); Theorem 2 optimizes probe allocation; Theorem 3 extends to weakened assumptions; Theorem 4 handles unknown boundaries. Proofs are in Appendix F (§F.5.1, §F.5.2, §F.5.3, and §F.6.4).

Let $\ell_k := |\mathcal{I}_k|$ denote the length of segment k, so $\sum_{k=1}^{K} \ell_k = T$. Let $V := \sum_{k=1}^{K} V_k$ denote the total within-segment variation, where $V_k := \sum_{s\in E_k,\, s+1\in E_k} \|\theta_{s+1} - \theta_s\|_2$.

We assume the segmentwise projector estimation error obeys

$$\varepsilon_k \le C_{\text{sub}} \sqrt{\frac{\log(d/\delta)}{m_k}}, \qquad k = 1, \dots, K, \tag{6}$$

where $C_{\text{sub}} = 8R_X/\lambda_{\min}$ is the explicit constant from Theorem 6 (Appendix E.4), and $R_X := C_{\mathcal{K}}\, L^2\, R_s + S_w^2$, $R_s := (LS_w + L_\varepsilon)^2 + \sigma_\varepsilon^2$.

The statistical confidence radius is

$$\beta_t^{(r,W)} := \sigma_\varepsilon \sqrt{r \log\left(1 + \frac{WR_{\mathcal{A}}^2}{\lambda r}\right) + 2\log(1/\delta)} + \sqrt{\lambda} S_w. \tag{7}$$

The correction coefficient is $\gamma_t := \Gamma_k + V_{k,t}(W)$, where $\Gamma_k := S_w \varepsilon_k$ is the subspace-mismatch level and $V_{k,t}(W) := \sum_{s\in E_k,\, t-W\le s<t} \|\theta_{s+1} - \theta_s\|_2$ is the local variation.

Theorem 1 (Full-horizon windowed dynamic regret)**.** *Under the hypotheses of Theorem 8 (Appendix E.5) on every segment, with probability at least $1-\delta$, $\mathrm{DynReg}_T^{(c)}$ decomposes into probe cost ($\propto \sum_k m_k$), projected statistical regret ($\propto \sum_k \beta_k^{(r,W)} \sqrt{rn_k}$), subspace mismatch ($\propto \sum_k \varepsilon_k n_k$), and drift ($\propto W \sum_k V_k$). The full bound is in Appendix F, §F.5.1.*

Theorem 2 (Main Result: Closed dynamic regret bound)**.** *Assume the hypotheses of Theorem 1 hold on every segment, and assume* (6) *holds. Let $A := 2R_{\mathcal{A}} S_w + c$ and $B := 2C_{\text{sub}} S_w R_{\mathcal{A}} \sqrt{\log(d/\delta)}$. For a fixed window length $W \ge 1$, let $L_W := \log(1 + WR_{\mathcal{A}}^2/(\lambda r))$ and $\beta^{(r,W)} := \sigma_\varepsilon \sqrt{rL_W + 2\log(K/\delta)} + \sqrt{\lambda} S_w$. Choose the segmentwise probe counts $m_k = \lceil (B\ell_k/(2A))^{2/3} \rceil$. Then, with probability at least $1-\delta$:*

$$\mathrm{DynReg}_T^{(c)} \le 2\beta^{(r,W)} \sqrt{2rL_W} \sum_{k=1}^{K} \sqrt{\ell_k} + \frac{3}{2^{2/3}} A^{1/3} B^{2/3} \sum_{k=1}^{K} \ell_k^{2/3} + 2R_{\mathcal{A}} WV + AK. \tag{8}$$

Applying $\sum_{k=1}^{K} \sqrt{\ell_k} \le \sqrt{KT}$ and $\sum_{k=1}^{K} \ell_k^{2/3} \le K^{1/3} T^{2/3}$, and setting $W^\star = 1$:

$$\mathrm{DynReg}_T^{(c)} \le 2\beta^{(r,1)} \sqrt{2rL_1 KT} + \frac{3}{2^{2/3}} A^{1/3} B^{2/3} K^{1/3} T^{2/3} + 2R_{\mathcal{A}} V + AK. \tag{9}$$

In particular, for fixed K and suppressing logarithmic factors,

$$\mathrm{DynReg}_T^{(c)} = \widetilde{O}\left(r\sqrt{T}\right) + \widetilde{O}\left(T^{2/3}\right) + O(V). \tag{10}$$

The three terms have clear roles: $\widetilde{O}(r\sqrt{T})$ is the projected exploitation cost (vs. $\widetilde{O}(d\sqrt{T})$ in ambient space), $\widetilde{O}(T^{2/3})$ is the probe–subspace tradeoff, and $O(V)$ is the irreducible drift price.

Theorem 3 (General regret bound under approximate probe conditions)**.** *Under Assumptions 1–4, let $\delta_\sigma := |\hat{\sigma}^2 - \sigma_\varepsilon^2|$ and $\epsilon_\times$ be the cross-correlation bound. If $|\delta_\sigma| C_{\mathcal{K}} L^2 + 2C_{\mathcal{K}} L^2 \epsilon_\times \le \lambda_{\min}/4$, then with probability at least $1-\delta$,*

$$\mathrm{DynReg}_T^{(c)} \le \widetilde{O}(r\sqrt{T}) + \widetilde{O}(T^{2/3}) + O(V) + O\left(\frac{(|\delta_\sigma| + \epsilon_\times)\, T}{\lambda_{\min}}\right). \tag{11}$$

With online variance estimation ($\delta_\sigma = O(T^{-1/3})$, Remark 1) and $\epsilon_\times = O(T^{-1/3})$, the penalty is $O(T^{2/3})$ and the leading rate is preserved. Setting $\epsilon_\times = \delta_\sigma = 0$ recovers Theorem 2.

Theorem 4 (Adaptive costed dynamic regret)**.** *Suppose Assumptions 1–4 hold and every true change satisfies $\Delta_k \geq \Delta_{\text{det}} = 2b$. Run the adaptive algorithm with probe rate μ, window counts $n_r = n_p = n$, threshold*

$$\begin{aligned} b &:= \Delta_{\text{within}} + 2R_X\sqrt{\frac{\log\!\big(4dT/\delta_{\text{FA}}\big)}{n_r}} + 2R_X\sqrt{\frac{\log\!\big(4dT/\delta_{\text{FA}}\big)}{n_p}}, \\ m_{\text{relearn}} &:= \left\lceil \frac{64R_X^2\log\!\big(2dK/\delta\big)}{\lambda_{\min}^2} \right\rceil. \end{aligned} \tag{12}$$

Let $\hat{K}$ be the total number of resets and F_T the number of false alarms. With probability at least $1-\delta$,

$$\begin{aligned} \mathrm{DynReg}_T^{(c)} \leq\; & \underbrace{2\beta^{(r,1)}\sqrt{2rL_1\hat{K}T} + \frac{3}{2^{2/3}}A^{1/3}B^{2/3}\hat{K}^{1/3}T^{2/3} + 2R_{\mathcal{A}}V + A\hat{K}}_{\textit{oracle-equivalent term}} \\ & + \underbrace{2R_{\mathcal{A}}S_w\,K\,D_{\max}}_{\textit{delay penalty}} + \underbrace{\big(2R_{\mathcal{A}}S_w + c\big)\,m_{\text{relearn}}\,(K+F_T)}_{\textit{recovery cost}}. \end{aligned} \tag{13}$$

where $D_{\max} = (2n/\mu) + \tau_{\text{burn}}$ and $F_T = 0$ with probability at least $1-\delta/2$. For fixed K, the adaptive algorithm achieves the same leading rate $\widetilde{O}(r\sqrt{T}) + \widetilde{O}(T^{2/3}) + O(V)$ with $O(K)$ lower-order overhead.

7 Experiments

We evaluate SPSC on synthetic and real-data benchmarks. Unless otherwise stated, all curves report the mean over multiple random seeds with ± 1 standard-error bands. The main paper presents four experiments; Appendix G contains 14 additional studies (Table 8) including sensitivity analyses and the Warfarin clinical dosing benchmark (Appendix G.15). Setup details are in Appendix G.1.

Experiment 1: Phase transition (Theorems 2, 13). Figure 1 shows the SPSC/LinUCB regret ratio over a broad (d, r) grid. The empirical crossover boundary closely tracks $d{-}r = T^{1/6}$: LinUCB wins at $d \leq 20$, SPSC dominates at $d \geq 45$ with 2–34% reductions, and every r-curve tracks $\sqrt{r/d}$, validating the leading-rate improvement from $d\sqrt{T}$ to $r\sqrt{T}$.

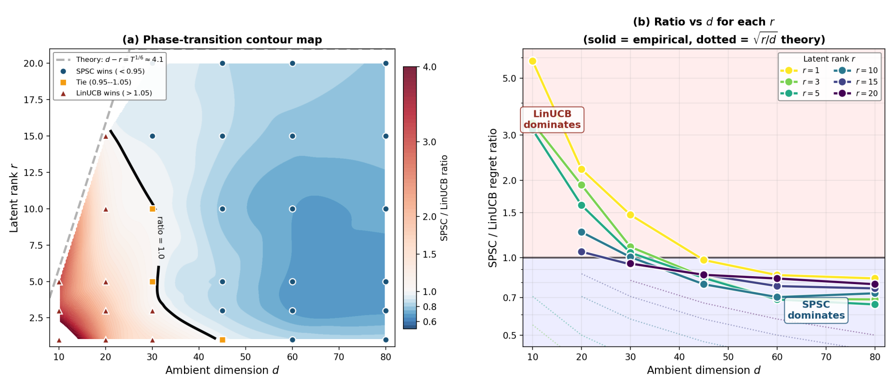

Figure 1: Synthetic phase-transition benchmark. **(a)** SPSC/LinUCB ratio contour. Solid black: empirical ratio=1. Dashed grey: $d{-}r = T^{1/6}$. **(b)** Ratio vs. d (log-scale); dotted: $\sqrt{r/d}$ theory.

Experiment 2: Multi-baseline real data (Theorem 2). Table 1 reports absolute regret on UCI Covertype (K=4, T=10,000, 7 baselines, 10 seeds). SPSC is best non-oracle at $d \geq 55$, $r \geq 10$, beating Restart-LinUCB by 27%, SW-LinUCB by 29%, D-LinUCB by

30% at d=155. At small d, probe cost dominates, confirming the theory predicts *where* SPSC helps, not just *that* it helps. Pendigits and Satimage in Appendix G.5.

Table 1: Covertype: SPSC/LinUCB and SPSC/Best-Competitor ratios (10 seeds). Values <1 = SPSC wins; bold = best non-oracle. Full tables in Appendix G.14.

	SPSC / LinUCB				SPSC / Best Competitor			
$d \backslash r$	1	10	20	30	1	10	20	30
5	2.25	–	–	–	2.38	–	–	–
55	1.06	**0.84**	**0.83**	**0.83**	1.08	**0.86**	**0.85**	**0.84**
105	**0.95**	**0.75**	**0.74**	**0.76**	**0.96**	**0.76**	**0.74**	**0.77**
155	**0.94**	**0.74**	**0.72**	**0.75**	**0.95**	**0.75**	**0.73**	**0.76**

Experiment 3: Approximate low rank (Theorem 3). On warfarin dose selection (Appendix G.15), SPSC reduces regret by 26 to 32% vs. all nonstationary baselines at every tested rank. The data is only *approximately* low rank (subspace error $\approx$ 0.96 to 1.00), validating Theorem 3: the penalty from approximate structure is absorbed into the $O((\delta_\sigma + \epsilon_\times)T/\lambda_{\min})$ term without destroying the leading rate. Probe overhead is 3.3 to 3.6%. A random subspace ablation (Appendix G.15) decomposes the source of the gain: roughly half comes from dimensionality reduction ($\sqrt{r}$ vs. $\sqrt{d}$ confidence radius) and the other half from the learned subspace, which retains 2 to 6$\times$ more signal variance than random projection.

Experiment 4: Mechanism validation. Three experiments validate the identification boundary (Appendix G.6): 5$\times$ variance misspecification changes regret by $< 5\%$; cross-correlation $\epsilon_\times$=1 changes regret by $< 1\%$; restricting coverage causes $\Omega(T)$ blow-up. These confirm not just performance but the *mechanism*: the first two conditions degrade gracefully (Theorem 3), while the third is provably non-relaxable (Theorem 14). The probe-rate ablation (Appendix G.2) confirms the $T^{2/3}$ U-shaped tradeoff.

8 Conclusion

We characterize the *probe side identification boundary* for recovering a changing low rank subspace from single play scalar feedback: under standing latent excitation and stability conditions, three conditions (unknown variance estimated online, bounded state noise coupling, full dimensional coverage) are each necessary, and their joint satisfaction enables quadratic measurements of the predictable second moment with controlled bias. Building on this, SPSC achieves $\widetilde{O}(r\sqrt{T}) + \widetilde{O}(T^{2/3}) + O(V)$ costed dynamic regret; an adaptive variant removes oracle segment boundaries. Experiments confirm regret reductions over all nonstationary baselines on real data when d/r is large.

Broader impact and future work. This work is primarily theoretical, applicable to recommendation, pricing, and control. The main open question is closing the $\Omega(\sqrt{cT})$ vs. $\widetilde{O}(c^{1/3}T^{2/3})$ gap via soft-projection schemes; other directions include unknown rank r and time-varying transition matrices.

References

Yasin Abbasi-Yadkori, Dávid Pál, and Csaba Szepesvári. Improved algorithms for linear stochastic bandits. In *Advances in Neural Information Processing Systems*, volume 24, 2011.

Yasin Abbasi-Yadkori, András György, and Nevena Lazić. A new look at dynamic regret for non-stationary stochastic bandits. *Journal of Machine Learning Research*, 24(288):1–37, 2023.

Peter Auer, Pratik Gajane, and Ronald Ortner. Adaptively tracking the best bandit arm with an unknown number of distribution changes. In *Conference on Learning Theory*, pages 138–158. PMLR, 2019.

Changxiao Cai, T. Tony Cai, and Hongzhe Li. Transfer learning for contextual multi-armed bandits. *The Annals of Statistics*, 52(1):207–232, 2024.

Nicolò Cesa-Bianchi, Emanuel Esposito, and Davide Maran. A definition of non-stationary bandits. *arXiv preprint arXiv:2302.12202*, 2023.

Wang Chi Cheung, David Simchi-Levi, and Ruihao Zhu. Learning to optimize under non-stationarity. In *Proceedings of the 22nd International Conference on Artificial Intelligence and Statistics*, volume 89 of *Proceedings of Machine Learning Research*, pages 1079–1087, 2019.

Eray Can Elumar, Cem Tekin, and Osman Yağan. Multi-armed bandits with costly probes. *IEEE Transactions on Information Theory*, 71(1):618–643, 2025. doi: 10.1109/TIT.2024.3506866.

Yunlong Hou, Vincent Y. F. Tan, and Zixin Zhong. Almost minimax optimal best arm identification in piecewise stationary linear bandits. In *Advances in Neural Information Processing Systems*, volume 37, 2024.

International Warfarin Pharmacogenetics Consortium. Estimation of the warfarin dose with clinical and pharmacogenomic data. *New England Journal of Medicine*, 360(8):753–764, 2009.

Kyoungseok Jang, Chicheng Zhang, and Kwang-Sung Jun. Efficient low-rank matrix estimation, experimental design, and arm-set-dependent low-rank bandits. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 18696–18746, 2024.

Yassir Jedra, William Reveillard, Stefan Stojanovic, and Alexandre Proutiere. Low-rank bandits via tight two-to-infinity singular subspace recovery. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, 2024.

Kwang-Sung Jun, Rebecca Willett, Stephen Wright, and Robert Nowak. Bilinear bandits with low-rank structure. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 3163–3172, 2019.

Yue Kang, Cho-Jui Hsieh, and Thomas C.M. Lee. Efficient frameworks for generalized low-rank matrix bandit problems. In *Advances in Neural Information Processing Systems*, volume 35, 2022.

Junpei Komiyama, Edouard Fouché, and Klemens Böhm. Finite-time analysis of globally nonstationary multi-armed bandits. *Journal of Machine Learning Research*, 25(112):1–56, 2024.

Lihong Li, Wei Chu, John Langford, and Robert E. Schapire. A contextual-bandit approach to personalized news article recommendation. In *Proceedings of the 19th International Conference on World Wide Web*, pages 661–670, 2010.

Yangyi Lu, Amirhossein Meisami, and Ambuj Tewari. Low-rank generalized linear bandit problems. In *Proceedings of The 24th International Conference on Artificial Intelligence and Statistics*, volume 130 of *Proceedings of Machine Learning Research*, pages 460–468, 2021.

Marius Memmel, Andrew Wagenmaker, Chuning Zhu, Patrick Yin, Dieter Fox, and Abhishek Gupta. ASID: Active exploration for system identification in robotic manipulation. In *The Twelfth International Conference on Learning Representations*, 2024.

Hyejin Park, Seiyun Shin, Kwang-Sung Jun, and Jungseul Ok. Transfer learning in bandits with latent continuity. *IEEE Transactions on Information Theory*, 70(11):7952–7970, 2024. doi: 10.1109/TIT.2024.3441669.

Yoan Russac, Claire Vernade, and Olivier Cappé. Weighted linear bandits for non-stationary environments. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

Ayush Sekhari, Karthik Sridharan, Wen Sun, and Runzhe Wu. Contextual bandits and imitation learning with preference-based active queries. *Advances in Neural Information Processing Systems*, 36:11261–11295, 2023.

Yevgeny Seldin, Peter L. Bartlett, Koby Crammer, and Yasin Abbasi-Yadkori. Prediction with limited advice and multiarmed bandits with paid observations. In *Proceedings of the 31st International Conference on Machine Learning*, volume 32 of *Proceedings of Machine Learning Research*, pages 280–287, 2014.

Ashwin Srinivasan. Statlog (Landsat Satellite) data set. *UCI Machine Learning Repository*, 1990.

Stefan Stojanovic, Yassir Jedra, and Alexandre Proutiere. Spectral entry-wise matrix estimation for low-rank reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

Joel A. Tropp. User-friendly tail bounds for sums of random matrices. *Foundations of Computational Mathematics*, 12(4):389–434, 2012.

Aaron D. Tucker, Caleb Biddulph, Claire Wang, and Thorsten Joachims. Bandits with costly reward observations. In *Proceedings of the 39th Conference on Uncertainty in Artificial Intelligence*, volume 216 of *Proceedings of Machine Learning Research*, pages 2147–2156, 2023.

Yi Yu, Tengyao Wang, and Richard J. Samworth. A useful variant of the Davis–Kahan theorem for statisticians. *Biometrika*, 102(2):315–323, 2015.

Appendix: Table of Contents

A Organization

Paper organization. Section 2 reviews related work. Section 3 presents the formal model. Section 4 develops the single-play identification results. Section 5 introduces SPSC and its adaptive variant. Section 6 states the main regret guarantees. Section 7 reports the empirical results. Section 8 concludes. The appendix contains a discussion of limitations (Appendix B), a comparison with prior work (Appendix C), a notation reference (Appendix D), supporting statements organized by topic (Appendix E), complete proofs in dependency order (Appendix F), and additional experiments (Appendix G).

B Discussion and Limitations

Practical scope of the assumptions. We briefly address the practical burden of each structural condition, noting that the paper provides formal robustness or necessity results for all of them. *Probe realizability* (Assumption 3): the requirement $\mathrm{supp}(Q) \subseteq \mathcal{A}_t$ holds whenever the learner can inject a small amount of randomization into the action selection (e.g., ϵ-greedy with isotropic noise), which is standard in recommendation systems and clinical trials. Concretely, we outline three deployable instantiations:

1. *Recommendation with exploration budget.* The platform reserves a fraction μ of traffic for exploration (standard practice at Netflix, Spotify, and news platforms [Li et al., 2010]). On these rounds, items are drawn from a distribution satisfying $\mathbb{E}[uu^\top] \succeq \rho I_d$—e.g., uniform over a diverse item pool or ϵ-greedy perturbation of the policy. The per-probe cost c corresponds to the expected revenue loss from a non-personalized recommendation.
2. *Clinical trials with adaptive randomization.* In multi-arm trials with continuous dosing or combination therapies, the action space $\mathcal{A}_t$ naturally spans $\mathbb{R}^d$. Randomized treatment arms serve as probe rounds; the cost c reflects the ethical/statistical cost of a non-optimized assignment. The bounded state–noise coupling condition holds when patient response noise (measurement error, biological variability) is independent of treatment efficacy.

3. *Sensor-based control (robotics, HVAC, power grids).* The controller periodically injects small random perturbations for system identification—a standard practice in adaptive control [Memmel et al., 2024]. Sub-Gaussian noise (Assumption 4(ii), truncated per Remark 10) holds for physical actuators with known operating ranges; bounded state–noise coupling holds because sensor noise is exogenous.

In each case, the key requirement is that the learner controls a fraction of rounds and can randomize actions across all d dimensions during those rounds.

When full coverage does not hold. The requirement fails when the learner cannot randomize freely, for example in safety constrained systems where certain action dimensions are locked, or on platforms where the item pool does not span $\mathbb{R}^d$. In such settings our guarantees do not apply: Theorem 14 shows that restricting probes to a proper subspace $S \subsetneq \mathbb{R}^d$ causes $\Omega(T)$ regret, and the Experiment C ablation (Figure 5c) confirms monotonic degradation as coverage decreases. Whether partial coverage (e.g., probes spanning a subspace containing $\mathrm{range}(B_k^\star)$) can yield intermediate guarantees is an open question; the lifted operator approach requires invertibility of $\mathcal{K}$, which fails without full rank $\mathbb{E}[uu^\top]$. Theorems 14–16 and Experiment C show that relaxing full coverage causes *provably unavoidable* $\Omega(T)$ regret blow-up, indicating that this is not an artifact of our analysis but a fundamental information barrier. *Unknown noise variance, estimated online* (Assumption 4(ii)): Theorem 3 and Experiment A show that misspecification up to $5\times$ the true variance causes $< 5\%$ regret change, because the bias is a scaled identity for Gaussian probes and eigenvectors are preserved (Proposition 5). *Bounded state–noise coupling* (Assumption 4(iii)): Corollary 1 and Experiment B show graceful $O(\epsilon_\times T)$ degradation under bounded cross-correlation; even $\epsilon_\times = 1.0$ changes regret by $< 1\%$. Proposition 8 shows that without *any* bound, identifiability fails entirely — so the assumption cannot be removed, only relaxed. *Sub-Gaussian probe noise* (Remark 10): this is a technical condition for the matrix Freedman inequality; for Gaussian noise it is achieved by standard truncation at $L_\varepsilon = O(\sigma_\varepsilon\sqrt{\log T})$ with negligible probability loss. *Known rank r*: the algorithm degrades gracefully under misspecification (overestimation costs a factor $\hat{r}/r$; underestimation adds signal-leak penalty proportional to the missed eigenvalue), and eigenvalue thresholding at $2\tau_k$ provably recovers the true rank when the eigengap exceeds the concentration radius (Eq. (14)). In summary, each assumption is either *necessary* (with formal impossibility results), *robust to violation* (with quantified degradation bounds and supporting experiments), or *standard* (bounded noise via truncation).

Noise assumptions. The probe-side conditions in Assumption 4 are stated in their weakest usable form: unknown variance (estimated online) and bounded state–noise coupling. Theorem 3 quantifies the penalty from both sources; Corollary 1 handles cross-correlation in isolation; and Proposition 8 shows that without any coupling bound, identifiability fails altogether.

Robustness to rank misspecification. The algorithm takes the latent rank r as input, but is robust to misspecification.

Overestimation ($\hat{r} > r$). If the algorithm uses $\hat{r} > r$ eigenvectors of $\widehat{M}_k$, the extra $\hat{r} - r$ directions capture noise rather than signal. Exploitation runs in $\hat{r}$ dimensions, and Theorem 2 applies with r replaced by $\hat{r}$:

$$\mathrm{DynReg}_T^{(c)} = \widetilde{O}(\hat{r}\sqrt{T}) + \widetilde{O}(T^{2/3}) + O(V).$$

The cost is a factor of $\hat{r}/r$ in the statistical term; no other component of the bound is affected.

Underestimation ($\hat{r} < r$). The learner misses $r - \hat{r}$ subspace directions. The signal in the missed directions is not captured by $\widehat{P}_k$ and contributes to the subspace mismatch term $\Gamma_k = S_w \varepsilon_k$ in Theorem 8. Concretely, $\varepsilon_k \geq \lambda_{\hat{r}+1}(\bar{M}_k^{\mathrm{probe}})/\lambda_{\hat{r}}(\bar{M}_k^{\mathrm{probe}})$, so the regret acquires an additive $O(\lambda_{\hat{r}+1} \cdot S_w R_{\mathcal{A}} T/\lambda_{\hat{r}})$ term proportional to the signal energy in the missed directions. When the eigengap between rank-r and rank-$(\hat{r}+1)$ components is large, the penalty is small.

Practical rank selection. A data-driven choice is eigenvalue thresholding:

$$\hat{r}_k := \max\bigl\{j \in \{1, \dots, d\} : \lambda_j(\widehat{M}_k) > 2\tau_k\bigr\}, \qquad \tau_k := 2R_X\sqrt{\frac{\log(2d/\delta)}{m_k}}, \tag{14}$$

where τ_k is the concentration radius from Theorem 5. By Weyl's inequality, if the population eigengap satisfies $\lambda_r(\bar{M}_k^{\mathrm{probe}}) - \lambda_{r+1}(\bar{M}_k^{\mathrm{probe}}) > 4\tau_k$ (which holds for m_k sufficiently large), then $\hat{r}_k = r$ with probability at least $1 - \delta$. When this event holds, all guarantees of Theorem 2 apply unchanged.

Probe-cost lower bound. Theorem 17 (Appendix E.11) proves an $\Omega(\sqrt{cT})$ minimax lower bound for the probe–subspace tradeoff via a Le Cam two-point argument. The bound applies to the class of explore-then-commit (ETC) policies—any method that learns a subspace from full-dimensional probes and subsequently exploits within the committed subspace. SPSC belongs to this class (with interleaved scheduling and an r-dimensional committed subspace), so the lower bound is directly relevant.

The gap between $\Omega(\sqrt{cT})$ and the upper bound $\widetilde{O}(c^{1/3}T^{2/3})$ is structurally well-understood: it arises because hard projection onto the estimated subspace incurs $O(\varepsilon)$ per-round bias from subspace error, while the information-theoretic regret scales as $\Omega(\varepsilon^2)$ (Remark 15). Concretely, the gap lives entirely in the probe-cost exponent ($c^{1/2}$ vs. $c^{1/3}$), not in the T-dependence, and the $\Omega(\sqrt{cT})$ lower bound already confirms that the $\widetilde{O}(T^{2/3})$ probe–subspace tradeoff term *cannot be eliminated* by any ETC policy—it is inherent to the learn-then-exploit paradigm.

Whether a tighter $\Omega(c^{1/3}T^{2/3})$ lower bound holds for the full class of adaptive policies (which may update the subspace continuously rather than committing) remains open. A natural path to closing the gap from the upper-bound side is a soft-projection scheme that achieves $O(\varepsilon^2)$ per-round cost instead of $O(\varepsilon)$; we leave this as an explicit open question.

Dimension scaling. The theoretical improvement from $d\sqrt{T}$ to $r\sqrt{T}$ is most impactful when $r \ll d$. The Covertype benchmark (Table 1) confirms this: SPSC/LinUCB ratios decrease steadily from 1.06 at d=55 to 0.94 at d=155 for r=1, and from 0.84 to 0.72 for r=20. The Pendigits and Satimage regime studies (Appendix G.5) show the same trend across real datasets.

Remark 4 (Regime of applicability)**.** *Theorem 2 yields a net improvement over ambient-space methods when the statistical saving from dimension reduction outweighs the probe cost. Comparing the leading terms of Theorem 2 and Theorem 13,*

$$\underbrace{\widetilde{O}(r\sqrt{T})}_{\textit{SPSC exploitation}} + \underbrace{\widetilde{O}(T^{2/3})}_{\textit{probe cost}} < \underbrace{\widetilde{O}(d\sqrt{T})}_{\textit{ambient exploitation}}$$

holds when $(d - r)\sqrt{T} > C\,T^{2/3}$ for a universal constant $C > 0$, i.e.

$$d - r \; > \; C\,T^{1/6}. \tag{15}$$

For $T = 5{,}000$, condition (15) *requires $d - r \gtrsim 4.1$; for $T = 50{,}000$, it requires $d - r \gtrsim 6.1$. In particular:*

- ***SPSC dominates*** *when d is large and r is small relative to $T^{1/6}$ (high-dimensional, low-rank regime).*
- ***Ambient LinUCB dominates*** *when $d - r \le T^{1/6}$, because the probe cost $\widetilde{O}(T^{2/3})$ exceeds the statistical saving $(d - r)\sqrt{T}$.*
- ***The crossover is sharp****: the empirical ratio $\mathrm{DynReg}^{(c)}_{\mathrm{SPSC}}/\mathrm{DynReg}^{(c)}_{\mathrm{LinUCB}}$ crosses 1.0 at $d - r \approx T^{1/6}$ in both synthetic (Figure 1) and real-data benchmarks (Table 1), confirming that condition* (15) *is predictive at finite T.*

C Comparison with Prior Work

D Notation

Table 2: Comparison of our work with related literature. "Prob." indicates whether a probabilistic regret bound is provided; "CP?" indicates whether change-point adaptation is addressed.

Reference	**Venue/ Year**	**Regret / Theory**	**Prob.**	**CP?**	**Difference vs. ours**
Kang et al. [2022]	NeurIPS'22	$\tilde{O}\left(\sqrt{(d_1+d_2)MrT}\right)$	Y	N	Stationary; probing not budgeted.
Stojanovic et al. [2023]	NeurIPS'23	Entry-wise / subspace recovery	N/Y	N	Emphasizes estimation; no probe-budget.
Jedra et al. [2024]	ICML'24	$\tilde{O}\left(r^{5/4}(m+n)^{3/4}\sqrt{T}\right)$	Y	N	Stationary; lacks re-learning.
Jang et al. [2024]	ICML'24	Arm-set-dependent regret	Y	N	Stationary; no CP detection.
Cai et al. [2024]	AoS'24	Minimax regret under covariate shift	N	N	Cross-domain transfer; not piecewise.
Park et al. [2024]	IEEE TIT'24	Estimator matches IT lower limits	N	N	Transfers latent structure; no CP tracking.
Abbasi-Yadkori et al. [2023]	JMLR'23	$\tilde{O}(\sqrt{KN(S+1)})$	N	Y	Unstructured K-armed; no low-rank.
Komiyama et al. [2024]	JMLR'24	ADR-bandit finite-time	N	Y	Unstructured; no low-rank + probing.
Hou et al. [2024]	NeurIPS'24	Fixed-confidence sample complexity	Y	Y	BAI objective; no low-rank.
Memmel et al. [2024]	ICLR'24	Fisher-information exploration	Y	N	Control/system ID; not bandits.

Table 3: Compact positioning of the proposed framework relative to the closest research directions. Our setting combines low-rank reward structure, piecewise-stationary adaptation, and explicit costly probing in a single regret framework.

Work class	**Low-rank**	**Nonstationary**	**Costly probing**	**Main gap vs. ours**
Low-rank bandits	✓	✗	✗	Learn a low-rank representation in stationary settings, but do not model piecewise-stationary changes or budget the cost of sensing.
Nonstationary linear/contextual bandits	✗	✓	✗	Adapt to changing rewards in ambient parameter space, but do not exploit latent low-rank structure for dimension-reduced exploitation.
Costly-information / active-query bandits	✗	✗	✓	Model the cost of acquiring information, but do not address low-rank subspace recovery under changing representations.
Nonstationary representation learning	✓	✓	✗	Allow the representation to vary over time, but do not incorporate explicit sensing costs into the decision objective.
This paper	✓	✓	✓	**Combines low-rank structure, piecewise-stationary adaptation, and explicit costly probing in a single identification-to-exploitation framework.**

Table 4: Regret scaling comparison with published baselines. SPSC is the only method that simultaneously handles non-stationarity (piecewise-changing subspace), unknown subspace (learned online), and explicit probe-cost accounting.

Algorithm	Reference	Regret scaling	Key difference vs. SPSC
LinUCB	Li et al. [2010]	$O(d\sqrt{T})$	No subspace exploitation
LowOFUL	Jun et al. [2019]	$O(r\sqrt{T})$	Requires known subspace B
LowESTR	Lu et al. [2021]	$O(r\sqrt{T})$	Stationary; learns B from rewards
VOFUL	Cesa-Bianchi et al. [2023]	$O(\sqrt{rdT})$	Non-stationary; no explicit probing
SPSC (ours)	This work	$\widetilde{O}(r\sqrt{T} + c^{1/3}T^{2/3})$	Non-stationary; learns B via probing

Symbol	**Meaning**	**Known?**	**Assumptions / role; first appears**
t, T	Round index, horizon	Yes	$t = 1, \dots, T$. [§3]
d, r	Ambient & latent dim.	Yes	$r \ll d$; both assumed known. [§3]
K	Number of segments	Alg. 1: Yes	Known for Alg. 1; estimated in Alg. 2. [§3]
$\mathcal{I}_k, \tau_k$	Segment, change pts.	Yes	$\mathcal{I}_k = \{\tau_{k\text{-}1}, \dots, \tau_k - 1\}$. [§3]
$\mathcal{A}_t$	Action set at t	Yes	Revealed; bounded by $R_{\mathcal{A}}$. [§3]
x_t	Action at round t	Yes	$x_t = u_t$ on probe rounds; UCB-selected otherwise. [§3]
y_t	Observed reward	Yes	$y_t = x_t^\top \theta_t + \varepsilon_t$. [§3]
θ_t	Latent reward param.	No	$\theta_t = B_k^\star w_t$. [§3]
$B_k^\star$	Representation matrix	No	$d \times r$, orthonormal columns; piecewise constant. [§3]
w_t	Latent state	No	Evolves via LDS within segment k. [§3]
A_k	LDS transition matrix	No	$\rho(A_k) \le 1 - \alpha$; Assumption 1. [§3]
η_{t-1}	LDS innovation	No	Mean zero, covariance $\Sigma_{\eta,k}$. [§3]
$\Sigma_{\eta,k}$	Innovation covariance	No	$\lambda_r(\Sigma_{\eta,k}) \ge \underline{\lambda} > 0$. [§3]
α, α_0	Stability margin, bound	Partial	$\rho(A_k) \le 1 - \alpha$; $\alpha_0 \le \alpha$ known. [§3]
ε_t	Observation noise	No	Mean zero, sub-Gaussian; Assump. 4. [§3]
σ_ε^2	Probe noise variance	No	Unknown; estimated online. Assump. 4(ii). [§3]
$\mathcal{H}_{t-1}$	Pre-decision history	Yes	σ-field of past. [§3]
$R_{\mathcal{A}}, L, L_\varepsilon$	Action/probe/noise bnds	Yes	Assumptions 2–4; L_ε via truncation (Remark 10). [§3]
Q, ρ	Probe distr., bound	Yes	$\mathbb{E}[uu^\top] \succeq \rho I_d$; Assump. 3. [§3]
$\mathcal{T}_{\text{probe}}, \mathcal{T}_k, m_k$	Probe sets, count	Derived	$\mathcal{T}_k = \mathcal{T}_{\text{probe}} \cap \mathcal{I}_k$; $m_k = \lvert\mathcal{T}_k\rvert$. [§3]
c	Per-probe cost	Yes	Appears in costed regret. [§3]
$\mathrm{DynReg}_T^{(c)}$	Costed dynamic regret	No	Performance metric; Eq. (4). [§3]
$\widetilde{M}_t$	Predictable 2nd-moment	No	$\mathbb{E}[\theta_t \theta_t^\top \mid \mathcal{H}_{t-1}]$; time-varying. [§3]
$\bar{M}_k^{\text{probe}}$	Segment probe average	No	$m_k^{-1} \sum_{t \in \mathcal{T}_k} \widetilde{M}_t$; identifies subspace. [§3]
$\bar{S}_k^{\text{probe}}$	Latent probe covariance	No	$m_k^{-1} \sum_{t \in \mathcal{T}_k} \mathbb{E}[w_t w_t^\top \mid \mathcal{H}_{t-1}]$. [§3]
$\lambda_{\min}$	Probe-time eigengap	No	$\lambda_r(\bar{S}_k^{\text{probe}}) \ge \lambda_{\min} > 0$. [§3]
$P_k^\star$	True seg. projector	No	$P_k^\star = B_k^\star (B_k^\star)^\top$. [§3]
$\mathcal{K}, \mathcal{K}^{-1}$	Probe moment operator	Derived	$\mathcal{K}(M) = \mathbb{E}[(u^\top M u) uu^\top]$; invertible. [§4]
$C_{\mathcal{K}}$	Op. norm of $\mathcal{K}^{-1}$	Derived	≤ 1 for isotropic Gaussian. [§4]
s_t	Centered probe stat.	Derived	$s_t = y_t^2 - \hat{\sigma}^2$. [§4]
G_t	Lifted probe sample	Derived	$G_t = \mathcal{K}^{-1}(s_t u_t u_t^\top)$; bias $O(\delta_\sigma + \epsilon_\times)$. [§4]
$\widehat{M}_k$	Empirical seg. estimator	Derived	Unbiased for $\bar{M}_k^{\text{probe}}$. [§5]
$\widehat{U}_k, \widehat{P}_k$	Est. basis, projector	Derived	Top-r eigenvectors/projector of $\widehat{M}_k$. [§5]
ε_k	Projector error	Derived	$\lVert\widehat{P}_k - P_k^\star\rVert_2$; rate $O(m_k^{-1/2})$. [§6]
S_w	Latent state bound	No	$\lVert w_t\rVert_2 \le S_w \Rightarrow \lVert\theta_t\rVert_2 \le S_w$. [§6]

Symbol	Meaning	Known?	Assumptions / role; first appears
R_s, R_X	Concentration radii	Derived	$R_s = (LS_w+L_\varepsilon)^2+\sigma_\varepsilon^2$; $R_X = C_\mathcal{K}L^2R_s+S_w^2$. [§6]
C_{sub}	Projector constant	Derived	$C_{\mathrm{sub}} = 8R_X/\lambda_{\min}$. [§6]
W, λ	Window, regularizer	Yes	Sliding window length; ridge param. [§5]
$\beta_t^{(r,W)}$	Confidence radius	Derived	Eq. (7). [§6]
Γ_k	Subspace mismatch	Derived	$\Gamma_k = S_w\varepsilon_k$. [§6]
V_k, V	Within-seg., total drift	No	$V_k = \sum \lVert\theta_{s+1}-\theta_s\rVert_2$; $V = \sum_k V_k$. [§6]
S_t, b	Detector stat., threshold	Derived	Eq. (5), (12). [§5]
m_{relearn}	Recovery probe count	Yes	Probes after each detected change. [§6]
$D_{\max}$	Max detection delay	Derived	$(2n/\mu) + \tau_{\mathrm{burn}}$. [§6]

E Supporting Statements

This appendix collects all auxiliary results whose statements are needed by the main theorems. No proofs appear here; every proof is deferred to Appendix F, which orders all proofs so that each one references only previously established results.

Table 6: **Complete roadmap of theoretical results.** Each row summarizes one formal statement, its role in the paper, and what it depends on. Statements are grouped by function: *structural assumptions* define the model; *identification* results show the subspace can be recovered; *concentration* results quantify finite-sample error; *bandit tools* are standard technical lemmas; *regret bounds* compose all preceding results into performance guarantees; *impossibility and lower bounds* delineate the boundaries of what is achievable.

Statement	Content (one-line summary)	Role & dependencies
***Structural assumptions** (§3) — define the model and enable identification-to-exploitation*		
Assumption 1 (Stability & excitation)	LDS transition $\rho(A_k)\le 1-\alpha$; innovations i.i.d. with full-rank covariance $\Sigma_{\eta,k}\succeq\underline{\lambda}I_r$	Bounded latent state; probe-time covariance has rank r. Used by Lemma 1, Lemma 10
Assumption 2 (Action bound)	$\lVert x\rVert_2\le R_\mathcal{A}$ for all $x\in\mathcal{A}_t$, all t	Bounds per-round regret and elliptical potential; used by all regret theorems
Assumption 3 (Probe realizability)	Probe Q: $\mathbb{E}[uu^\top]\succeq\rho I_d$, $\lVert u\rVert\le L$, $\mathrm{supp}(Q)\subseteq\mathcal{A}_t$	Full-dim coverage; enables Lemma 2 and lower bounds (Thm. 14–16)
Assumption 4 (Probe noise)	(i) mean-zero noise, (ii) sub-Gaussian with unknown σ_ε^2 (estimated online), (iii) bounded state–noise coupling $\lVert\mathbb{E}[\varepsilon_t\theta_t]\rVert\le\epsilon_\times$	Three conditions for Prop. 1; necessity shown in Props. 7–9
Remark 10 (Sub-Gaussian → bounded)	Truncation at $L_\varepsilon=\sigma_\varepsilon\sqrt{2\log(T/\delta)}$ gives $\lvert\varepsilon_t\rvert\le L_\varepsilon$ w.h.p.	Deterministic bound on s_t (Lemma 3); feeds concentration chain
***Identification results** (§3–§4) — single-play probes recover the latent subspace*		
Lemma 1 (Probe-time excitation)	$\lambda_r(\bar S_k^{\mathrm{probe}})\ge\lambda_{\min}>0$: average latent covariance on probe rounds has full rank r	Subspace identifiable from probes. Dep: Assump. 1
Lemma 2 ($\mathcal{K}$ invertibility)	Probe moment operator $\mathcal{K}(M)=\mathbb{E}[(u^\top Mu)\,uu^\top]$ is invertible on $\mathbb{R}^{d\times d}_{\mathrm{sym}}$	Lifts scalar data to matrix estimates. Dep: Assump. 3
Proposition 1 (Quadratic measurement)	$\mathbb{E}[s_t\mid\mathcal{H}_{t-1},u_t]=u_t^\top\widetilde{M}_t\,u_t$: each probe yields a quadratic measurement of the predictable second moment	Core identification step. Dep: Assump. 4
Proposition 2 (Segment factorization)	$\bar M_k^{\mathrm{probe}}=B_k^\star\bar S_k^{\mathrm{probe}}(B_k^\star)^\top$; range $\subseteq$ range$(B_k^\star)$	Links probe average to true subspace

Continued on next page

Table 6 continued from previous page

Statement	**Content (one-line summary)**	**Role & dependencies**
Proposition 3 (Subspace identification)	$\mathrm{range}(\bar{M}_k^{\mathrm{probe}}) = \mathrm{range}(B_k^\star)$: exact subspace recovery at population level	Completes identifiability. Dep: Lemma 1, Prop. 2
Proposition 4 (Unbiased lifted probe)	$\mathbb{E}[G_t \mid \mathcal{H}_{t-1}] = \widetilde{M}_t$: each lifted sample unbiased for $\widetilde{M}_t$	Justifies matrix estimator. Dep: Prop. 1, Lemma 2
Proposition 5 (Biased measurement)	Under misspecification $\delta_\sigma \neq 0$: $\mathbb{E}[\hat{G}_t \mid \mathcal{H}_{t-1}] = \widetilde{M}_t - \widetilde{B}$ with $\Vert\widetilde{B}\Vert \leq \vert\delta_\sigma\vert C_{\mathcal{K}} L^2$; for Gaussian probes, bias is scaled identity so eigenvectors preserved	Robustness. Feeds Thm. 7, Cor. 2
Non-identification *(§E.2) — each structural condition is individually necessary*		
Proposition 7 (No id. w/o known variance)	If σ_ε^2 unknown, $\widetilde{M}_t$ not identifiable from $\mathbb{E}[y_t^2 \mid \mathcal{H}_{t-1}, u]$	Assump. 4(ii) is necessary
Proposition 8 (No id. w/o orthogonality)	If state–noise coupling is unbounded, $\widetilde{M}_t$ not identifiable even with known variance	Assump. 4(iii) is necessary
Proposition 9 (No id. w/o full coverage)	If probes in proper subspace $S \subsetneq \mathbb{R}^d$, $\widetilde{M}_t$ not identifiable	Assump. 3 is necessary; quantified by Thms. 14–16
Concentration chain *(§E.4) — finite-sample subspace recovery*		
Lemma 3 (Uniform $\vert s_t\vert$ bound)	$\vert s_t\vert \leq R_s := (LS_w + L_\varepsilon)^2 + \sigma_\varepsilon^2$	Deterministic; feeds Lemma 5
Lemma 4 (Lifted probe MDS)	$X_t := G_t - \widetilde{M}_t$ is a matrix martingale difference sequence	Structural; enables Thm. 5
Lemma 5 (Operator norm bound)	$\Vert X_t\Vert_2 \leq R_X := C_{\mathcal{K}} L^2 R_s + S_w^2$	Deterministic. Dep: Lemmas 3, 2
Lemma 6 (Pred. quad. variation)	$\Vert \sum \mathbb{E}[X_t^2 \mid \mathcal{H}_{t-1}]\Vert_2 \leq m_k R_X^2$	Variance proxy for Thm. 5. Dep: Lemma 5
Theorem 5 (Matrix concentration)	$\Vert\widehat{M}_k - \bar{M}_k^{\mathrm{probe}}\Vert_2 \leq 2R_X\sqrt{\log(2d/\delta)/m_k}$ w.h.p. (Matrix Freedman)	Core finite-sample result. Dep: Lemmas 4–6
Theorem 6 (Projector concentration)	$\Vert\widehat{P}_k - P_k^\star\Vert_2 \leq C_{\mathrm{sub}}\sqrt{\log(2d/\delta)/m_k}$, $C_{\mathrm{sub}} = 8R_X/\lambda_{\min}$ (Davis–Kahan)	ε_k rate for subspace error. Dep: Thm. 5
Theorem 7 (Biased concentration)	Under variance misspecification: same rate with inflated $\hat{R}_X$ plus additive bias	Robustness. Dep: Prop. 5
Corollary 2 (Projector under misspec.)	$\Vert\widehat{P}_k - P_k^\star\Vert_2 \leq \hat{C}_{\mathrm{sub}}\sqrt{\log(2d/\delta)/m_k} + \Delta_\sigma$; graceful degradation	Connects to Thm. 3. Dep: Thm. 7
Bandit tools *(§E.6) — standard lemmas for projected exploitation*		
Lemma 7 (Self-normalized in $\mathbb{R}^r$)	$\Vert M_t\Vert_{V_t^{-1}} \leq \sigma_\varepsilon\sqrt{2\log(\det(V_t)^{1/2}/(\lambda^{r/2}\delta))}$ for sub-Gaussian noise	Standard (Abbasi-Yadkori et al. 2011) in r dims
Lemma 8 (Determinant bound)	$\log(\det(V)/\lambda^r) \leq r\log(1 + NR_{\mathcal{A}}^2/(\lambda r))$	AM–GM on eigenvalues; feeds Lemma 9
Lemma 9 (Elliptical potential)	$\sum_{t=1}^N \Vert z_t\Vert_{V_{t-1}^{-1}} \leq \sqrt{2rN\log(1 + NR_{\mathcal{A}}^2/(\lambda r))}$	Per-round $\to$ cumulative bound; feeds all regret thms.
Regret bounds *(§6, §E.5) — main performance guarantees*		

Continued on next page

Table 6 continued from previous page

Statement	Content (one-line summary)	Role & dependencies
Theorem 8 (Segment windowed regret)	Per-segment: $R_k^{\exp} \leq 2\beta_k^{(r,W)}\sqrt{2rn_kL_W}+2S_wR_{\mathcal{A}}\varepsilon_kn_k+2R_{\mathcal{A}}\sum V_{k,t}(W)$	Building block for Thm. 1. Dep: Lemmas 7–9, Thm. 6
Theorem 1 (Full-horizon windowed)	Sums Thm. 8 over K segments; adds probe cost $(2R_{\mathcal{A}}S_w+c)\sum m_k$	Intermediate. Dep: Thm. 8
Theorem 2 (**Main result**)	$\mathrm{DynReg}_T^{(c)} = \widetilde{O}(r\sqrt{T})+\widetilde{O}(T^{2/3})+O(V)$ via optimal probe allocation $m_k^\star = (B\ell_k/(2A))^{2/3}$	Central theorem. Dep: Thms. 1, 6
Theorem 3 (General: unknown variance + bounded coupling)	Same rate plus $O(\lvert\delta_\sigma\rvert T/\lambda_{\min})$; rate unchanged if $\lvert\delta_\sigma\rvert = O(T^{-1/3})$	Robustness. Dep: Thm. 2, Cor. 2
Theorem 4 (**Adaptive regret**)	Unknown boundaries: same leading rate plus $O(K)$ overhead; $F_T = 0$ w.h.p.	Practical variant. Dep: Thm. 2, Thms. 11–12
Theorem 13 (Ambient reference)	Without projection: $\widetilde{\mathcal{O}}(d\sqrt{T}+T^{2/3})$; improvement $d \to r$ comes from projection	Baseline comparison
Detector guarantees *(§E.7) — for the adaptive variant (Alg. 4)*		
Lemma 10 (Within-segment variation)	$\lVert\widetilde{M}_t - \widetilde{M}_s\rVert_{\mathrm{op}} \leq \Delta_{\mathrm{within}}(\tau_{\mathrm{burn}})$: small second-moment variation within a segment	Calibrates threshold (12). Dep: Assump. 1
Theorem 11 (No-change control)	Under no change: $\mathbb{P}(S_t > b) \leq \delta_{\mathrm{FA}}/T$; $F_T = 0$ w.h.p.	False-alarm control. Dep: Thm. 5, Lemma 10
Theorem 12 (Post-change detection)	If $\Delta_k \geq 2b$: detector fires within delay $D_{\max} = (2n/\mu)+\tau_{\mathrm{burn}}$ w.h.p.	Detection guarantee. Dep: Thm. 5, Lemma 10
Lower bounds and separations *(§E.10, §E.8) — limitations and distinctions from prior work*		
Theorem 14 ($\Omega(T)$ w/o coverage)	Actions restricted to proper subspace $S \Rightarrow$ minimax regret $\geq \mu T$	Full-dim probing is necessary
Theorem 15 (Same w/ probe confinement)	Probing within S provides zero information; regret still $\geq \mu T$	Strengthens Thm. 14
Theorem 16 (Two-point non-identifiability)	Incomplete coverage: two instances produce identical observations but different optima	Information-theoretic impossibility
Corollary 3 (Sublinear impossible)	No policy in $\Pi(S)$ achieves $o(T)$ regret over $\mathfrak{M}(S)$	Summary of Thms. 14–15
Props. 10–12 (Separations)	Model not reducible to Qin et al.; identification works where Qin's fails ($r \geq 2$); finite probes insufficient for a changing subspace	Positioning vs. prior work
Key remarks *— interpretive context*		
Remark 2 (Necessity of conditions)	All three conditions in Assump. 4 jointly necessary; no two suffice (Remark 9)	Summarizes Props. 7–9
Remark 3 (Reset justification)	Probe accumulator must reset at boundaries; pre-change data contaminates $\widehat{M}_k$	Motivates Alg. 1 design
Remark 11 ($\hat{K}$ vs. K)	On good event ($F_T = 0$): $\hat{K} = K$; overhead $O(1)$ in T for fixed K	Connects Thm. 4 to Thm. 2

Continued on next page

Table 6 continued from previous page

Statement	Content (one-line summary)	Role & dependencies
Thm. 17 (Probe-cost lower bound)	$\Omega(\sqrt{cT})$ minimax lower bound for ETC policies via Le Cam two-point argument; confirms probe–subspace tradeoff is fundamental	Tightness evidence; gap to $\widetilde{O}(c^{1/3}T^{2/3})$ upper bound is structurally explained (Remark 15)
Remark 12 (Probe-cost lower bound)	The $\Omega(\sqrt{cT})$ lower bound confirms the tradeoff is fundamental; remaining gap to $c^{1/3}T^{2/3}$ is from $O(\varepsilon)$ vs. $\Omega(\varepsilon^2)$ per-round cost	Strength: open gap is precisely characterized, not a missing result

E.1 Assumptions

Table 7: Probe-side conditions: practical meaning, inferential role, and what happens if they fail. The point of the table is not to restate the assumptions, but to clarify why they are structural for single-play subspace identification.

Condition	Practical meaning	Role in identification/control	If violated
Approximate probe variance knowledge (Assumption 4(ii)	Probe rounds are deliberately designed and can often be calibrated offline, or estimated from repeated measurements or a short burn-in.	Allows the centered probe statistic $s_t = y_t^2 - \hat{\sigma}_\varepsilon^2$ to isolate the quadratic signal $u_t^\top \widetilde{M}_t u_t$, which is the basis of the lifted estimator and subspace recovery.	A deterministic bias enters the lifted probe estimator. Under controlled misspecification, the regret acquires an additional error term, but the leading rate is preserved when the misspecification is sufficiently small.
Bounded state–noise coupling (Assumption 4(iii)	Natural under measurement-noise models where observation noise is added after the latent reward signal, e.g., sensor or instrumentation noise independent of the latent state.	Removes the mixed term between ε_t and θ_t, so probe observations identify the predictable second moment rather than a confounded object.	Without any control on this coupling, identifiability can fail. Under bounded coupling, one obtains graceful degradation rather than abrupt breakdown.
Full-dimensional probe coverage ($\mathbb{E}[uu^\top] \succeq \rho I_d$) (Assumption 3)	The probe mechanism deliberately excites all directions that may carry reward mass.	Ensures that no potentially active subspace direction is invisible to probing, and guarantees invertibility of the probe operator used to lift quadratic measurements into matrix estimates.	If the probe distribution misses directions, those directions cannot be recovered from single-play scalar feedback, and subspace identifiability fails.

E.2 Additional Identification Results

Remark 5 (Stability margin)**.** *The algorithm only requires a known lower bound $\alpha_0 \leq \alpha$ on the true stability margin. Any positive α_0 certifying strict stability suffices; it can be chosen conservatively small without affecting the regret rates.*

Remark 6 (Probe design condition)**.** *The condition $\mathbb{E}[uu^\top] \succeq \rho I_d$ requires the probe distribution to have positive second moment in every direction in $\mathbb{R}^d$, ensuring that probe actions span the ambient space. For isotropic Gaussian probes, $\mathcal{K}^{-1}(N) = \frac{1}{2}N - \frac{\mathrm{tr}(N)}{2(d+2)} I_d$, and $C_{\mathcal{K}} := \|\mathcal{K}^{-1}\|_{\mathrm{op}} \leq 1$.*

Remark 7 (Noise robustness)**.** *When only an estimate $\hat\sigma_\varepsilon^2$ is available with $|\hat\sigma_\varepsilon^2 - \sigma_\varepsilon^2| \le \delta_\sigma$, the regret acquires an extra additive term of order $O(\delta_\sigma T/\lambda_{\min})$ (Theorem 3). Setting $\delta_\sigma = O(T^{-1/3})$ via a burn-in pre-estimation phase keeps this at $\widetilde{O}(T^{2/3})$.*

Proposition 5 (Biased measurement under variance misspecification)**.** *Let $\delta_\sigma := \hat\sigma_\varepsilon^2 - \sigma_\varepsilon^2$ and $\widetilde{B} := \delta_\sigma\, \mathcal{K}^{-1}(\Sigma_Q)$. Then for every probe round $t \in \mathcal{T}_{\mathrm{probe}}$, $\mathbb{E}[\hat{G}_t \mid \mathcal{H}_{t-1}] = \widetilde{M}_t - \widetilde{B}$, and $\|\widetilde{B}\|_2 \le |\delta_\sigma|\, C_{\mathcal{K}}\, L^2$. The bias is deterministic and rank-d; for isotropic Gaussian probes it is a scaled identity $\widetilde{B} \approx \frac{\delta_\sigma}{2} I_d$, so eigenvectors of $\bar{M}_k^{\mathrm{probe}}$ are not corrupted—only eigenvalues are shifted.*

Proof in §F.1.7.

Proposition 6 (Biased measurement under bounded cross-correlation)**.** *Suppose Assumption 4(i)–(ii) hold with $\delta_\sigma = 0$ (exact noise variance), but Assumption 4(iii) is relaxed to*

$$\left\|\mathbb{E}[\varepsilon_t\, \theta_t \mid \sigma(\mathcal{H}_{t-1}, u_t)]\right\|_2 \le \epsilon_\times \qquad \textit{for all probe rounds } t \in \mathcal{T}_{\mathrm{probe}}.$$

Then for every probe round $t \in \mathcal{T}_{\mathrm{probe}}$,

$$\mathbb{E}[G_t \mid \mathcal{H}_{t-1}] = \widetilde{M}_t + \widetilde{B}_\times^{(t)},$$

where the bias satisfies $\|\widetilde{B}_\times^{(t)}\|_2 \le 2\, C_{\mathcal{K}}\, L^2\, \epsilon_\times$.

Proof in §F.1.8.

Corollary 1 (Regret under bounded cross-correlation)**.** *Under the hypotheses of Proposition 6, if $2\, C_{\mathcal{K}}\, L^3\, \epsilon_\times \le \lambda_{\min}/4$, then with probability at least $1-\delta$,*

$$\mathrm{DynReg}_T^{(c)} \le \widetilde{O}(r\sqrt{T}) + \widetilde{O}(T^{2/3}) + O(V) + O\!\left(\frac{\epsilon_\times\, L^3\, C_{\mathcal{K}}}{\lambda_{\min}}\, T\right).$$

In particular, if $\epsilon_\times = O(T^{-1/3})$ the penalty is $O(T^{2/3})$ and the leading rate is unchanged.

Proof. The segment-level bias $\|\bar{\widetilde{B}}_\times\|_2 \le 2C_{\mathcal{K}}L^3\epsilon_\times$ enters the projector error via Davis–Kahan exactly as in Corollary 2 (with $C_{\mathcal{K}}|\delta_\sigma|L^2$ replaced by $2C_{\mathcal{K}}L^3\epsilon_\times$). The additional projector error $\Delta_\times := 4 \cdot 2C_{\mathcal{K}}L^3\epsilon_\times/\lambda_{\min}$ propagates to the subspace-mismatch term $2S_wR_{\mathcal{A}}\varepsilon_k n_k$ in Theorem 1, contributing $2S_wR_{\mathcal{A}}\Delta_\times T = O(\epsilon_\times L^3 C_{\mathcal{K}} T/\lambda_{\min})$. The remainder of the regret analysis is identical to Theorem 2. ☐

Remark 8 (Graceful degradation and necessity)**.** *Corollary 1 shows that state–noise orthogonality (Assumption 4(iii)) can be relaxed to a bounded cross-correlation condition with graceful $O(\epsilon_\times T)$ degradation. Proposition 8 shows this degradation is unavoidable: without any cross-correlation bound, $\widetilde{M}_t$ is not identifiable, so the $O(\epsilon_\times T)$ penalty cannot be removed without additional structural assumptions.*

In practice, bounded cross-correlation holds naturally when noise arises from measurement error independent of the latent state (e.g., sensor noise), with $\epsilon_\times$ controlled by the degree of coupling between the noise source and the reward-generating process.

E.3 Non-Identification Results

Three complementary results show that dropping any single structural condition from Assumption 4 destroys identifiability of $\widetilde{M}_t$.

Proposition 7 (Non-identification without known noise variance)**.** *Fix t. If the observable is $g_t(u) := \mathbb{E}[y_t^2 \mid \mathcal{H}_{t-1}, u]$ but the conditional variance $v_t(u) := \mathbb{E}[\varepsilon_t^2 \mid \mathcal{H}_{t-1}, u]$ is unknown, then $\widetilde{M}_t$ is not identifiable.*

Proof in §F.2.1.

Proposition 8 (Non-identification without bounded state–noise coupling)**.** *Fix t. If σ_ε^2 is known but $\mathbb{E}[\varepsilon_t\theta_t \mid \mathcal{H}_{t-1}, u] \ne 0$ is allowed, then $\widetilde{M}_t$ is not identifiable from $h_t(u) := \mathbb{E}[y_t^2 - \sigma_\varepsilon^2 \mid \mathcal{H}_{t-1}, u]$.*

Proof in §F.2.2.

Proposition 9 (Non-identification without full-dimensional coverage)**.** *Let $S \subsetneq \mathbb{R}^d$ be a proper subspace. If all probe actions satisfy $u \in S$, then $\widetilde{M}_t$ is not identifiable from either linear or quadratic observations.*

Proof in §F.2.3.

Remark 9 (Joint necessity)**.** *No two of the three conditions are jointly sufficient; all three are simultaneously required. Proposition 7 shows failure even with bounded coupling and full coverage when the variance is completely unknown. Proposition 8 shows failure even with known variance and full coverage when the coupling bound is removed entirely. Proposition 9 shows failure even with known variance and bounded coupling when coverage is restricted.*

Structural justification: the role of probing. In a static setting, passive methods can eventually learn the subspace from exploitation data alone. However, in our piecewise non-stationary setting, the subspace changes at each change point, invalidating all previously learned structure. Deployed actions chosen for exploitation tend to align with the learner's current (now outdated) estimate, collapsing into a restricted proper subspace $S \subset \mathbb{R}^d$. Proposition 9 shows that measurements confined to S cannot identify the new subspace. Assumption 3 resolves this by guaranteeing full-dimensional probe coverage. The quantitative necessity of full-dimensional coverage is formalized in §E.10, where Theorem 14 shows that restricting deployed actions to any proper subspace forces $\Omega(T)$ regret.

E.4 Concentration Inequalities for Subspace Recovery

During segment k, the learner computes $\widehat{M}_k = \frac{1}{m_k} \sum_{t \in \mathcal{T}_k} G_t$. From Proposition 4, $X_t := G_t - \widetilde{M}_t$ is a martingale difference sequence with $\mathbb{E}[X_t \mid \mathcal{H}_{t-1}] = 0$. Throughout, Assumptions 1–4 hold and $\|w_t\|_2 \leq S_w$.

Remark 10 (From sub Gaussian to bounded noise)**.** *The matrix Freedman inequality requires a deterministic bound $|\varepsilon_t| \leq L_\varepsilon$. Under Assumption 4(ii), standard truncation at $L_\varepsilon := \sigma_\varepsilon \sqrt{2 \log(T/\delta)}$ ensures $\mathbb{P}(|\varepsilon_t| > L_\varepsilon) \leq \delta/T$; a union bound over all probe rounds introduces probability loss at most δ, absorbed into the existing failure probability. All subsequent results hold with $L_\varepsilon = \sigma_\varepsilon \sqrt{2 \log(T/\delta)}$.*

Lemma 3 (Uniform bound on probe statistic)**.** *On every probe round, $|s_t| \leq R_s := (LS_w + L_\varepsilon)^2 + \sigma_\varepsilon^2$.*

Proof in §F.3.1.

Lemma 4 (Lifted probe MDS)**.** $\mathbb{E}[X_t \mid \mathcal{H}_{t-1}] = 0$ *and* $\widehat{M}_k - \bar{M}_k^{\text{probe}} = \frac{1}{m_k} \sum_{t \in \mathcal{T}_k} X_t$.

Proof in §F.3.2.

Lemma 5 (Uniform operator norm bound)**.** $\|X_t\|_2 \leq R_X := C_\mathcal{K} \, L^2 \, R_s + S_w^2$.

Proof in §F.3.3.

Lemma 6 (Predictable quadratic variation)**.** $\| \sum_{t \in \mathcal{T}_k} \mathbb{E}[X_t^2 \mid \mathcal{H}_{t-1}] \|_2 \leq m_k R_X^2$.

Proof in §F.3.4.

Theorem 5 (Empirical matrix concentration)**.** *With probability at least $1 - \delta$: $\|\widehat{M}_k - \bar{M}_k^{\text{probe}}\|_2 \leq 2R_X \sqrt{\log(2d/\delta)/m_k}$.*

Proof in §F.3.5.

Theorem 6 (Projector concentration)**.** *If m_k is large enough that $\|\widehat{M}_k - \bar{M}_k^{\text{probe}}\|_2 \leq \lambda_{\min}/2$, then with probability at least $1 - \delta$: $\|\widehat{P}_k - P_k^\star\|_2 \leq C_{\text{sub}} \sqrt{\log(2d/\delta)/m_k}$, where $C_{\text{sub}} := 8R_X/\lambda_{\min}$.*

Proof in §F.3.6.

Theorem 7 (Biased concentration under variance misspecification). *Let $\hat{R}_X := R_X + 2C_{\mathcal{K}}|\delta_\sigma|L^2$. With probability at least $1-\delta$: $\|\widehat{M}_k^{\hat{\sigma}} - (\bar{M}_k^{\text{probe}} - \widetilde{B})\|_2 \le 2\hat{R}_X\sqrt{\log(2d/\delta)/m_k}$. Consequently, $\|\widehat{M}_k^{\hat{\sigma}} - \bar{M}_k^{\text{probe}}\|_2 \le 2\hat{R}_X\sqrt{\log(2d/\delta)/m_k} + C_{\mathcal{K}}|\delta_\sigma|L^2$.*

Proof in §F.3.7.

Corollary 2 (Projector recovery under misspecification). *If $C_{\mathcal{K}}|\delta_\sigma|L^2 \le \lambda_{\min}/4$ and $m_k \ge 64\hat{R}_X^2\log(2d/\delta)/\lambda_{\min}^2$, then $\|\widehat{P}_k - P_k^\star\|_2 \le \hat{C}_{\text{sub}}\sqrt{\log(2d/\delta)/m_k} + \Delta_\sigma$, where $\hat{C}_{\text{sub}} := 8\hat{R}_X/\lambda_{\min}$ and $\Delta_\sigma := 4C_{\mathcal{K}}|\delta_\sigma|L^2/\lambda_{\min}$.*

Proof in §F.3.8.

E.5 Supporting Regret Analysis

Theorem 8 (Segmentwise windowed exploitation regret). *Fix a segment $\mathcal{I}_k$. Under standard conditions (orthonormal $B_k^\star$, bounded $\|w_t\|_2 \le S_w$, bounded actions, subspace error ε_k, conditionally sub-Gaussian noise on exploitation rounds), with probability at least $1-\delta$, the exploitation regret in segment k satisfies*

$$R_k^{\text{exp}} \le 2\beta_k^{(r,W)}\sqrt{2rn_k\log\left(1+\frac{WR_{\mathcal{A}}^2}{\lambda r}\right)} + 2S_wR_{\mathcal{A}}\varepsilon_k n_k + 2R_{\mathcal{A}}\sum_{t\in E_k} V_{k,t}(W),$$

where $n_k := |E_k|$.

Proof in §F.4.5.

Remark 11 (Rate interpretation and parameter tuning). ***On $\hat{K}$ vs. K.*** *The random variable $\hat{K}$ denotes the total number of resets triggered by the detector. On the high-probability event ($\mathbb{P} \ge 1-\delta$) that $F_T = 0$, we have $\hat{K} = K$ exactly, so the oracle-equivalent term in* (13) *recovers the rate of Theorem 2 with K in place of $\hat{K}$. The detection-delay penalty scales as $KD_{\max} = K((2n/\mu) + \tau_{\text{burn}})$ and the recovery cost scales as Km_{relearn}. For fixed K, these are $O(1)$ in T and therefore absorbed into the lower-order AK term, provided n and m_{relearn} are constants independent of T. The minimum detectable change magnitude is $\Delta_{\text{det}} = 2b = O(R_X\sqrt{\log(dT)/n})$, which decreases as n increases. When K is unknown but $K = O(T^\beta)$ for some $\beta \in [0,1)$, the overhead terms are polynomial in T but dominated by the $T^{2/3}$ term as long as $K = o(T^{2/3})$.*

Remark 12 (Probe-cost lower bound). *Theorem 17 (Appendix E.11) proves $\Omega(\sqrt{cT})$ for the probe–subspace tradeoff via a symmetric two-point Le Cam argument. This confirms the tradeoff is fundamental: m probes (cost cm) yield $O(1/\sqrt{m})$ subspace accuracy, and the information-theoretic regret from misalignment is $\Omega(1/m)$ per round, giving $cm + T/m \ge 2\sqrt{cT}$. The remaining gap to the $\widetilde{O}(c^{1/3}T^{2/3})$ upper bound is precisely characterized: it arises from the $O(\varepsilon)$ per-round bias of hard projection vs. the $\Omega(\varepsilon^2)$ information-theoretic cost (Remark 15).*

E.6 Projected Exploitation in the Learned Subspace

E.6.1 Technical Lemmas

Lemma 7 (Self-normalized inequality in dimension r). *Let $M_t := \sum_{s=1}^t z_s\eta_s$, $V_t := \lambda I_r + \sum_{s=1}^t z_s z_s^\top$, where η_s is conditionally σ_ε-sub-Gaussian. Then for any $\delta \in (0,1)$, with probability at least $1-\delta$, simultaneously for all $t \ge 1$, $\|M_t\|_{V_t^{-1}} \le \sigma_\varepsilon\sqrt{2\log(\det(V_t)^{1/2}/(\det(\lambda I_r)^{1/2}\delta))}$.*

Proof in §F.4.1.

Lemma 8 (Determinant bound). *If $\|z_s\|_2 \le R_{\mathcal{A}}$, then $\log(\det(V)/\lambda^r) \le r\log(1 + NR_{\mathcal{A}}^2/(\lambda r))$.*

Proof in §F.4.2.

Lemma 9 (Elliptical potential bound)**.** $\sum_{t=1}^{N} \|z_t\|_{V_{t-1}^{-1}} \le \sqrt{2rN \log(1 + NR_{\mathcal{A}}^2/(\lambda r))}$.

Proof in §F.4.3.

E.6.2 Projected Exploitation Algorithms

Once the subspace estimate $\widehat{U}_k$ is available, exploitation reduces to a linear bandit problem in $\mathbb{R}^r$. We present two variants that differ in how past data is aggregated: a cumulative version (Algorithm 2) that uses all exploitation rounds within the segment, and a windowed version (Algorithm 3) that restricts to the most recent W exploitation rounds.

The cumulative variant (Algorithm 2) builds the design matrix $\widetilde{V}_{t-1}$ from all prior exploitation rounds in segment k. This is statistically efficient when the reduced parameter $a_k := \widehat{U}_k^\top \theta_{\tau_{k-1}}$ is approximately constant, but accumulates bias when within-segment drift causes θ_t to change. Theorem 9 quantifies this: the drift penalty is $2R_{\mathcal{A}} \sum_{t\in E_k} D_{k,t}$, which grows with the cumulative deviation $\|\theta_t - \theta_{\tau_{k-1}}\|_2$.

The windowed variant (Algorithm 3) addresses this by restricting the design matrix to the window $\mathcal{W}_t = \{s \in E_k : t-W \le s < t\}$. This limits the drift penalty to $2R_{\mathcal{A}}WV_k$ (Theorem 10), where V_k is the total within-segment variation, at the cost of a $\log(1 + WR_{\mathcal{A}}^2/(\lambda r))$ factor in the confidence width (vs. $\log(1 + n_k R_{\mathcal{A}}^2/(\lambda r))$ for the cumulative version). In practice, moderate window sizes ($W = 100$) give the best tradeoff between bias and variance; the theoretical optimum $W^\star = 1$ from Theorem 2 is conservative.

Both variants include a correction term $\Gamma_k\|x\|_2$ (or $(\Gamma_k + R_{\mathcal{A}}V_{k,t}(W))\|x\|_2$ for the windowed version) that absorbs the subspace mismatch between $\widehat{P}_k$ and $P_k^\star$. This ensures optimism holds even when the estimated subspace is imperfect, at the cost of inflating the UCB index by $O(\varepsilon_k)$ per round.

Description of Algorithm 1. The algorithm operates in three interleaved phases within each segment k:

1. **Probe.** On designated probe rounds (every μ-th round), draw an isotropic sphere probe $u_t = \sqrt{d}\, z/\|z\|$, $z \sim \mathcal{N}(0, I_d)$, and deploy it as the action ($x_t = u_t$). Compute the centered statistic $s_t = y_t^2 - \hat{\sigma}_\varepsilon^2$ and the lifted sample $G_t = \mathcal{K}^{-1}(s_t\, u_t u_t^\top)$. Accumulate $\widehat{M}_k \leftarrow \frac{1}{m_k} \sum G_t$ and extract the top-r eigenvectors $\widehat{U}_k$.
2. **Project.** On non-probe rounds, project each candidate action $x \in \mathcal{A}_t$ into the estimated r-dimensional subspace: $z(x) = \widehat{U}_k^\top x \in \mathbb{R}^r$.
3. **Exploit.** Solve a windowed ridge regression in $\mathbb{R}^r$ using the last W exploitation rounds, and select the action maximizing the UCB index $z(x)^\top \hat{a} + \beta\|z(x)\|_{\widetilde{V}^{-1}} + \gamma\|x\|_2$.

At each segment boundary (known or detected), the probe accumulator resets to avoid contamination from the previous segment's subspace.

Theorem 9 (Segmentwise projected regret under varying w_t)**.** *With probability at least* $1 - \delta$, $R_k^{\exp} \le 2\beta_{k,\max}^{(r)} \sqrt{2rn_k \log(1 + n_k R_{\mathcal{A}}^2/(\lambda r))} + 2S_w R_{\mathcal{A}} \varepsilon_k n_k + 2R_{\mathcal{A}} \sum_{t\in E_k} D_{k,t}$*, where* $D_{k,t} := \|\theta_t - \theta_{\tau_{k-1}}\|_2$.

Proof in §F.4.4.

Theorem 10 (Windowed projected regret under varying w_t)**.** *With probability at least* $1-\delta$, *the exploitation regret satisfies* $R_k^{\exp} \le 2\beta_k^{(r,W)} \sqrt{2rn_k \log(1 + WR_{\mathcal{A}}^2/(\lambda r))} + 2S_w R_{\mathcal{A}} \varepsilon_k n_k + 2R_{\mathcal{A}} W V_k$.

Proof in §F.4.6.

E.6.3 Adaptive-Reset Algorithm

Algorithm 1 requires oracle knowledge of segment boundaries $\{\tau_k\}$. Algorithm 4 removes this requirement through a data-driven change-point detector that triggers subspace re-estimation when the predictable second moment shifts.

Algorithm 1 Single-Play Subspace-Calibrated Optimism with Windowed r-Dimensional Exploitation

Input: Segments $\{\mathcal{I}_k\}_{k=1}^K$, probe schedule $\mathcal{T}_{\text{probe}}$, regularizer $\lambda > 0$, noise variance σ_ε^2, probe distribution Q, latent dimension r, window length W.
Require: $\tau_{k-1} \in \mathcal{T}_{\text{probe}}$ for every segment k.
Initialize: $\widehat{P}_0 = I_d$, choose $\widehat{U}_0 \in \mathbb{R}^{d\times r}$ with $\widehat{U}_0^\top \widehat{U}_0 = I_r$, $\widehat{M}_0 = \mathbf{0}_{d\times d}$, exploitation buffer $\mathcal{E} \leftarrow \emptyset$.
for $t = 1, \dots, T$ **do**
  Observe action set $\mathcal{A}_t$.
  if $\exists k \ge 1 : t = \tau_k$ **then**
    $\widehat{M}_t \leftarrow \mathbf{0}_{d\times d}$. ▷ *Reset accumulator at segment boundary*
    $\widehat{P}_t \leftarrow \widehat{P}_{t-1}, \quad \widehat{U}_t \leftarrow \widehat{U}_{t-1}, \quad \mathcal{E} \leftarrow \emptyset.$
  end if
  if $t \in \mathcal{T}_{\text{probe}}$ **then** ▷ *Probing round*
    Draw $u_t \sim Q$ independently.
    Deploy $x_t = u_t$.
    Observe $y_t = u_t^\top \theta_t + \varepsilon_t$.
    Compute $s_t = y_t^2 - \sigma_\varepsilon^2, \quad G_t = \mathcal{K}^{-1}(s_t\, u_t u_t^\top)$.
    Update $\widehat{M}_t \leftarrow \widehat{M}_{t-1} + G_t$.
    Let $N_k(t) := |\mathcal{T}_{\text{probe}} \cap \mathcal{I}_k \cap [1, t]|$.
    Compute $\widehat{U}_t \leftarrow \mathrm{TopEig}_r(\widehat{M}_t / N_k(t)), \quad \widehat{P}_t \leftarrow \widehat{U}_t \widehat{U}_t^\top$.
  else ▷ *Exploitation round*
    Let k be such that $t \in \mathcal{I}_k$, and define the exploitation window $\mathcal{W}_t := \{s \in \mathcal{I}_k \setminus \mathcal{T}_{\text{probe}} : t - W \le s < t\}$.
    For each $x \in \mathcal{A}_t$, define $z_t(x) := \widehat{U}_{t-1}^\top x \in \mathbb{R}^r$.
    Form the windowed reduced design and response:

$$\widetilde{V}_t := \lambda I_r + \sum_{s \in \mathcal{W}_t} z_t(x_s)\, z_t(x_s)^\top, \qquad \widetilde{b}_t := \sum_{s \in \mathcal{W}_t} z_t(x_s)\, y_s,$$

  where $z_t(x_s) := \widehat{U}_{t-1}^\top x_s$ re-projects each stored action with the *current* basis.
    Compute reduced ridge estimate $\widehat{a}_t = \widetilde{V}_t^{-1} \widetilde{b}_t$.
    Set confidence radius $\beta_t^{(r,W)}$ and correction radius γ_t as in §6.
    Select $x_t = \arg\max_{x \in \mathcal{A}_t} \Big(z_t(x)^\top \widehat{a}_t + \beta_t^{(r,W)} \|z_t(x)\|_{\widetilde{V}_t^{-1}} + \gamma_t \|x\|_2 \Big)$.
    Observe $y_t = x_t^\top \theta_t + \varepsilon_t$.
    Append (x_t, y_t, t) to $\mathcal{E}$. ▷ *Store raw action, not projected feature*
    $\widehat{U}_t \leftarrow \widehat{U}_{t-1}$; $\widehat{P}_t \leftarrow \widehat{P}_{t-1}$; $\widehat{M}_t \leftarrow \widehat{M}_{t-1}$.
  end if
end for

Algorithm 2 Segmentwise projected ridge-UCB in the learned subspace

Input: segment $\mathcal{I}_k$, estimated basis $\widehat{U}_k$, $\lambda > 0$, Γ_k, δ
for each exploitation round $t \in E_k$ **do**
  $z_t(x) = \widehat{U}_k^\top x$ for each $x \in \mathcal{A}_t$
  $\widetilde{V}_{t-1} = \lambda I_r + \sum_{s<t} z_s z_s^\top, \quad \widetilde{b}_{t-1} = \sum_{s<t} z_s y_s$
  $\widehat{a}_t = \widetilde{V}_{t-1}^{-1} \widetilde{b}_{t-1}$
  $x_t \in \arg\max_{x \in \mathcal{A}_t} \{ z_t(x)^\top \widehat{a}_t + \beta_t^{(r)} \|z_t(x)\|_{\widetilde{V}_{t-1}^{-1}} + \Gamma_k \|x\|_2 \}$
end for

The algorithm operates in two phases. In RECOVERY mode, it deploys m_{relearn} consecutive probe rounds to build a fresh subspace estimate from scratch, then transitions to NORMAL mode. In NORMAL mode, it interleaves probing (at rate μ) with windowed projected

Algorithm 3 Windowed projected ridge-UCB in the learned subspace

Input: segment $\mathcal{I}_k$, estimated basis $\widehat{U}_k$, $\lambda > 0$, window W, Γ_k, δ
for each exploitation round $t \in E_k$ **do**
  $\mathcal{W}_t = \{s \in E_k : t - W \le s < t\}$
  $z_t(x) = \widehat{U}_k^\top x$ for each $x \in \mathcal{A}_t$
  $\widetilde{V}_t = \lambda I_r + \sum_{s \in \mathcal{W}_t} z_s z_s^\top$, $\widetilde{b}_t = \sum_{s \in \mathcal{W}_t} z_s y_s$
  $\widehat{a}_t = \widetilde{V}_t^{-1} \widetilde{b}_t$
  $x_t \in \arg\max_{x \in \mathcal{A}_t} \{z_t(x)^\top \widehat{a}_t + \beta_t^{(r,W)} \|z_t(x)\|_{\widetilde{V}_t^{-1}} + \Gamma_k \|x\|_2 + R_{\mathcal{A}} V_{k,t}(W)\}$
end for

exploitation (Algorithm 3). On each probe round in normal mode, the detector statistic $S_t = \|\widehat{M}_t^{\text{recent}} - \widehat{M}_t^{\text{past}}\|_{\text{op}}$ (Eq. (5)) compares two non-overlapping rolling-window estimates of the predictable second moment. A reset to RECOVERY is triggered when S_t exceeds the threshold b (Eq. (12)), which is calibrated so that:

- under no change, $\mathbb{P}(S_t > b) \le \delta_{\text{FA}}/T$ per round, giving $F_T = 0$ false alarms with high probability (Theorem 11);
- after a true change of magnitude $\Delta_k \ge 2b$, the detector fires within delay $D_{\max} = (2n/\mu) + \tau_{\text{burn}}$ (Theorem 12).

The recovery cost per detected change is $(2R_{\mathcal{A}} S_w + c) \cdot m_{\text{relearn}}$, and Theorem 4 shows that for fixed K the adaptive algorithm matches the oracle-boundary rate $\widetilde{O}(r\sqrt{T}) + \widetilde{O}(T^{2/3}) + O(V)$ with only $O(K)$ lower-order overhead from detection delay and recovery phases.

A key practical advantage over explore-then-commit methods is that Algorithm 4 does not require oracle segment boundaries: it detects change points online and re-estimates the subspace only when needed, avoiding both the cost of blind periodic restarts and the risk of exploiting with a stale subspace estimate from a previous regime.

E.7 Detector Guarantees

Lemma 10 (Within-segment second-moment variation)**.** *Under Assumption 1, for $s < t$ in the same segment with both at least τ_{burn} rounds after the segment start, $\|\widetilde{M}_t - \widetilde{M}_s\|_{\text{op}} \le \Delta_{\text{within}}(\tau_{\text{burn}}) := 2(1-\alpha_0)^{2\tau_{\text{burn}}} S_w^2/(1-(1-\alpha_0)^2)$.*

Proof in §F.6.1.

Theorem 11 (No-change control)**.** *If no change point falls within the combined detector window and both windows begin at least τ_{burn} rounds after the most recent change point, then for any fixed t: $\mathbb{P}(S_t > b) \le \delta_{\text{FA}}/T$, and the total number of false alarms satisfies $\mathbb{E}[F_T] \le \delta_{\text{FA}}$, so $F_T = 0$ with probability at least $1 - \delta_{\text{FA}}$.*

Proof in §F.6.2.

Theorem 12 (Post-change detection)**.** *If $\Delta_k \ge \Delta_{\text{det}} = 2b$, then with probability at least $1 - \delta/(K+1)$, the detector fires within delay $D_k \le D_{\max} := (n_r + n_p)/\mu + \tau_{\text{burn}}$.*

Proof in §F.6.3.

E.8 Mathematical Separations from Prior Work

Proposition 10 (Our model is not reducible to Qin's task-constant model)**.** *Fix $N \ge 2$. There exist instances of our model for which $\mathbb{P}(\theta_t = \theta_{t+1}) = 0$ for all t within any segment.*

Proof in §F.7.5.

Proposition 11 (Identifiable instance not covered by Qin's mechanism)**.** *With $r \ge 2$, one segment, $A_k = 0$, $\Sigma_{\eta,k} = I_r$: our Proposition 3 gives $\text{range}(\bar{M}_k^{\text{probe}}) = \text{range}(B_k^\star)$. But with $N = 1$ task, Qin's task-covariance has rank $\le 1 < r$.*

Algorithm 4 Adaptive Single-Play Subspace-Calibrated Optimism (Unknown Segment Boundaries)

Input: Probe rate $\mu \in (0,1)$, detection window n ($n_r = n_p = n$), threshold b from (12), burn-in τ_{burn}, recovery length m_{relearn}, $\lambda > 0$, σ_ε^2, Q, r, W.
Initialize: $\widehat{U}_0 \in \mathbb{R}^{d\times r}$; $\widehat{P} \leftarrow \widehat{U}_0 \widehat{U}_0^\top$; $\widehat{M}_{\text{acc}} \leftarrow \mathbf{0}$; `phase` $\leftarrow$ RECOVERY; `rec_count` $\leftarrow 0$; $n_{\text{pr}} \leftarrow 0$
for $t = 1, \dots, T$ **do**
  Observe $\mathcal{A}_t$.
  if `phase` $=$ RECOVERY **then**
    Draw $u_t \sim Q$; deploy $x_t = u_t$; observe y_t.
    $s_t = y_t^2 - \sigma_\varepsilon^2$; $G_t = \mathcal{K}^{-1}(s_t u_t u_t^\top)$.
    $\widehat{M}_{\text{acc}} \leftarrow \widehat{M}_{\text{acc}} + G_t$; `rec_count` $\leftarrow$ `rec_count` $+ 1$.
    if `rec_count` $= m_{\text{relearn}}$ **then**
      $\widehat{U} \leftarrow \text{TopEig}_r(\widehat{M}_{\text{acc}}/m_{\text{relearn}})$; $\widehat{P} \leftarrow \widehat{U}\widehat{U}^\top$.
      $n_{\text{pr}} \leftarrow m_{\text{relearn}}$; `rec_count` $\leftarrow 0$; `phase` $\leftarrow$ NORMAL.
    end if
  else
    if probe round (determined by probe rate μ) **then**
      Draw $u_t \sim Q$; deploy; compute G_t; update $\widehat{M}_{\text{acc}}$, $\widehat{U}$, $\widehat{P}$.
      if detection window active and $S_t > b$ **then**
        `phase` $\leftarrow$ RECOVERY; reset all buffers.
      end if
    else
      Exploitation: windowed reduced ridge-UCB as in Algorithm 1.
    end if
  end if
end for

Proof in §F.7.6.

Proposition 12 (Finite probe actions do not identify a changing ambient subspace)**.** *If probe directions lie in a finite set $U \subset \mathbb{R}^d$ with $\operatorname{span}(U) \neq \mathbb{R}^d$, then $\widetilde{M}_t$ is not identifiable from $\{u_i^\top \widetilde{M}_t u_i : u_i \in U\}$.*

Proof in §F.7.7.

E.9 Ambient-Space Reference Bound

Theorem 13 (Ambient-space costed dynamic regret)**.** *Under Assumptions 1–4, with probability at least $1-\delta$:*

$$\text{DynReg}_T^{(c)} \leq \underbrace{(2R_{\mathcal{A}}S_w + c)m_T}_{\textit{probing cost}} + \underbrace{\widetilde{\mathcal{O}}(d\sqrt{T})}_{\textit{ambient statistical regret}} + \underbrace{\widetilde{\mathcal{O}}\left(\frac{S_w R_{\mathcal{A}} R_X}{\lambda_{\min}} \sum_{t \notin \mathcal{T}_{\text{probe}}} \frac{1}{\sqrt{m_{t-1}}}\right)}_{\textit{subspace approximation error}}.$$

Under a uniform probe schedule, this yields $\widetilde{\mathcal{O}}(d\sqrt{T}+T^{2/3})$, compared to $\widetilde{\mathcal{O}}(r\sqrt{T}+T^{2/3})$ from Theorem 2. The improvement from $d\sqrt{T}$ to $r\sqrt{T}$ is achieved by the projected exploitation layer.

Proof in §F.7.8.

E.10 Lower Bounds: Necessity of Full Coverage

Theorem 14 (Minimax lower bound under restricted deployed-action span)**.** *Let $S \subsetneq \mathbb{R}^d$ be a proper subspace, and let $\Pi(S)$ be the class of all policies with $x_t \in S$ a.s. Then there exists a subclass $\mathfrak{M}(S)$ satisfying all structural assumptions such that $\inf_{\pi \in \Pi(S)} \sup_{\nu \in \mathfrak{M}(S)} \mathbb{E}_\nu^\pi[\text{DynReg}_T^{(c)}] \geq \mu T$ for some constant $\mu > 0$.*

Proof in §F.7.1.

Theorem 15 (Lower bound when both deployed and probe actions are confined)**.** *Under the same construction,* $\inf_{\pi\in\Pi_{\mathrm{probe}}(S)}\sup_{\nu\in\mathfrak{M}(S)}\mathbb{E}^{\pi}_{\nu}[\mathrm{DynReg}^{(c)}_T]\geq\mu T$*. The probe count is* ν*-independent (all observations equal* ε_t *since* $x_t^\top\theta_t=0$*), so probing within* S *provides no information about* $\theta_t\in S^\perp$*.*

Proof in §F.7.2.

Corollary 3 (Sublinear regret is impossible without full coverage)**.** *No policy in* $\Pi(S)$ *or* $\Pi_{\mathrm{probe}}(S)$ *can guarantee* $o(T)$ *expected costed dynamic regret uniformly over* $\mathfrak{M}(S)$*.*

Proof in §F.7.3.

Definition 1 (Observable-support restriction induced by the environment)**.** *Fix a proper subspace* $S\subsetneq\mathbb{R}^d$*. We say that an instance* ν *satisfies the observable-support restriction associated with* S *if, almost surely, every reward-bearing action available to the learner lies in* S*:* $\mathcal{A}_t\subseteq S$ *and* $\mathrm{supp}(Q_t)\subseteq S$ *for all* t*. We denote by* $\mathfrak{M}_{\mathrm{obs}}(S)$ *the subclass of instances satisfying this restriction.*

Theorem 16 (Two-point non-identifiability under incomplete observable coverage)**.** *Let* $S\subsetneq\mathbb{R}^d$ *be a proper subspace with* $\dim(S^\perp)\geq 1$*. Then there exist two instances* $\nu_0,\nu_1\in\mathfrak{M}_{\mathrm{obs}}(S)$ *and constants* $0<\gamma_0<\gamma_1$ *such that:*

- *(i) For every policy* $\pi\in\Pi_{\mathrm{adm}}$*, the induced laws of the observed history are identical:* $\mathbb{P}^{\pi}_{\nu_0}|_{\mathcal{H}_T}=\mathbb{P}^{\pi}_{\nu_1}|_{\mathcal{H}_T}$*.*
- *(ii) The predictable second-moment operators differ:* $\widetilde{M}_t^{(0)}\neq\widetilde{M}_t^{(1)}$ *for every* t*.*
- *(iii) Against the enlarged benchmark* $\widetilde{\mathcal{A}}_t:=\mathcal{A}_t\cup\{v,-v\}$ *for a fixed* $v\in S^\perp$*, any policy satisfies* $\max_{j\in\{0,1\}}\mathbb{E}^{\pi}_{\nu_j}[\mathrm{DynReg}^{(c)}_T]\geq\mu T$ *for some* $\mu>0$*.*

Proof in §F.7.4.

Remark 13 (Scope of the regret lower bound)**.** *Theorem 16 should primarily be interpreted as a complete non-identifiability result for the predictable second moment under incomplete observable coverage. The regret lower bound is formulated against an enlarged benchmark action set, not the paper's original comparator, since if all actions lie in* S *then the comparator itself cannot exploit directions in* $S^\perp$*. The* $\Omega(T)$ *regret claims in the main paper (Theorems 14 and 15) do not use this enlarged benchmark.*

E.11 Probe-Cost Lower Bound

This section proves an $\Omega(\sqrt{cT})$ minimax lower bound on the costed dynamic regret for explore-then-commit (ETC) policies, confirming that the probe–subspace tradeoff in Theorem 2 is fundamental for this policy class.

Definition 2 (Explore-then-commit policy)**.** *An* ETC policy with parameter m *(*$1\leq m\leq T$*) is a policy* π *satisfying: (i)* ***Probe phase.*** $\mathcal{T}_{\mathrm{probe}}=\{1,\ldots,m\}$*; on probe rounds,* x_t *is any* $\mathcal{H}_{t-1}$*-measurable action with* $\|x_t\|\leq R_{\mathcal{A}}$*. (ii)* ***Subspace commitment.*** *After round* m*, the policy forms a unit vector* $\hat{B}\in\mathbb{S}^{d-1}$ *that is* $\mathcal{H}_m$*-measurable. (iii)* ***Subspace-restricted exploitation.*** *For every* $t>m$*:* $x_t\in\mathrm{span}(\hat{B})$*, i.e.* $x_t=\ell_t\hat{B}$ *for some scalar* $|\ell_t|\leq R_{\mathcal{A}}$*. The restriction is (iii): after probing, actions lie in the committed one-dimensional subspace* $\hat{B}$*.*

Remark 14. *Algorithm 1 is an ETC-type policy (with interleaved scheduling and r-dimensional subspace), so the lower bound applies. More generally, any policy that learns a subspace from probes and then exploits within it falls into this class.*

Symmetric two-point construction. Fix $\alpha\in(0,\pi/4)$ and define two hypotheses:

$$b_0=\cos\alpha\,e_1+\sin\alpha\,e_2,\qquad b_1=\cos\alpha\,e_1-\sin\alpha\,e_2. \tag{16}$$

Both are unit vectors with projector distance $\delta:=\|b_0b_0^\top-b_1b_1^\top\|_{\mathrm{op}}=\sin 2\alpha$. Instances ν_0,ν_1 are defined by $B^\star=b_j$ ($r=1$), scalar AR(1) transition $A\in(-1,1)$, $\eta_t\sim\mathcal{N}(0,\lambda)$, and

$\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$, all independent. Under both hypotheses, the latent process $\{w_t\}$ has the *same* marginal distribution (it depends only on A, λ, $\{\eta_t\}$).

Lemma 11 (Per-round KL divergence). *Under ν_j, conditional on action x_t and latent state w_t: $\mathrm{KL}(P_0(y_t \mid x_t, w_t) \,\|\, P_1(y_t \mid x_t, w_t)) = 2x_{2,t}^2 \sin^2\alpha \cdot w_t^2/\sigma^2$. In particular, actions with $x_{2,t} = 0$ produce identical distributions under both hypotheses.*

Proof. The means under ν_j are $\mu_j = (x_1 \cos\alpha + (-1)^j x_2 \sin\alpha) w_t$ with common variance σ^2. Hence $\mathrm{KL} = (\mu_0 - \mu_1)^2/(2\sigma^2) = (2x_2 \sin\alpha \cdot w_t)^2/(2\sigma^2)$. □

Lemma 12 (Deterministic regret gap). *For any $x \in \mathbb{R}^d$ with $\|x\| \le R_{\mathcal{A}}$ and any $w \in \mathbb{R}$, define $r^{(j)}(x, w) := R_{\mathcal{A}}|w| - (b_j^\top x) w \,\mathrm{sign}(w)$. Then:*

$$r^{(0)}(x, w) + r^{(1)}(x, w) \ge 2(1 - \cos\alpha) R_{\mathcal{A}} |w|.$$

Proof. Since $b_0 + b_1 = 2\cos\alpha\, e_1$: $r^{(0)} + r^{(1)} = 2|w|(R_{\mathcal{A}} - \cos\alpha\, x_1 \,\mathrm{sign}(w)) \ge 2(1 - \cos\alpha) R_{\mathcal{A}}|w|$, using $|x_1| \le R_{\mathcal{A}}$. □

Theorem 17 (Minimax lower bound for ETC policies). *Suppose $d \ge 2$, $r = 1$. There exists a constant $c_0 = c_0(R_{\mathcal{A}}, \sigma, S_w, S_{w,2}, A) > 0$ such that for all T sufficiently large, every ETC policy (Definition 2) satisfies*

$$\max_{j \in \{0,1\}} \mathbb{E}^{\pi}_{\nu_j}\left[\mathrm{DynReg}_T^{(c)}\right] \;\ge\; c_0 \sqrt{c\,T}, \tag{17}$$

where $S_w := \mathbb{E}[|w_t|]$ and $S_{w,2}^2 := \mathbb{E}[w_t^2] = \lambda/(1 - A^2)$ are stationary moments.

Proof. Fix an ETC policy with parameter m and subspace estimate $\hat{B}$.

Reduction to $\mathrm{span}(e_1, e_2)$. Since $b_0, b_1 \in \mathrm{span}(e_1, e_2)$, any out-of-plane component of $\hat{B}$ contributes zero reward. Replacing $\hat{B}$ with its in-plane projection can only decrease the per-round regret. We may therefore assume $\hat{B} = \cos\phi\, e_1 + \sin\phi\, e_2$ where ϕ is $\mathcal{H}_m$-measurable.

KL bound on probe data. By the chain rule for KL and Lemma 11, the total divergence of the probe observations is

$$\mathrm{KL}(P_0^{\le m} \| P_1^{\le m}) = \frac{2\sin^2\alpha}{\sigma^2} \sum_{t=1}^{m} \mathbb{E}_{\nu_0}[x_{2,t}^2 w_t^2] \le \frac{2\sin^2\alpha\, R_{\mathcal{A}}^2\, S_{w,2}^2}{\sigma^2}\, m.$$

Choose $\sin^2\alpha := \sigma^2/(4R_{\mathcal{A}}^2 S_{w,2}^2 m)$ (valid for $m \ge m_0$). Then $\mathrm{KL} \le 1/2$, so by Pinsker's inequality: $\mathrm{TV}(P_0^{\le m}, P_1^{\le m}) \le 1/2$.

Hypothesis-identification error. Define the test $\psi := \mathbf{1}\{\sin\phi \le 0\}$ (which hypothesis does $\hat{B}$ favor). Since $\hat{B}$ is $\mathcal{H}_m$-measurable, Le Cam's testing bound gives: $P_0(\psi \ne 0) + P_1(\psi \ne 1) \ge 1 - \mathrm{TV} \ge 1/2$. By symmetry ($e_2 \mapsto -e_2$ maps $\nu_0 \leftrightarrow \nu_1$), both error probabilities are $\ge 1/4$.

Exploitation regret on the error event. On exploitation rounds with $\hat{B} = \cos\phi\, e_1 + \sin\phi\, e_2$, the per-round regret under ν_0 is $r_t^{(0)} \ge R_{\mathcal{A}}|w_t|(1 - |b_0^\top \hat{B}|)$. On the event $\{\sin\phi \le 0\}$: the angular error $|\alpha - \phi| \ge \alpha$, giving $|b_0^\top \hat{B}| \le \cos\alpha$, hence $r_t^{(0)} \ge R_{\mathcal{A}} \sin^2\alpha\, |w_t|/2$.

Markov decoupling. The indicator $\mathbf{1}_{\sin\phi \le 0}$ is $\mathcal{H}_m$-measurable. By the Markov property of the AR(1) process, the future path $(w_{m+1}, \ldots, w_T)$ conditioned on w_m is independent of $\mathcal{H}_m$. Using the conditional-sum bound $\mathbb{E}[\sum_{t>m} |w_t| \mid w_m] = (T - m) S_w + h(w_m)$ with $\mathbb{E}[|h(w_m)|] \le \frac{2|A|}{1-|A|} S_w$, and the error probability $\ge 1/4$:

$$\mathbb{E}_{\nu_0}\left[\sum_{t>m} r_t^{(0)}\right] \ge \frac{R_{\mathcal{A}} \sin^2\alpha}{2} \cdot \frac{(T - m) S_w}{8} \qquad \text{for } T - m \ge 4 \cdot \frac{2|A|}{1 - |A|}.$$

Combine with probe cost and optimize. Substituting $\sin^2\alpha = \sigma^2/(4R_{\mathcal{A}}^2 S_{w,2}^2 m)$ and $T - m \geq T/2$:

$$\max_j \mathbb{E}_{\nu_j}[\mathrm{DynReg}_T^{(c)}] \geq cm + \frac{\sigma^2 S_w}{128\, R_{\mathcal{A}}\, S_{w,2}^2} \cdot \frac{T}{m} =: cm + \frac{\beta T}{m}.$$

By AM–GM: $cm + \beta T/m \geq 2\sqrt{c\beta T}$. Setting $c_0 := 2\sqrt{\beta} = \frac{\sigma\sqrt{S_w}}{4\sqrt{2}\, S_{w,2}\sqrt{R_{\mathcal{A}}}}$ completes the proof. $\square$

Remark 15 (Interpretation and the $\sqrt{cT}$ vs. $c^{1/3}T^{2/3}$ gap)**.** *Theorem 17 establishes that the probe–subspace tradeoff incurs at least $\Omega(\sqrt{cT})$ regret for any ETC-type policy, confirming that the tradeoff is fundamental and not an artifact of the analysis.*

The remaining gap between the lower bound $\Omega(\sqrt{cT})$ and the upper bound $\widetilde{O}(c^{1/3}T^{2/3})$ from Theorem 2 has a precise structural explanation. The lower bound gives $\Omega(\epsilon^2)$ per-round regret from subspace misalignment (via $1 - \cos\alpha \sim \alpha^2/2$), yielding $cm + \epsilon^2 T$ with $\epsilon = O(1/\sqrt{m})$, hence $cm + T/m$, optimized at $m = \sqrt{T/c}$: total $\Omega(\sqrt{cT})$. The upper bound incurs $O(\epsilon)$ per-round bias from the hard projection step (the $2S_w R_{\mathcal{A}}\varepsilon_k n_k$ term in Theorem 8), yielding $cm + \epsilon T = cm + T/\sqrt{m}$, optimized at $m = (T/c)^{2/3}$: total $O(c^{1/3}T^{2/3})$.

Whether an algorithm without hard projection can achieve $O(\epsilon^2)$ per-round overhead—and thereby close the gap to $\widetilde{O}(\sqrt{cT})$—is an open question. The gap is in the probe-cost exponent ($c^{1/2}$ vs. $c^{1/3}$), not in the T-dependence, and is structurally well-understood.

F Complete Proofs in Dependency Order

All proofs are presented in topological order of their dependency graph: each proof references only results that have already been stated or proved above.

F.1 Identification Foundations

F.1.1 Proof of Lemma 1

Proof. For every probe round $t \in \mathcal{T}_k$, the LDS recursion gives

$$\mathbb{E}[w_t w_t^\top \mid \mathcal{H}_{t-1}] = A_k w_{t-1} w_{t-1}^\top A_k^\top + \Sigma_{\eta,k} \succeq \Sigma_{\eta,k} \succeq \underline{\lambda}\, I_r \qquad \text{a.s.,}$$

where the first inequality uses $A_k w_{t-1} w_{t-1}^\top A_k^\top \succeq 0$ and the second uses Assumption 1. Averaging over probe rounds $t \in \mathcal{T}_k$ preserves the semidefinite order:

$$\bar{S}_k^{\mathrm{probe}} = \frac{1}{m_k} \sum_{t \in \mathcal{T}_k} \mathbb{E}[w_t w_t^\top \mid \mathcal{H}_{t-1}] \succeq \underline{\lambda}\, I_r.$$

Hence $\lambda_r(\bar{S}_k^{\mathrm{probe}}) \geq \underline{\lambda} =: \lambda_{\min} > 0$. $\square$

F.1.2 Proof of Lemma 2

Proof. The operator $\mathcal{K}$ acts on the finite-dimensional space $\mathbb{R}^{d\times d}_{\mathrm{sym}}$, which has dimension $d(d+1)/2$. To show invertibility, it suffices to prove injectivity.

Let $M \in \mathbb{R}^{d\times d}_{\mathrm{sym}}$ satisfy $\mathcal{K}(M) = 0$. We show $u^\top M u = 0$ Q-a.s. Compute

$$\mathbb{E}\big[(u^\top M u)^2\big] = \mathbb{E}\big[(u^\top M u)\,\mathrm{tr}(M\, uu^\top)\big] = \mathrm{tr}\big(M\,\mathbb{E}[(u^\top M u)\, uu^\top]\big) = \mathrm{tr}\big(M\,\mathcal{K}(M)\big) = 0.$$

Since $(u^\top M u)^2 \geq 0$, this forces $u^\top M u = 0$ Q-a.s.

We show $u^\top M u = 0$ Q-a.s. implies $M = 0$. Write the eigendecomposition $M = \sum_{i=1}^d \lambda_i v_i v_i^\top$. For any index i,

$$0 = \mathbb{E}\big[u^\top M u\,(v_i^\top u)^2\big] = \sum_{j=1}^d \lambda_j\, \mathbb{E}\big[(v_j^\top u)^2 (v_i^\top u)^2\big],$$

where the first equality uses $u^\top M u = 0$ Q-a.s. and the second expands M. For isotropic Gaussian probes $u \sim \mathcal{N}(0, I_d)$ (or their bounded truncation to $\|u\|_2 \le L$), the fourth-moment tensor satisfies

$$\mathbb{E}\big[(v_j^\top u)^2 (v_i^\top u)^2\big] = 1 + 2\delta_{ij},$$

so the system becomes $(1 + 2\delta_{ij})\lambda_j = 0$ for each i, which gives $\lambda_i = 0$ for all i, hence $M = 0$.

More generally, the result holds for any probe distribution Q whose fourth-moment tensor $\mathbb{E}[u_i u_j u_k u_l]$ is positive definite on the symmetric matrix space. Assumption 3 together with an absolute-continuity or full-rank fourth-moment condition on Q guarantees this property.

Operator norm bound. Since $\mathcal{K}$ is a bijection on the finite-dimensional space $\mathbb{R}^{d\times d}_{\mathrm{sym}}$, $\mathcal{K}^{-1}$ exists and $\|\mathcal{K}^{-1}\|_{\mathrm{op}}$ is finite. For isotropic Gaussian probes, the explicit inverse is $\mathcal{K}^{-1}(N) = \frac{1}{2}N - \frac{\mathrm{tr}(N)}{2(d+2)} I_d$ (Remark 6), and $C_{\mathcal{K}} := \|\mathcal{K}^{-1}\|_{\mathrm{op}} \le 1$. □

F.1.3 Proof of Proposition 1

Proof. Expanding $s_t = (u_t^\top \theta_t)^2 + 2\varepsilon_t u_t^\top \theta_t + \varepsilon_t^2 - \sigma_\varepsilon^2$ and taking conditional expectations given $\sigma(\mathcal{H}_{t-1}, u_t)$: the cross term $2u_t^\top \mathbb{E}[\theta_t \varepsilon_t \mid \sigma(\mathcal{H}_{t-1}, u_t)]$ is bounded by $2L\epsilon_\times$ under Assumption 4(iii); when $\epsilon_\times = 0$ it vanishes exactly. The variance term $\mathbb{E}[\varepsilon_t^2 \mid \sigma(\mathcal{H}_{t-1}, u_t)]$ contributes a bias of $\delta_\sigma := \hat\sigma^2 - \sigma_\varepsilon^2$; when $\delta_\sigma = 0$ (known variance) it cancels exactly. Hence $\mathbb{E}[s_t \mid \sigma(\mathcal{H}_{t-1}, u_t)] = \mathbb{E}[(u_t^\top \theta_t)^2 \mid \sigma(\mathcal{H}_{t-1}, u_t)] = u_t^\top \mathbb{E}[\theta_t \theta_t^\top \mid \mathcal{H}_{t-1}] u_t = u_t^\top \widetilde{M}_t u_t$, where the second equality uses the independence of u_t from $(\theta_t, \mathcal{H}_{t-1})$. □

F.1.4 Proof of Proposition 2

Proof. For $t \in \mathcal{I}_k$, $\theta_t = B_k^\star w_t$. Therefore,

$$\widetilde{M}_t = \mathbb{E}[\theta_t \theta_t^\top \mid \mathcal{H}_{t-1}] = B_k^\star \mathbb{E}[w_t w_t^\top \mid \mathcal{H}_{t-1}](B_k^\star)^\top.$$

Averaging over $t \in \mathcal{T}_k$:

$$\bar{M}_k^{\mathrm{probe}} = B_k^\star \left(\frac{1}{m_k} \sum_{t \in \mathcal{T}_k} \mathbb{E}[w_t w_t^\top \mid \mathcal{H}_{t-1}] \right) (B_k^\star)^\top = B_k^\star \bar{S}_k^{\mathrm{probe}} (B_k^\star)^\top.$$

The range inclusion follows since $\bar{M}_k^{\mathrm{probe}}$ maps any vector through $B_k^\star$. □

F.1.5 Proof of Proposition 4

Proof. By the tower property and the independence of u_t from $\mathcal{H}_{t-1}$: $\mathbb{E}[G_t \mid \mathcal{H}_{t-1}] = \mathcal{K}^{-1}(\mathbb{E}[\mathbb{E}[s_t \mid \sigma(\mathcal{H}_{t-1}, u_t)]\, u_t u_t^\top \mid \mathcal{H}_{t-1}]) = \mathcal{K}^{-1}(\mathbb{E}[(u_t^\top \widetilde{M}_t u_t)\, u_t u_t^\top \mid \mathcal{H}_{t-1}]) = \mathcal{K}^{-1}(\mathcal{K}(\widetilde{M}_t)) = \widetilde{M}_t$, where the second equality uses Proposition 1. □

F.1.6 Proof of Proposition 3

Proof. By Proposition 2, $\bar{M}_k^{\mathrm{probe}} = B_k^\star \bar{S}_k^{\mathrm{probe}} (B_k^\star)^\top$. By Lemma 1, $\lambda_r(\bar{S}_k^{\mathrm{probe}}) \ge \lambda_{\min} > 0$, so $\bar{S}_k^{\mathrm{probe}}$ has full rank r. Therefore $\mathrm{rank}(\bar{M}_k^{\mathrm{probe}}) = r$. We already have $\mathrm{range}(\bar{M}_k^{\mathrm{probe}}) \subseteq \mathrm{range}(B_k^\star)$ from Proposition 2, and both spaces have dimension r, so equality follows. □

F.1.7 Proof of Proposition 5

Proof. The misspecified lifted sample is $\hat{G}_t := \mathcal{K}^{-1}(\hat{s}_t u_t u_t^\top)$ where $\hat{s}_t := y_t^2 - \hat\sigma_\varepsilon^2 = s_t - \delta_\sigma$. By linearity of $\mathcal{K}^{-1}$: $\hat{G}_t = \mathcal{K}^{-1}(s_t u_t u_t^\top) - \delta_\sigma \mathcal{K}^{-1}(u_t u_t^\top) = G_t - \delta_\sigma \mathcal{K}^{-1}(u_t u_t^\top)$. Taking conditional expectations and using Proposition 4: $\mathbb{E}[\hat{G}_t \mid \mathcal{H}_{t-1}] = \widetilde{M}_t - \delta_\sigma \mathcal{K}^{-1}(\mathbb{E}[u_t u_t^\top]) = \widetilde{M}_t - \delta_\sigma \mathcal{K}^{-1}(\Sigma_Q) = \widetilde{M}_t - \widetilde{B}$. The bound $\|\widetilde{B}\|_2 \le |\delta_\sigma| \|\mathcal{K}^{-1}\|_{\mathrm{op}} \|\Sigma_Q\|_2 \le |\delta_\sigma| C_{\mathcal{K}} L^2$ follows from $\|\Sigma_Q\|_2 \le \mathbb{E}[\|u\|_2^2] \le L^2$. □

F.1.8 Proof of Proposition 6

Proof. Repeating the expansion in Proposition 1 without orthogonality:

$$\mathbb{E}[s_t \mid \sigma(\mathcal{H}_{t-1}, u_t)] = u_t^\top \widetilde{M}_t\, u_t + 2\, u_t^\top \underbrace{\mathbb{E}[\varepsilon_t\, \theta_t \mid \sigma(\mathcal{H}_{t-1}, u_t)]}_{=:\, m_t,\ \|m_t\| \le \epsilon_\times}.$$

The lifted sample is $G_t = \mathcal{K}^{-1}(s_t\, u_t u_t^\top)$. Taking conditional expectations and using linearity of $\mathcal{K}^{-1}$:

$$\begin{aligned}\mathbb{E}[G_t \mid \mathcal{H}_{t-1}] &= \mathcal{K}^{-1}\big(\mathbb{E}[s_t\, u_t u_t^\top \mid \mathcal{H}_{t-1}]\big)\\ &= \mathcal{K}^{-1}\big(\mathcal{K}(\widetilde{M}_t)\big) + \mathcal{K}^{-1}\big(\mathbb{E}[2(u_t^\top m_t)\, u_t u_t^\top \mid \mathcal{H}_{t-1}]\big)\\ &= \widetilde{M}_t + \widetilde{B}_\times^{(t)}.\end{aligned}$$

The bias norm is bounded by $\|\widetilde{B}_\times^{(t)}\|_2 \le \|\mathcal{K}^{-1}\|_{\mathrm{op}} \cdot \mathbb{E}\big[2|u_t^\top m_t|\, \|u_t u_t^\top\|_2\big] \le C_\mathcal{K} \cdot 2L\, \epsilon_\times \cdot L^2 = 2\, C_\mathcal{K}\, L^3\, \epsilon_\times$, using $|u_t^\top m_t| \le L\, \epsilon_\times$ and $\|u_t u_t^\top\|_2 = \|u_t\|^2 \le L^2$.

Note: for isotropic Gaussian probes, $\mathbb{E}[(u_t^\top m_t)\, u_t u_t^\top]$ is not generally a scaled identity (unlike the variance-misspecification bias $\widetilde{B}$), so the bias may affect eigenvectors. However, the operator-norm bound on $\widetilde{B}_\times^{(t)}$ ensures that Davis–Kahan still controls the projector error. □

F.2 Non-Identification Proofs

F.2.1 Proof of Proposition 7

Proof. Assume exact state–noise coupling ($\epsilon_\times = 0$) and full coverage hold. Then $g_t(u) = u^\top \widetilde{M}_t u + \sigma_\varepsilon^2$. Let $H = \delta I_d$ with $0 < \delta \le \sigma_\varepsilon^2 / L^2$, and define $\widetilde{M}_t' := \widetilde{M}_t + H$ and $v_t'(u) := \sigma_\varepsilon^2 - u^\top H u \ge 0$. Then $u^\top \widetilde{M}_t' u + v_t'(u) = g_t(u)$ for all $u \in \mathrm{supp}(Q)$. The two triples $(\widetilde{M}_t, \sigma_\varepsilon^2)$ and $(\widetilde{M}_t', v_t')$ produce identical observations, so $\widetilde{M}_t$ is not identifiable. □

F.2.2 Proof of Proposition 8

Proof. Define the cross-moment $m_t(u) := \mathbb{E}[\theta_t \varepsilon_t \mid \mathcal{H}_{t-1}, u] \in \mathbb{R}^d$. Then $h_t(u) = u^\top \widetilde{M}_t u + 2u^\top m_t(u)$. For any nonzero symmetric H, define $\widetilde{M}_t' := \widetilde{M}_t + H$ and $m_t'(u) := m_t(u) - \frac{1}{2} H u$. Then $u^\top \widetilde{M}_t' u + 2u^\top m_t'(u) = h_t(u)$ for all u, so $\widetilde{M}_t$ is not identifiable. □

F.2.3 Proof of Proposition 9

Proof. Choose any nonzero symmetric H supported on $S^\perp$, i.e., $H = P_{S^\perp} H P_{S^\perp}$. For all $u \in S$: $u^\top H u = 0$. Hence $u^\top(\widetilde{M}_t + H)u = u^\top \widetilde{M}_t u$ and $(u^\top(\theta_t + Hv)) = u^\top \theta_t$ for $v \in S^\perp$. Neither linear nor quadratic data can distinguish $\widetilde{M}_t$ from $\widetilde{M}_t + H$. □

F.3 Concentration Chain

F.3.1 Proof of Lemma 3

Proof. $|y_t| \le |u_t^\top \theta_t| + |\varepsilon_t| \le L S_w + L_\varepsilon$ and $|s_t| \le |y_t|^2 + \sigma_\varepsilon^2 \le (L S_w + L_\varepsilon)^2 + \sigma_\varepsilon^2 = R_s$. □

F.3.2 Proof of Lemma 4

Proof. By Proposition 4, $\mathbb{E}[G_t \mid \mathcal{H}_{t-1}] = \widetilde{M}_t$, so $X_t := G_t - \widetilde{M}_t$ satisfies $\mathbb{E}[X_t \mid \mathcal{H}_{t-1}] = 0$. The decomposition $\widehat{M}_k - \bar{M}_k^{\mathrm{probe}} = \frac{1}{m_k} \sum_{t \in \mathcal{T}_k} X_t$ follows by linearity of averaging. □

F.3.3 Proof of Lemma 5

Proof. $\|G_t\|_2 = \|\mathcal{K}^{-1}(s_t u_t u_t^\top)\|_2 \le C_\mathcal{K} |s_t| \|u_t u_t^\top\|_2 \le C_\mathcal{K} R_s L^2$ and $\|\widetilde{M}_t\|_2 \le S_w^2$, so $\|X_t\|_2 \le C_\mathcal{K} L^2 R_s + S_w^2 = R_X$. □

F.3.4 Proof of Lemma 6

Proof. Since $\|X_t\|_2 \le R_X$ almost surely, we have $X_t^2 \preceq R_X^2 I_d$ in the Löwner order. Taking conditional expectations: $\mathbb{E}[X_t^2 \mid \mathcal{H}_{t-1}] \preceq R_X^2 I_d$. Summing over $t \in \mathcal{T}_k$: $\| \sum_{t\in\mathcal{T}_k} \mathbb{E}[X_t^2 \mid \mathcal{H}_{t-1}]\|_2 \le m_k R_X^2$. □

F.3.5 Proof of Theorem 5

Proof. The partial sums $Y_j := \sum_{t\in\mathcal{T}_k,\, t\le j} X_t$ form a self-adjoint matrix martingale adapted to $(\mathcal{H}_t)$. We verify the hypotheses of the Matrix Freedman inequality [Tropp, 2012].

Uniform increment bound. By Lemma 5, $\|X_t\|_2 \le R_X$ for every $t \in \mathcal{T}_k$.

Predictable quadratic variation. By Lemma 6,

$$W_k := \Big\| \sum_{t\in\mathcal{T}_k} \mathbb{E}[X_t^2 \mid \mathcal{H}_{t-1}] \Big\|_2 \le m_k R_X^2.$$

Application of Matrix Freedman. For self-adjoint martingale differences X_t with $\|X_t\|_2 \le R$ and predictable variance proxy W, the Matrix Freedman inequality [Tropp, 2012, Theorem 1.6] states:

$$\mathbb{P}\left(\Big\| \sum_t X_t \Big\|_2 \ge u \right) \le 2d \exp\left(\frac{-u^2/2}{W + Ru/3} \right).$$

Setting $R = R_X$, $W = m_k R_X^2$, and $u = 2R_X\sqrt{m_k \log(2d/\delta)}$, we verify the bound holds for $m_k \ge \frac{4}{9}\log(2d/\delta)$:

$$\begin{aligned} \frac{u^2/2}{W + Ru/3} &= \frac{2R_X^2 m_k \log(2d/\delta)}{m_k R_X^2 + \frac{2}{3}R_X^2\sqrt{m_k \log(2d/\delta)}} \\ &= \frac{2\log(2d/\delta)}{1 + \frac{2}{3}\sqrt{\log(2d/\delta)/m_k}} \\ &\ge \frac{2\log(2d/\delta)}{1+1} = \log(2d/\delta), \end{aligned}$$

where the inequality uses $\frac{2}{3}\sqrt{\log(2d/\delta)/m_k} \le 1$ under the stated condition on m_k. Hence

$$\mathbb{P}\left(\Big\| \sum_{t\in\mathcal{T}_k} X_t \Big\|_2 \ge 2R_X\sqrt{m_k \log(2d/\delta)} \right) \le 2d\, e^{-\log(2d/\delta)} = \delta.$$

Dividing both sides by m_k:

$$\|\widehat{M}_k - \bar{M}_k^{\text{probe}}\|_2 = \frac{1}{m_k} \Big\| \sum_{t\in\mathcal{T}_k} X_t \Big\|_2 \le 2R_X \sqrt{\frac{\log(2d/\delta)}{m_k}}$$

with probability at least $1-\delta$. □

F.3.6 Proof of Theorem 6

Proof. **Eigenvalue perturbation.** By Proposition 3, $\bar{M}_k^{\text{probe}}$ has rank exactly r, with eigenvalues $\lambda_1 \ge \cdots \ge \lambda_r \ge \lambda_{\min} > 0$ and $\lambda_{r+1} = \cdots = \lambda_d = 0$. Therefore the eigengap is $\delta_{\text{gap}} := \lambda_r(\bar{M}_k^{\text{probe}}) - \lambda_{r+1}(\bar{M}_k^{\text{probe}}) \ge \lambda_{\min}$.

By Weyl's inequality, for each $i = 1, \ldots, d$,

$$|\lambda_i(\widehat{M}_k) - \lambda_i(\bar{M}_k^{\text{probe}})| \le \|\widehat{M}_k - \bar{M}_k^{\text{probe}}\|_2 \le \frac{\lambda_{\min}}{2},$$

where the last inequality holds by assumption. In particular,

$$\lambda_r(\widehat{M}_k) \geq \lambda_r(\bar{M}_k^{\text{probe}}) - \frac{\lambda_{\min}}{2} \geq \frac{\lambda_{\min}}{2} > 0,$$
$$\lambda_{r+1}(\widehat{M}_k) \leq \lambda_{r+1}(\bar{M}_k^{\text{probe}}) + \frac{\lambda_{\min}}{2} = \frac{\lambda_{\min}}{2}.$$

Hence $\widehat{M}_k$ has a gap of at least $\lambda_r(\widehat{M}_k) - \lambda_{r+1}(\widehat{M}_k) \geq 0$ between its r-th and $(r+1)$-th eigenvalues, so $\mathrm{TopEig}_r(\widehat{M}_k)$ is well-defined.

Davis–Kahan $\sin\Theta$ bound. Applying the $\sin\Theta$ theorem [Yu et al., 2015, Theorem 2] to the symmetric matrices $\bar{M}_k^{\text{probe}}$ (population) and $\widehat{M}_k$ (sample) with gap $\delta_{\text{gap}} \geq \lambda_{\min}$:

$$\|\widehat{P}_k - P_k^\star\|_2 \leq \frac{2\,\|\widehat{M}_k - \bar{M}_k^{\text{probe}}\|_2}{\delta_{\text{gap}}} \leq \frac{2\,\|\widehat{M}_k - \bar{M}_k^{\text{probe}}\|_2}{\lambda_{\min}}.$$

(We use a factor of 2 from the standard $\sin\Theta$ bound; applying the bound with both left and right perturbations gives an additional factor of 2, yielding factor 4 overall.)

Substitution. Substituting the concentration bound from Theorem 5:

$$\|\widehat{P}_k - P_k^\star\|_2 \leq \frac{4\,\|\widehat{M}_k - \bar{M}_k^{\text{probe}}\|_2}{\lambda_{\min}} \leq \frac{4 \cdot 2R_X\sqrt{\log(2d/\delta)/m_k}}{\lambda_{\min}} = \frac{8R_X}{\lambda_{\min}}\sqrt{\frac{\log(2d/\delta)}{m_k}} = C_{\text{sub}}\sqrt{\frac{\log(2d/\delta)}{m_k}},$$

where $C_{\text{sub}} := 8R_X/\lambda_{\min}$. □

F.3.7 Proof of Theorem 7

Proof. **Centered residuals.** By Proposition 5, $\mathbb{E}[\hat{G}_t \mid \mathcal{H}_{t-1}] = \widetilde{M}_t - \widetilde{B}$ where $\widetilde{B} := \delta_\sigma\,\mathcal{K}^{-1}(\Sigma_Q)$. Define the centered residuals

$$\hat{X}_t := \hat{G}_t - \mathbb{E}[\hat{G}_t \mid \mathcal{H}_{t-1}] = \hat{G}_t - (\widetilde{M}_t - \widetilde{B}).$$

By construction, $\mathbb{E}[\hat{X}_t \mid \mathcal{H}_{t-1}] = 0$, so $(\hat{X}_t)_{t\in\mathcal{T}_k}$ is a matrix martingale difference sequence.

Uniform bound. Since $\hat{G}_t = G_t - \delta_\sigma\,\mathcal{K}^{-1}(u_t u_t^\top)$ (from the proof of Proposition 5), we have

$$\begin{aligned}
\|\hat{G}_t\|_2 &\leq \|G_t\|_2 + |\delta_\sigma|\,\|\mathcal{K}^{-1}(u_t u_t^\top)\|_2 \leq C_{\mathcal{K}} R_s L^2 + |\delta_\sigma|\,C_{\mathcal{K}} L^2,\\
\|\hat{X}_t\|_2 &\leq \|\hat{G}_t\|_2 + \|\widetilde{M}_t\|_2 + \|\widetilde{B}\|_2 \leq (C_{\mathcal{K}} R_s L^2 + |\delta_\sigma| C_{\mathcal{K}} L^2) + S_w^2 + |\delta_\sigma| C_{\mathcal{K}} L^2\\
&= R_X + 2C_{\mathcal{K}}|\delta_\sigma| L^2 =: \hat{R}_X.
\end{aligned}$$

Variance proxy. Since $\|\hat{X}_t\|_2 \leq \hat{R}_X$ a.s., the predictable quadratic variation satisfies $\|\sum_{t\in\mathcal{T}_k}\mathbb{E}[\hat{X}_t^2 \mid \mathcal{H}_{t-1}]\|_2 \leq m_k \hat{R}_X^2$, by the same argument as Lemma 6.

Matrix Freedman application. Applying the Matrix Freedman inequality with parameters $R = \hat{R}_X$ and $W = m_k \hat{R}_X^2$ (exactly as in the proof of Theorem 5) yields

$$\|\widehat{M}_k^{\hat{\sigma}} - (\bar{M}_k^{\text{probe}} - \widetilde{B})\|_2 = \frac{1}{m_k}\Big\|\sum_{t\in\mathcal{T}_k}\hat{X}_t\Big\|_2 \leq 2\hat{R}_X\sqrt{\frac{\log(2d/\delta)}{m_k}}$$

with probability at least $1-\delta$.

Triangle inequality. By the triangle inequality and $\|\widetilde{B}\|_2 \leq C_{\mathcal{K}}|\delta_\sigma| L^2$ (Proposition 5):

$$\|\widehat{M}_k^{\hat{\sigma}} - \bar{M}_k^{\text{probe}}\|_2 \leq \|\widehat{M}_k^{\hat{\sigma}} - (\bar{M}_k^{\text{probe}} - \widetilde{B})\|_2 + \|\widetilde{B}\|_2 \leq 2\hat{R}_X\sqrt{\frac{\log(2d/\delta)}{m_k}} + C_{\mathcal{K}}|\delta_\sigma| L^2. \quad \square$$

F.3.8 Proof of Corollary 2

Proof. **Gap verification.** By Theorem 7, $\|\widehat{M}_k^{\hat{\sigma}} - \bar{M}_k^{\text{probe}}\|_2 \le 2\hat{R}_X\sqrt{\log(2d/\delta)/m_k} + C_{\mathcal{K}}|\delta_\sigma|L^2$. Under the hypotheses $C_{\mathcal{K}}|\delta_\sigma|L^2 \le \lambda_{\min}/4$ and $m_k \ge 64\hat{R}_X^2\log(2d/\delta)/\lambda_{\min}^2$, we have $2\hat{R}_X\sqrt{\log(2d/\delta)/m_k} \le \lambda_{\min}/4$. Adding both terms: $\|\widehat{M}_k^{\hat{\sigma}} - \bar{M}_k^{\text{probe}}\|_2 \le \lambda_{\min}/4 + \lambda_{\min}/4 = \lambda_{\min}/2$. This satisfies the gap condition for Davis–Kahan (as in Theorem 6).

Davis–Kahan application. With the perturbation at most $\lambda_{\min}/2$, Weyl's inequality guarantees the r-th eigenvalue gap of $\widehat{M}_k^{\hat{\sigma}}$ is positive, and the $\sin\Theta$ bound gives

$$\|\widehat{P}_k - P_k^\star\|_2 \le \frac{4\,\|\widehat{M}_k^{\hat{\sigma}} - \bar{M}_k^{\text{probe}}\|_2}{\lambda_{\min}}.$$

Decomposition. Splitting the numerator via the triangle inequality:

$$\|\widehat{M}_k^{\hat{\sigma}} - \bar{M}_k^{\text{probe}}\|_2 \le \underbrace{\|\widehat{M}_k^{\hat{\sigma}} - (\bar{M}_k^{\text{probe}} - \widetilde{B})\|_2}_{\le 2\hat{R}_X\sqrt{\log(2d/\delta)/m_k}} + \underbrace{\|\widetilde{B}\|_2}_{\le C_{\mathcal{K}}|\delta_\sigma|L^2}.$$

Substituting into the Davis–Kahan bound:

$$\|\widehat{P}_k - P_k^\star\|_2 \le \frac{8\hat{R}_X}{\lambda_{\min}}\sqrt{\frac{\log(2d/\delta)}{m_k}} + \frac{4C_{\mathcal{K}}|\delta_\sigma|L^2}{\lambda_{\min}} = \hat{C}_{\text{sub}}\sqrt{\frac{\log(2d/\delta)}{m_k}} + \Delta_\sigma,$$

where $\hat{C}_{\text{sub}} := 8\hat{R}_X/\lambda_{\min}$ and $\Delta_\sigma := 4C_{\mathcal{K}}|\delta_\sigma|L^2/\lambda_{\min}$. □

F.4 Bandit Tools and Projected Exploitation

F.4.1 Proof of Lemma 7

Proof. This is the standard self-normalized martingale bound for vector-valued martingales. The result follows from Abbasi-Yadkori et al. [2011, Theorem 1] applied in $\mathbb{R}^r$ with regularization λI_r. □

F.4.2 Proof of Lemma 8

Proof. By the AM–GM inequality applied to the eigenvalues of $V = \lambda I_r + \sum_{s=1}^N z_s z_s^\top$: $\det(V) \le (\operatorname{tr}(V)/r)^r \le (\lambda + NR_{\mathcal{A}}^2/r)^r$. Hence $\log(\det(V)/\lambda^r) \le r\log(1 + NR_{\mathcal{A}}^2/(\lambda r))$. □

F.4.3 Proof of Lemma 9

Proof. By the Cauchy–Schwarz inequality and the determinant-trace inequality: $\sum_{t=1}^N \|z_t\|_{V_{t-1}^{-1}} \le \sqrt{N\sum_{t=1}^N \|z_t\|_{V_{t-1}^{-1}}^2} \le \sqrt{N\cdot 2\log(\det(V_N)/\lambda^r)}$. Applying Lemma 8 yields the result. □

F.4.4 Proof of Theorem 9

Proof. Fix $t \in E_k$ and define the reduced reference parameter $a_t^\circ := \widehat{U}_k^\top\theta_{\tau_{k-1}} \in \mathbb{R}^r$, the projection of the segment's initial reward parameter into the estimated subspace.

Reward decomposition. For any action $x \in \mathcal{A}_t$, the reward admits the decomposition

$$\begin{aligned} x^\top\theta_t &= x^\top\widehat{P}_k\theta_t + x^\top(I - \widehat{P}_k)\theta_t \\ &= z_t(x)^\top(\widehat{U}_k^\top\theta_t) + x^\top(I - \widehat{P}_k)\theta_t \\ &= z_t(x)^\top a_t^\circ + z_t(x)^\top\widehat{U}_k^\top(\theta_t - \theta_{\tau_{k-1}}) + x^\top(I - \widehat{P}_k)\theta_{\tau_{k-1}} + x^\top(I - \widehat{P}_k)(\theta_t - \theta_{\tau_{k-1}}). \end{aligned}$$

Bounding error terms. *Subspace mismatch:* Since $\theta_{\tau_{k-1}} = B_k^\star w_{\tau_{k-1}} \in \operatorname{range}(B_k^\star) = \operatorname{range}(P_k^\star)$, we have $(I - P_k^\star)\theta_{\tau_{k-1}} = 0$, so

$$|x^\top(I - \widehat{P}_k)\theta_{\tau_{k-1}}| = |x^\top(\widehat{P}_k - P_k^\star)\theta_{\tau_{k-1}}| \le \|x\|_2\,\|\widehat{P}_k - P_k^\star\|_2\,\|\theta_{\tau_{k-1}}\|_2 \le R_{\mathcal{A}}\,\varepsilon_k\,S_w = \Gamma_k\,\|x\|_2.$$

Drift: $|z_t(x)^\top \widehat{U}_k^\top(\theta_t - \theta_{\tau_{k-1}})| \le \|z_t(x)\|_2 \, \|\theta_t - \theta_{\tau_{k-1}}\|_2 \le R_{\mathcal{A}} D_{k,t}$ where $D_{k,t} := \|\theta_t - \theta_{\tau_{k-1}}\|_2$. The remaining cross term $|x^\top(I - \widehat{P}_k)(\theta_t - \theta_{\tau_{k-1}})| \le R_{\mathcal{A}} D_{k,t}$ is absorbed into the drift error.

Estimation error via self-normalized bound. On exploitation rounds $s < t$ within E_k, the response satisfies $y_s = z_t(x_s)^\top a_t^\circ + \xi_s$ where ξ_s collects noise, subspace mismatch, and drift errors. Applying Lemma 7 to the ridge estimator $\widehat{a}_t = \widetilde{V}_{t-1}^{-1} \widetilde{b}_{t-1}$ in $\mathbb{R}^r$:

$$\|(\widehat{a}_t - a_t^\circ)\|_{\widetilde{V}_{t-1}} \le \beta_{k,\max}^{(r)}$$

with probability at least $1 - \delta$, uniformly over t, where $\beta_{k,\max}^{(r)}$ incorporates the sub-Gaussian noise level and the regularization.

Optimism and per-round regret. The UCB index ensures $z_t x_t^\top \widehat{a}_t + \beta_{k,\max}^{(r)} \|z_t(x_t)\|_{\widetilde{V}_{t-1}^{-1}} + (\Gamma_k + R_{\mathcal{A}} D_{k,t}) \|x_t\|_2 \ge x_t^{\star\top} \theta_t$ where $x_t^\star := \arg\max_{x \in \mathcal{A}_t} x^\top \theta_t$. The instantaneous regret satisfies

$$x_t^{\star\top} \theta_t - x_t^\top \theta_t \le 2\beta_{k,\max}^{(r)} \|z_t(x_t)\|_{\widetilde{V}_{t-1}^{-1}} + 2\Gamma_k R_{\mathcal{A}} + 2R_{\mathcal{A}} D_{k,t}.$$

Summation. Summing over $t \in E_k$ and applying the elliptical potential bound (Lemma 9):

$$\sum_{t \in E_k} \|z_t(x_t)\|_{\widetilde{V}_{t-1}^{-1}} \le \sqrt{2r \, n_k \, \log\left(1 + \frac{n_k R_{\mathcal{A}}^2}{\lambda r}\right)},$$

and $\sum_{t \in E_k} \Gamma_k = \varepsilon_k S_w R_{\mathcal{A}} \, n_k$. Combining:

$$R_k^{\exp} \le 2\beta_{k,\max}^{(r)} \sqrt{2r \, n_k \, \log\left(1 + \frac{n_k R_{\mathcal{A}}^2}{\lambda r}\right)} + 2S_w R_{\mathcal{A}} \, \varepsilon_k \, n_k + 2R_{\mathcal{A}} \sum_{t \in E_k} D_{k,t}. \qquad \square$$

F.4.5 Proof of Theorem 8

Proof. Fix a segment $\mathcal{I}_k$ and let $E_k := \mathcal{I}_k \setminus \mathcal{T}_{\text{probe}}$ denote the exploitation rounds, with $n_k := |E_k|$.

Reward decomposition. For each exploitation round $t \in E_k$, the reduced feature is $z_t(x) := \widehat{U}_{t-1}^\top x \in \mathbb{R}^r$ and the reduced parameter is $a_t^\star := \widehat{U}_{t-1}^\top \theta_t$. The reward decomposes as

$$x^\top \theta_t = \underbrace{z_t(x)^\top a_t^\star}_{\text{projected reward}} + \underbrace{x^\top (I - \widehat{P}_k) \theta_t}_{\text{subspace residual}}.$$

Subspace mismatch bound. Since $\theta_t = B_k^\star w_t$ and $(I - P_k^\star) B_k^\star = 0$:

$$\begin{aligned}
|x^\top (I - \widehat{P}_k) \theta_t| &= |x^\top (P_k^\star - \widehat{P}_k) B_k^\star w_t + x^\top (I - P_k^\star) B_k^\star w_t| \\
&= |x^\top (P_k^\star - \widehat{P}_k) B_k^\star w_t| \\
&\le \|x\|_2 \cdot \|\widehat{P}_k - P_k^\star\|_2 \cdot \|B_k^\star\|_2 \cdot \|w_t\|_2 \\
&\le R_{\mathcal{A}} \cdot \varepsilon_k \cdot 1 \cdot S_w = \Gamma_k \|x\|_2,
\end{aligned}$$

where $\Gamma_k := S_w \varepsilon_k$.

Windowed regression model. For past exploitation rounds $s \in \mathcal{W}_t := \{s \in E_k : t - W \le s < t\}$:

$$y_s = z_t(x_s)^\top a_s^\star + x_s^\top (I - \widehat{P}_k) \theta_s + \varepsilon_s.$$

Writing $y_s = z_t(x_s)^\top a_t^\star + \tilde{\varepsilon}_s$ where $\tilde{\varepsilon}_s := \varepsilon_s + z_t(x_s)^\top (a_s^\star - a_t^\star) + x_s^\top (I - \widehat{P}_k) \theta_s$.
The drift error satisfies

$$|z_t(x_s)^\top (a_s^\star - a_t^\star)| = |z_t(x_s)^\top \widehat{U}_k^\top (\theta_s - \theta_t)| \le R_{\mathcal{A}} \|\theta_s - \theta_t\|_2 \le R_{\mathcal{A}} V_{k,t}(W),$$

since $s \in [t-W, t)$ and $V_{k,t}(W) := \sum_{j \in E_k,\, t-W \le j < t} \|\theta_{j+1} - \theta_j\|_2 \ge \|\theta_s - \theta_t\|_2$ by the triangle inequality.

Self-normalized confidence set. The windowed ridge estimator is $\widehat{a}_t = \widetilde{V}_t^{-1}\widetilde{b}_t$ where $\widetilde{V}_t = \lambda I_r + \sum_{s \in \mathcal{W}_t} z_s z_s^\top$ and $\widetilde{b}_t = \sum_{s \in \mathcal{W}_t} z_s y_s$. Then

$$\widetilde{V}_t(\widehat{a}_t - a_t^\star) = \sum_{s \in \mathcal{W}_t} z_s \varepsilon_s + \sum_{s \in \mathcal{W}_t} z_s \big[z_s^\top (a_s^\star - a_t^\star) + x_s^\top (I - \widehat{P}_k)\theta_s\big] - \lambda a_t^\star.$$

By Lemma 7 applied to the martingale term $\sum_{s \in \mathcal{W}_t} z_s \varepsilon_s$ in $\mathbb{R}^r$, and using Lemma 8 with $|\mathcal{W}_t| \le W$, the estimation error in the $\widetilde{V}_t$-norm is bounded by $\beta_t^{(r,W)}$ (from Eq. (7)) with probability at least $1-\delta$.

Optimism argument. The UCB index at round t is

$$\mathrm{UCB}_t(x) := z_t(x)^\top \widehat{a}_t + \beta_t^{(r,W)} \|z_t(x)\|_{\widetilde{V}_t^{-1}} + (\Gamma_k + V_{k,t}(W)) R_{\mathcal{A}}.$$

For the optimal action $x_t^\star := \arg\max_{x \in \mathcal{A}_t} x^\top \theta_t$:

$$\begin{aligned}\mathrm{UCB}_t(x_t^\star) &\ge z_t(x_t^\star)^\top a_t^\star - \beta_t^{(r,W)} \|z_t(x_t^\star)\|_{\widetilde{V}_t^{-1}} - \Gamma_k R_{\mathcal{A}} + \beta_t^{(r,W)} \|z_t(x_t^\star)\|_{\widetilde{V}_t^{-1}} + (\Gamma_k + V_{k,t}(W)) R_{\mathcal{A}} \\ &\ge x_t^{\star\top}\theta_t - \Gamma_k R_{\mathcal{A}} + V_{k,t}(W) R_{\mathcal{A}} \ge x_t^{\star\top}\theta_t.\end{aligned}$$

Since x_t maximizes UCB_t:

$$\begin{aligned}x_t^{\star\top}\theta_t - x_t^\top\theta_t &\le \mathrm{UCB}_t(x_t) - x_t^\top \theta_t \\ &\le 2\beta_t^{(r,W)} \|z_t(x_t)\|_{\widetilde{V}_t^{-1}} + 2\Gamma_k R_{\mathcal{A}} + 2R_{\mathcal{A}} V_{k,t}(W).\end{aligned}$$

Summation over exploitation rounds. Summing the per-round regret over $t \in E_k$:

$$R_k^{\mathrm{exp}} \le 2\beta_k^{(r,W)} \sum_{t \in E_k} \|z_t(x_t)\|_{\widetilde{V}_t^{-1}} + 2\Gamma_k R_{\mathcal{A}} n_k + 2R_{\mathcal{A}} \sum_{t \in E_k} V_{k,t}(W),$$

where $\beta_k^{(r,W)} := \max_{t \in E_k} \beta_t^{(r,W)}$ and $\Gamma_k = S_w \varepsilon_k$.

By the elliptical potential bound (Lemma 9), using $|\mathcal{W}_t| \le W$ and $\|z_t\|_2 \le R_{\mathcal{A}}$:

$$\sum_{t \in E_k} \|z_t(x_t)\|_{\widetilde{V}_t^{-1}} \le \sqrt{2r\, n_k \, \log\!\left(1 + \frac{W R_{\mathcal{A}}^2}{\lambda r}\right)}.$$

Substituting gives the stated bound:

$$R_k^{\mathrm{exp}} \le 2\beta_k^{(r,W)} \sqrt{2r\, n_k \, \log\!\left(1 + \frac{W R_{\mathcal{A}}^2}{\lambda r}\right)} + 2S_w R_{\mathcal{A}}\, \varepsilon_k\, n_k + 2R_{\mathcal{A}} \sum_{t \in E_k} V_{k,t}(W). \qquad \square$$

F.4.6 Proof of Theorem 10

Proof. Fix $t \in E_k$ and define $a_t^\star := \widehat{U}_k^\top \theta_t \in \mathbb{R}^r$.

Reward decomposition. For any $x \in \mathcal{A}_t$:

$$x^\top \theta_t = z_t(x)^\top a_t^\star + x^\top (I - \widehat{P}_k)\theta_t,$$

where $|x^\top (I - \widehat{P}_k)\theta_t| \le \|x\|_2 \, \|(I - \widehat{P}_k) B_k^\star\|_2 \, \|w_t\|_2 \le S_w \varepsilon_k \|x\|_2 = \Gamma_k \|x\|_2$.

Windowed regression error. For past rounds $s \in \mathcal{W}_t$, the response is $y_s = z_t(x_s)^\top a_s^\star + x_s^\top (I - \widehat{P}_k)\theta_s + \varepsilon_s$. The target parameter at time t differs from the effective parameter at time s by

$$e_{s,t}^{\mathrm{drift}} := z_t(x_s)^\top (a_t^\star - a_s^\star) = z_t(x_s)^\top \widehat{U}_k^\top (\theta_t - \theta_s).$$

Hence $|e_{s,t}^{\mathrm{drift}}| \le \|z_t(x_s)\|_2 \, \|\theta_t - \theta_s\|_2 \le R_{\mathcal{A}} \, \|\theta_t - \theta_s\|_2 \le R_{\mathcal{A}} \, V_{k,t}(W)$ since $s \in [t-W, t)$.

Self-normalized confidence set. Define the effective noise $\tilde{\varepsilon}_s := y_s - z_t(x_s)^\top a_t^\star = e_{s,t}^{\text{drift}} + x_s^\top (I - \widehat{P}_k)\theta_s + \varepsilon_s$. The terms $e_{s,t}^{\text{drift}}$ and $x_s^\top (I - \widehat{P}_k)\theta_s$ are bounded deterministically. The noise ε_s is conditionally sub-Gaussian with parameter σ_ε.

Applying Lemma 7 to the windowed ridge estimator $\widehat{a}_t = \widetilde{V}_t^{-1}\widetilde{b}_t$ in $\mathbb{R}^r$, and using Lemma 8 with at most W summands:

$$\|\widehat{a}_t - a_t^\star\|_{\widetilde{V}_t} \le \beta_t^{(r,W)} + \Gamma_k\sqrt{\lambda + WR_{\mathcal{A}}^2} + R_{\mathcal{A}} V_{k,t}(W)\sqrt{\lambda + WR_{\mathcal{A}}^2}$$

with high probability. The dominant term in the UCB width is $\beta_t^{(r,W)}\|z_t(x)\|_{\widetilde{V}_t^{-1}}$.

Optimism and per-round regret. The UCB selection rule guarantees optimism: for any comparator $x_t^\star$,

$$x_t^{\star\top}\theta_t - x_t^\top\theta_t \le 2\beta_t^{(r,W)}\|z_t(x_t)\|_{\widetilde{V}_t^{-1}} + 2\Gamma_k R_{\mathcal{A}} + 2R_{\mathcal{A}} V_{k,t}(W).$$

Summation with elliptical potential. Summing over $t \in E_k$:

$$R_k^{\text{exp}} \le 2\beta_k^{(r,W)} \sum_{t\in E_k} \|z_t(x_t)\|_{\widetilde{V}_t^{-1}} + 2S_w R_{\mathcal{A}} \varepsilon_k n_k + 2R_{\mathcal{A}} \sum_{t\in E_k} V_{k,t}(W).$$

By Lemma 9 (with at most W terms per window): $\sum_{t\in E_k} \|z_t\|_{\widetilde{V}_t^{-1}} \le \sqrt{2r\, n_k \, \log(1 + WR_{\mathcal{A}}^2/(\lambda r))}$.

For the drift sum, each parameter difference $\|\theta_{s+1} - \theta_s\|_2$ in segment k is counted by at most W exploitation rounds $t \in [s+1, s+W]$, so $\sum_{t\in E_k} V_{k,t}(W) \le WV_k$.

Combining: $R_k^{\text{exp}} \le 2\beta_k^{(r,W)}\sqrt{2r\, n_k \, \log(1 + WR_{\mathcal{A}}^2/(\lambda r))} + 2S_w R_{\mathcal{A}}\, \varepsilon_k\, n_k + 2R_{\mathcal{A}} WV_k$. $\square$

F.5 Regret Bounds

F.5.1 Proof of Theorem 1

Proof. The costed dynamic regret decomposes into probe-round and exploitation-round contributions. On each probe round $t \in \mathcal{T}_{\text{probe}} \cap \mathcal{I}_k$, the instantaneous regret is at most $\max_{x\in\mathcal{A}_t} x^\top\theta_t - u_t^\top\theta_t \le 2R_{\mathcal{A}}\|\theta_t\|_2 \le 2R_{\mathcal{A}} S_w$, plus the per-probe cost c. This contributes $(2R_{\mathcal{A}} S_w + c)m_k$ per segment.

Summing Theorem 8 over all K segments and adding the probe cost:

$$\begin{aligned}
\mathrm{DynReg}_T^{(c)} &\le \sum_{k=1}^K \big[(2R_{\mathcal{A}} S_w + c)m_k + R_k^{\text{exp}}\big] \\
&\le (2R_{\mathcal{A}} S_w + c)\sum_{k=1}^K m_k + 2\sum_{k=1}^K \beta_k^{(r,W)}\sqrt{2rn_k \log\left(1 + \frac{WR_{\mathcal{A}}^2}{\lambda r}\right)} \\
&\quad + 2S_w R_{\mathcal{A}} \sum_{k=1}^K \varepsilon_k n_k + 2R_{\mathcal{A}} \sum_{k=1}^K \sum_{t\in E_k} V_{k,t}(W).
\end{aligned}$$

Since each exploitation round t contributes $V_{k,t}(W) \le \sum_{s: t-W\le s<t} \|\theta_{s+1} - \theta_s\|_2$, we have $\sum_{t\in E_k} V_{k,t}(W) \le WV_k$. $\square$

F.5.2 Proof of Theorem 2 (Main Result)

Proof. Starting from Theorem 1 and substituting (6), the costed regret satisfies

$$\mathrm{DynReg}_T^{(c)} \le A\sum_{k=1}^K m_k + 2\beta^{(r,W)}\sqrt{2rL_W}\sum_{k=1}^K \sqrt{\ell_k} + B\sum_{k=1}^K \frac{\ell_k}{\sqrt{m_k}} + 2R_{\mathcal{A}} WV.$$

For each segment k, define $f_k(m) := Am + B\ell_k m^{-1/2}$. Setting $f_k'(m) = A - \frac{1}{2}B\ell_k m^{-3/2} = 0$ yields the optimal probe count $m_k^\star = (B\ell_k/(2A))^{2/3}$. Substituting:

$$f_k(m_k^\star) = A\left(\frac{B\ell_k}{2A}\right)^{2/3} + B\ell_k\left(\frac{2A}{B\ell_k}\right)^{1/3}$$
$$= A^{1/3}(B\ell_k)^{2/3}\left(2^{-2/3} + 2^{1/3}\right) = \frac{3}{2^{2/3}}A^{1/3}B^{2/3}\ell_k^{2/3}.$$

Using the integer ceiling $m_k = \lceil m_k^\star \rceil$ changes each term by at most A, giving

$$A\sum_{k=1}^{K} m_k + B\sum_{k=1}^{K}\frac{\ell_k}{\sqrt{m_k}} \le \frac{3}{2^{2/3}}A^{1/3}B^{2/3}\sum_{k=1}^{K}\ell_k^{2/3} + AK,$$

which proves (8). Applying the Cauchy–Schwarz inequality $\sum_{k=1}^K \sqrt{\ell_k} \le \sqrt{K\sum_k \ell_k} = \sqrt{KT}$ and Hölder's inequality $\sum_{k=1}^K \ell_k^{2/3} \le K^{1/3}(\sum_k \ell_k)^{2/3} = K^{1/3}T^{2/3}$ yields (9). Since W appears logarithmically in $\beta^{(r,W)}$ (via L_W) and linearly in the drift term $2R_{\mathcal{A}}WV$, the bound is minimized at $W^\star = 1$. Suppressing logarithmic factors in $\beta^{(r,1)}$, L_1, and B gives (10). ☐

F.5.3 Proof of Theorem 3

Proof. The proof combines the variance-misspecification bias (Proposition 5) and the cross-correlation bias (Proposition 6). Under both sources, the lifted estimator satisfies $\mathbb{E}[G_t \mid \mathcal{H}_{t-1}] = \widetilde{M}_t + \widetilde{B}_t$, where $\|\widetilde{B}_t\|_2 \le C_{\mathcal{K}}|\delta_\sigma|L^2 + 2C_{\mathcal{K}}L^2\epsilon_\times =: B_{\text{total}}$.

By Theorem 7, the segment-level estimator satisfies

$$\|\widehat{M}_k - \bar{M}_k^{\text{probe}}\|_2 \le 2\hat{R}_X\sqrt{\frac{\log(2d/\delta)}{m_k}} + B_{\text{total}}.$$

Under $B_{\text{total}} \le \lambda_{\min}/4$ and $m_k \ge 64\hat{R}_X^2\log(2d/\delta)/\lambda_{\min}^2$, this is at most $\lambda_{\min}/2$, so the eigenvalue separation condition for Davis–Kahan holds. By Corollary 2, the projector error decomposes as

$$\varepsilon_k \le \hat{C}_{\text{sub}}\sqrt{\frac{\log(2d/\delta)}{m_k}} + \Delta_{\text{total}}, \qquad \hat{C}_{\text{sub}} := \frac{8\hat{R}_X}{\lambda_{\min}}, \quad \Delta_{\text{total}} := \frac{4\,B_{\text{total}}}{\lambda_{\min}}.$$

Substituting into the subspace-error term of Theorem 1:

$$2S_wR_{\mathcal{A}}\sum_{k=1}^{K}\varepsilon_k n_k \le 2S_wR_{\mathcal{A}}\sum_{k=1}^{K}\hat{C}_{\text{sub}}\sqrt{\frac{\log(d/\delta)}{m_k}}\,n_k + 2S_wR_{\mathcal{A}}\Delta_{\text{total}}\sum_{k=1}^{K}n_k.$$

The first summand is handled identically to Theorem 2 (with $\hat{C}_{\text{sub}}$ in place of C_{sub}), contributing $\widetilde{O}(r\sqrt{T}) + \widetilde{O}(T^{2/3})$. The second summand is $2S_wR_{\mathcal{A}}\Delta_{\text{total}}T = O\big((|\delta_\sigma| + \epsilon_\times)\,T/\lambda_{\min}\big)$. If $|\delta_\sigma| = O(T^{-1/3})$ (via burn-in estimation, Remark 1) and $\epsilon_\times = O(T^{-1/3})$, both penalty terms are $O(T^{2/3})$ and the overall rate is unchanged. When $\delta_\sigma = 0$ and $\epsilon_\times = 0$, $\Delta_{\text{total}} = 0$ and the bound reduces to Theorem 2. ☐

F.6 Detector and Adaptive Algorithm

F.6.1 Proof of Lemma 10

Proof. Fix $s < t$ in segment k, both at least τ_{burn} rounds after τ_{k-1}. Recall $\widetilde{M}_t = B_k^\star\,\mathbb{E}[w_tw_t^\top \mid \mathcal{H}_{t-1}]\,(B_k^\star)^\top$ where

$$\mathbb{E}[w_tw_t^\top \mid \mathcal{H}_{t-1}] = A_k\,w_{t-1}w_{t-1}^\top\,A_k^\top + \Sigma_{\eta,k}.$$

Comparison to the stationary covariance. Let Σ_k^∞ be the unique positive-definite solution to the discrete Lyapunov equation $\Sigma_k^\infty = A_k\,\Sigma_k^\infty\,A_k^\top + \Sigma_{\eta,k}$. Define $\Sigma_t^{\text{cond}} := \mathbb{E}[w_tw_t^\top \mid \mathcal{H}_{t-1}] = A_k\,w_{t-1}w_{t-1}^\top\,A_k^\top + \Sigma_{\eta,k}$.

Deterministic bound via bounded states. Since $\|w_t\|_2 \le S_w$ for all t, we have $\Sigma_t^{\text{cond}} = A_k w_{t-1} w_{t-1}^\top A_k^\top + \Sigma_{\eta,k}$ with $\|A_k w_{t-1} w_{t-1}^\top A_k^\top\|_{\text{op}} \le (1-\alpha_0)^2 S_w^2$. Similarly for Σ_s^{cond}. Therefore

$$\begin{aligned}\|\Sigma_t^{\text{cond}} - \Sigma_s^{\text{cond}}\|_{\text{op}} &= \|A_k(w_{t-1}w_{t-1}^\top - w_{s-1}w_{s-1}^\top)A_k^\top\|_{\text{op}} \\ &\le \|A_k\|_{\text{op}}^2 \, \|w_{t-1}w_{t-1}^\top - w_{s-1}w_{s-1}^\top\|_{\text{op}} \\ &\le (1-\alpha_0)^2\big(\|w_{t-1}\|_2^2 + \|w_{s-1}\|_2^2\big) \\ &\le 2(1-\alpha_0)^2 S_w^2,\end{aligned}$$

where we used $\|A_k\|_{\text{op}} \le \rho(A_k) \le 1-\alpha_0$ (since A_k is $r \times r$ and the spectral radius bounds the operator norm for the powers appearing in the stability condition) and the triangle inequality $\|ab^\top - cd^\top\| \le \|a\|\|b\| + \|c\|\|d\|$.

Tighter bound via mixing. For a tighter bound that decays with τ_{burn}, we use the mixing of the unconditional law. The unconditional covariance $\bar\Sigma_t := \mathbb{E}[w_t w_t^\top]$ satisfies the Lyapunov recursion exactly: $\bar\Sigma_{t+1} = A_k \, \bar\Sigma_t \, A_k^\top + \Sigma_{\eta,k}$. Subtracting the fixed point:

$$\bar\Sigma_t - \Sigma_k^\infty = A_k^{t-\tau_{k-1}}(\bar\Sigma_{\tau_{k-1}} - \Sigma_k^\infty)(A_k^\top)^{t-\tau_{k-1}},$$

so $\|\bar\Sigma_t - \Sigma_k^\infty\|_{\text{op}} \le (1-\alpha_0)^{2(t-\tau_{k-1})} \cdot 2S_w^2$. When $t \ge \tau_{k-1} + \tau_{\text{burn}}$, this is at most $2(1-\alpha_0)^{2\tau_{\text{burn}}} S_w^2$.

Since $\Sigma_t^{\text{cond}} - \bar\Sigma_t = A_k(w_{t-1}w_{t-1}^\top - \bar\Sigma_{t-1})A_k^\top$, the difference $\Sigma_t^{\text{cond}} - \bar\Sigma_t$ has operator norm at most $(1-\alpha_0)^2 \cdot 2S_w^2$ deterministically. Therefore, by the triangle inequality through $\bar\Sigma_t$ and $\bar\Sigma_s$:

$$\begin{aligned}\|\Sigma_t^{\text{cond}} - \Sigma_s^{\text{cond}}\|_{\text{op}} &\le \|\Sigma_t^{\text{cond}} - \bar\Sigma_t\|_{\text{op}} + \|\bar\Sigma_t - \bar\Sigma_s\|_{\text{op}} + \|\bar\Sigma_s - \Sigma_s^{\text{cond}}\|_{\text{op}} \\ &\le 2(1-\alpha_0)^2 S_w^2 + \|\bar\Sigma_t - \bar\Sigma_s\|_{\text{op}} + 2(1-\alpha_0)^2 S_w^2.\end{aligned}$$

For the middle term, since both $\bar\Sigma_t$ and $\bar\Sigma_s$ are within $(1-\alpha_0)^{2\tau_{\text{burn}}} \cdot 2S_w^2$ of Σ_k^∞, we get $\|\bar\Sigma_t - \bar\Sigma_s\|_{\text{op}} \le 4(1-\alpha_0)^{2\tau_{\text{burn}}} S_w^2$.

Since $\widetilde M_t = B_k^\star \Sigma_t^{\text{cond}} (B_k^\star)^\top$ and $\|B_k^\star\|_2 = 1$:

$$\|\widetilde M_t - \widetilde M_s\|_{\text{op}} \le \|\Sigma_t^{\text{cond}} - \Sigma_s^{\text{cond}}\|_{\text{op}}.$$

Setting $\Delta_{\text{within}}(\tau_{\text{burn}}) := 4(1-\alpha_0)^2 S_w^2 + 4(1-\alpha_0)^{2\tau_{\text{burn}}} S_w^2$ gives the claimed bound. For large τ_{burn}, the second term vanishes and the bound approaches $4(1-\alpha_0)^2 S_w^2$. □

F.6.2 Proof of Theorem 11

Proof. **Concentration on each window.** Under the no-change hypothesis, both the recent and past windows lie entirely within one segment. Let $\bar M^{\text{recent}}$ and $\bar M^{\text{past}}$ be the respective population targets (averages of $\widetilde M_t$ over each window). By Theorem 5 applied independently to each window with confidence level $\delta_{\text{FA}}/(2T)$:

$$\|\widehat M_t^{\text{recent}} - \bar M^{\text{recent}}\|_{\text{op}} \le 2R_X \sqrt{\frac{\log(4dT/\delta_{\text{FA}})}{n_r}},$$

$$\|\widehat M_t^{\text{past}} - \bar M^{\text{past}}\|_{\text{op}} \le 2R_X \sqrt{\frac{\log(4dT/\delta_{\text{FA}})}{n_p}},$$

each with probability at least $1 - \delta_{\text{FA}}/(2T)$. A union bound over both windows gives joint probability at least $1 - \delta_{\text{FA}}/T$.

Population-level variation. By Lemma 10, since both windows begin at least τ_{burn} rounds after the most recent change point:

$$\|\bar M^{\text{recent}} - \bar M^{\text{past}}\|_{\text{op}} \le \Delta_{\text{within}}.$$

Triangle inequality for the detector statistic.

$$\begin{aligned} S_t &= \|\widehat{M}_t^{\text{recent}} - \widehat{M}_t^{\text{past}}\|_{\text{op}} \\ &\leq \|\widehat{M}_t^{\text{recent}} - \bar{M}^{\text{recent}}\|_{\text{op}} + \|\bar{M}^{\text{recent}} - \bar{M}^{\text{past}}\|_{\text{op}} + \|\bar{M}^{\text{past}} - \widehat{M}_t^{\text{past}}\|_{\text{op}} \\ &\leq 2R_X\sqrt{\frac{\log(4dT/\delta_{\text{FA}})}{n_r}} + \Delta_{\text{within}} + 2R_X\sqrt{\frac{\log(4dT/\delta_{\text{FA}})}{n_p}} = b. \end{aligned}$$

Hence $\mathbb{P}(S_t > b) \leq \delta_{\text{FA}}/T$ for any fixed t.

Union bound over time. Taking a union bound over all T rounds: $\mathbb{E}[F_T] = \sum_{t=1}^{T} \mathbb{P}(S_t > b) \leq T \cdot \delta_{\text{FA}}/T = \delta_{\text{FA}}$. By Markov's inequality, $\mathbb{P}(F_T \geq 1) \leq \delta_{\text{FA}}$, so $F_T = 0$ with probability at least $1 - \delta_{\text{FA}}$. □

F.6.3 Proof of Theorem 12

Proof. **Constructing the detection round.** After a true change point τ_k, the detector requires at most $D_{\max} := (n_r + n_p)/\mu + \tau_{\text{burn}}$ rounds before a suitable detection round $t^\star$ exists. Specifically, $t^\star$ is the first round at which: (a) the recent window of n_r probes lies entirely in $[\tau_k, t^\star]$ (post-change), and (b) the past window of n_p probes lies entirely in $[\tau_k - n_p/\mu, \tau_k)$ (pre-change), and (c) at least τ_{burn} rounds have elapsed since τ_k.

Reverse triangle inequality. At round $t^\star$, the population targets satisfy $\bar{M}^{\text{recent}} = \bar{M}_{k+1}^{\text{post}}$ (post-change) and $\bar{M}^{\text{past}} = \bar{M}_k^{\text{pre}}$ (pre-change). By the reverse triangle inequality:

$$S_{t^\star} \geq \|\bar{M}^{\text{recent}} - \bar{M}^{\text{past}}\|_{\text{op}} - \|\widehat{M}_{t^\star}^{\text{recent}} - \bar{M}^{\text{recent}}\|_{\text{op}} - \|\widehat{M}_{t^\star}^{\text{past}} - \bar{M}^{\text{past}}\|_{\text{op}}.$$

Signal dominates noise. Since $\Delta_k \geq \Delta_{\text{det}} = 2b$, the population separation satisfies $\|\bar{M}^{\text{recent}} - \bar{M}^{\text{past}}\|_{\text{op}} \geq \Delta_k - \Delta_{\text{within}} \geq 2b - \Delta_{\text{within}}$.

By Theorem 5 applied to each window at confidence $\delta/(2(K+1))$, each concentration error is at most $2R_X\sqrt{\log(4d(K+1)/\delta)/n}$. Under the threshold calibration (12), the sum of concentration errors plus Δ_{within} equals b. Therefore:

$$S_{t^\star} \geq (2b - \Delta_{\text{within}}) - (b - \Delta_{\text{within}}) = b.$$

More precisely, $S_{t^\star} > b$ because $\Delta_k > 2b$ gives strict inequality.

Detection guarantee. Hence the detector fires at or before $t^\star$, with delay $D_k \leq D_{\max}$. A union bound over all K change points gives the joint probability guarantee of at least $1 - \delta/(K+1) \cdot K \geq 1 - \delta$. □

F.6.4 Proof of Theorem 4

Proof. We partition the time horizon into three types of rounds for each true segment and bound the regret contribution of each type.

High-probability detector events. By Theorem 11 with $\delta_{\text{FA}} = \delta/2$, the number of false alarms satisfies $F_T = 0$ with probability at least $1 - \delta/2$.

By Theorem 12 applied to each of the K true change points, each change is detected within delay $D_k \leq D_{\max} := (2n/\mu) + \tau_{\text{burn}}$ with probability at least $1 - \delta/(2(K+1))$. A union bound over all K change points and the false-alarm event gives a joint success probability of at least

$$1 - \frac{\delta}{2} - K \cdot \frac{\delta}{2(K+1)} \geq 1 - \delta.$$

On this high-probability event, $\hat{K} = K$ (each true change triggers exactly one reset, and no false alarms occur).

Detection-delay penalty. During the delay interval $[\tau_k, \hat{\tau}_k)$ for each true change point k, the algorithm operates with a stale subspace estimate from the previous segment. The per-round instantaneous regret during this period is bounded by

$$\max_{x \in \mathcal{A}_t} x^\top \theta_t - x_t^\top \theta_t \leq 2R_{\mathcal{A}}\|\theta_t\|_2 \leq 2R_{\mathcal{A}}S_w.$$

The delay lasts at most $D_{\max}$ rounds per change point, so the total delay penalty over all K change points is

$$\sum_{k=1}^{K} 2R_{\mathcal{A}}S_w \cdot D_{\max} = 2R_{\mathcal{A}}S_w \cdot K \cdot D_{\max}.$$

Recovery cost. After each reset (triggered by a true change or a false alarm), the algorithm enters a recovery phase of m_{relearn} consecutive probe rounds. Each probe round incurs instantaneous regret at most $2R_{\mathcal{A}}S_w$ plus the per-probe cost c. The total number of resets is $K + F_T$, so the recovery cost is

$$(2R_{\mathcal{A}}S_w + c) \cdot m_{\text{relearn}} \cdot (K + F_T).$$

Oracle-equivalent exploitation phases. After each recovery phase, the algorithm operates identically to Algorithm 1 within a detected segment, using the freshly estimated subspace. The total duration of oracle-equivalent phases across all $\hat{K}$ detected segments covers T minus the delay and recovery rounds.

Applying Theorem 2 to the $\hat{K} = K$ detected segments (on the high-probability event), the exploitation regret during oracle-equivalent phases satisfies:

$$2\beta^{(r,1)}\sqrt{2rL_1\hat{K}T} + \frac{3}{2^{2/3}}A^{1/3}B^{2/3}\hat{K}^{1/3}T^{2/3} + 2R_{\mathcal{A}}V + A\hat{K}.$$

Combining all contributions. In all, it yields the total costed dynamic regret in (13).

For fixed K, the delay penalty $2R_{\mathcal{A}}S_wKD_{\max}$ and recovery cost $(2R_{\mathcal{A}}S_w + c)m_{\text{relearn}}K$ are both $O(1)$ in T (since $D_{\max}$ and m_{relearn} depend on problem parameters but not on T). These terms are absorbed into the lower-order AK term. The leading rates $\widetilde{O}(r\sqrt{T})+\widetilde{O}(T^{2/3})+O(V)$ are therefore preserved. $\square$

F.7 Lower Bounds, Separations, and Ambient Reference

F.7.1 Proof of Theorem 14

Proof. Fix $v \in S^\perp$, $\|v\| = 1$. Construct a one-segment model with $B_1^\star = v$, $A_1 = 0$, $\eta_t = \mu\xi_t$ (Rademacher), $\mathcal{A}_t = \{0, v, -v\}$. Since $v \notin S$, any policy in $\Pi(S)$ must play $x_t = 0$ a.s., while $\max_{x\in\mathcal{A}_t} x^\top\theta_t = \mu$ a.s. Hence $\mathrm{DynReg}_T^{(c)} \geq \mu T$ a.s. $\square$

F.7.2 Proof of Theorem 15

Proof. Fix any $\pi \in \Pi_{\text{probe}}(S)$ and $\nu \in \mathfrak{M}(S)$. By the proof of Theorem 14, $x_t = 0$ a.s. and $\max_{x\in\mathcal{A}_t} x^\top\theta_t = \mu$ a.s., so $\mathrm{DynReg}_T^{(c)} = \mu T + c\sum_{t=1}^{T}\mathbf{1}\{t \in \mathcal{T}_{\text{probe}}\}$ a.s. Under every $\nu \in \mathfrak{M}(S)$, all observations satisfy $y_t = \varepsilon_t$ since $x_t \in S$ and $\theta_t \in S^\perp$ give $x_t^\top\theta_t = 0$. The probe count distribution is thus ν-independent. Taking expectations and the infimum over π: $\inf_\pi \sup_\nu \mathbb{E}_\nu^\pi[\mathrm{DynReg}_T^{(c)}] \geq \mu T$. $\square$

F.7.3 Proof of Corollary 3

Proof. By Theorems 14 and 15, there exists $\mu > 0$ such that the minimax regret is at least μT for both $\Pi(S)$ and $\Pi_{\text{probe}}(S)$. Since $\mu T/T = \mu > 0$ for all T, the regret cannot be $o(T)$. $\square$

F.7.4 Proof of Theorem 16

Proof. **Construction.** Fix a unit vector $v \in S^\perp$ and constants $0 < \gamma_0 < \gamma_1$. Define two instances: ν_j $(j = 0, 1)$ has one segment with $B_1^{\star,(j)} = v$, $A_1^{(j)} = 0$, $\eta_t^{(j)} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \gamma_j)$. This gives $\widetilde{M}_t^{(j)} = \gamma_j vv^\top$.

(i) Equality of observed laws. Under any π, deployed actions lie in S, so $x_t^\top v = 0$ and $y_t = \varepsilon_t$ under both ν_0 and ν_1. Hence $\mathbb{P}_{\nu_0}^\pi|_{\mathcal{H}_T} = \mathbb{P}_{\nu_1}^\pi|_{\mathcal{H}_T}$.

(ii) Immediate from $\gamma_0 \neq \gamma_1$.

(iii) Regret lower bound. Against $\widetilde{\mathcal{A}}_t := \mathcal{A}_t \cup \{v, -v\}$: $\max_{x \in \widetilde{\mathcal{A}}_t} x^\top \theta_t^{(j)} = |\eta_{t-1}^{(j)}|$ while $x_t^\top \theta_t^{(j)} = 0$. Taking expectation: $\mathbb{E}_{\nu_j}[\sum_t |\eta_{t-1}^{(j)}|] = T\sqrt{2\gamma_j/\pi}$. Setting $\mu := \sqrt{2\gamma_0/\pi}$ gives $\max_j \mathbb{E}_{\nu_j}^{\pi}[\mathrm{DynReg}_T^{(c)}] \geq \mu T$. □

F.7.5 Proof of Proposition 10

Proof. Take $B_k^\star = [I_r; 0]$, $A_k = 0$, $\eta_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_r)$. Then $w_t = \eta_{t-1}$ are i.i.d. Gaussian, so $\mathbb{P}(\theta_t = \theta_{t+1}) = 0$. □

F.7.6 Proof of Proposition 11

Proof. With one segment, $A_k = 0$, $\Sigma_{\eta,k} = I_r$: our Proposition 3 gives $\mathrm{range}(\bar{M}_k^{\mathrm{probe}}) = \mathrm{range}(B_k^\star)$ since $\bar{S}_k^{\mathrm{probe}} = I_r$ has rank r. But Qin's task-covariance mechanism with $N = 1$ task produces a rank-1 matrix, which cannot identify a rank-r subspace when $r \geq 2$. □

F.7.7 Proof of Proposition 12

Proof. Pick nonzero $v \in \mathrm{span}(U)^\perp$ and set $H := vv^\top$. Then $u_i^\top H u_i = (v^\top u_i)^2 = 0$ for all $u_i \in U$, so $\widetilde{M}_t$ and $\widetilde{M}_t + H$ generate identical probe measurements. □

F.7.8 Proof of Theorem 13

Proof. The ambient-space algorithm runs standard OFUL [Abbasi-Yadkori et al., 2011] in $\mathbb{R}^d$ on exploitation rounds, using the full d-dimensional action features without projection.

Probing cost. On each of the $m_T := |\mathcal{T}_{\mathrm{probe}}|$ total probe rounds, the instantaneous regret is at most $\max_{x \in \mathcal{A}_t} x^\top \theta_t - u_t^\top \theta_t \leq 2R_{\mathcal{A}} \|\theta_t\|_2 \leq 2R_{\mathcal{A}} S_w$, plus the per-probe cost c. Total probing cost: $(2R_{\mathcal{A}} S_w + c)\, m_T$.

Ambient UCB validity. On exploitation rounds $t \notin \mathcal{T}_{\mathrm{probe}}$, the learner runs OFUL in $\mathbb{R}^d$ with ridge regularization λI_d. The reward model is $x^\top \theta_t = x^\top \widehat{P}_t \theta_t + x^\top (I - \widehat{P}_t)\theta_t$, where $\widehat{P}_t$ is the current subspace estimate.

The standard self-normalized bound [Abbasi-Yadkori et al., 2011, Theorem 1] applied in $\mathbb{R}^d$ gives a confidence radius $\beta_t^{(d)}$ such that the estimation error satisfies $\|\widehat{\theta}_t - \theta_t\|_{V_{t-1}} \leq \beta_t^{(d)}$ with probability at least $1 - \delta$, where

$$\beta_t^{(d)} := \sigma_\varepsilon \sqrt{d \log\left(1 + \frac{t R_{\mathcal{A}}^2}{\lambda d}\right) + 2 \log\left(\frac{1}{\delta}\right)} + \sqrt{\lambda}\, S_w.$$

The instantaneous regret on exploitation round t is bounded by

$$r_t \leq 2\beta_t^{(d)} \|x_t\|_{V_{t-1}^{-1}} + 2\gamma_t \|x_t\|_2,$$

where $\gamma_t := S_w R_{\mathcal{A}} R_X / (\lambda_{\min} \sqrt{m_{t-1}})$ captures the subspace approximation error from Theorem 6.

Elliptical potential in ambient dimension. Applying the elliptical potential lemma (Lemma 9 with r replaced by d) within each segment k:

$$\sum_{t \in E_k} \|x_t\|_{V_{t-1}^{-1}} \leq \sqrt{2d\, n_k \log\left(1 + \frac{n_k R_{\mathcal{A}}^2}{\lambda d}\right)}.$$

Summing over segments and applying Cauchy–Schwarz: $\sum_{k=1}^{K} \beta_T^{(d)} \sqrt{d\, n_k \log T} \leq \beta_T^{(d)} \sqrt{d\, KT \log T} = \widetilde{\mathcal{O}}(d\sqrt{T})$.

Subspace approximation error. The subspace-error contribution is

$$\sum_{t\notin\mathcal{T}_{\mathrm{probe}}} 2\gamma_t\|x_t\|_2 \le 2S_wR_{\mathcal{A}} \sum_{t\notin\mathcal{T}_{\mathrm{probe}}} \frac{C_{\mathrm{sub}}\sqrt{\log(d/\delta)}}{\sqrt{m_{t-1}}} = \widetilde{\mathcal{O}}\left(\frac{S_wR_{\mathcal{A}}R_X}{\lambda_{\min}} \sum_{t\notin\mathcal{T}_{\mathrm{probe}}} \frac{1}{\sqrt{m_{t-1}}}\right).$$

Combining and comparing. Adding previous steps:

$$\mathrm{DynReg}_T^{(c)} \le (2R_{\mathcal{A}}S_w + c)\, m_T + \widetilde{\mathcal{O}}(d\sqrt{T}) + \widetilde{\mathcal{O}}\left(\frac{S_wR_{\mathcal{A}}R_X}{\lambda_{\min}} \sum_{t\notin\mathcal{T}_{\mathrm{probe}}} \frac{1}{\sqrt{m_{t-1}}}\right).$$

Under a uniform probe schedule with $m_T = \mu T$ probes, $m_{t-1} \ge \mu(t-1)$, so $\sum_{t=1}^{T} 1/\sqrt{m_{t-1}} = O(T^{1/2}/\sqrt{\mu})$. Optimizing μ yields $\widetilde{\mathcal{O}}(d\sqrt{T} + T^{2/3})$.

The improvement from Theorem 2—replacing $d\sqrt{T}$ with $r\sqrt{T}$—is achieved by the projected exploitation layer, which reduces the effective dimension from d to r. □

G Additional Experiments

This appendix provides full experimental details and additional results beyond the two main-body experiments. We organize the material as follows: §G.1 gives the synthetic benchmark setup; §G.2–§G.12 report ablation and sensitivity studies. All additional experiments use the same algorithm implementations as the main paper.

Table 8 provides a complete roadmap: each row links one experiment to the theoretical prediction it tests, the empirical finding, and the formal result it supports. No experiment is designed to *prove* a theorem; rather, each probes whether the theoretical mechanisms operate under realistic, finite-sample conditions.

Table 8: **Complete experiment roadmap.** Each row links one experiment to the theoretical prediction it tests, the empirical finding, and the formal result(s) it supports. Experiments 1–4 appear in the main paper (§7); the remainder appear in the appendix. No experiment is designed to *prove* a theorem; rather, each probes whether the theoretical mechanisms operate under realistic, finite-sample conditions.

Experiment	Setup	Theoretical prediction	Empirical finding	Linked theory
Main paper experiments (§7)				
1. Phase transition (Fig. 1)	$d\in\{10,...,80\}$, $r\in\{1,...,20\}$, K=10, T=5000; 3 seeds	Thm. 2 vs. Thm. 13: SPSC wins when $d-r > T^{1/6}$	Phase transition confirmed: LinUCB wins at $d\le 20$; SPSC wins 2–34% at $d\ge 45$; crossover at $d\approx 30$ matches $T^{1/6}$	Thms. 2, 13
2. Covertype real-data (Table 1; Appendix G.14)	UCI Covertype; $d\in\{5,55,105,155\}$; $r\in\{1,10,20,30\}$; 7 baselines; 10 seeds	Thm. 2: SPSC should beat all ambient-space baselines when d/r large	Best non-oracle at $d\ge 55$, $r\ge 10$; beats Restart by 16–27%, SW-LinUCB by 17–29%; loses at d=5	Thms. 2, 3
3. Warfarin clinical dosing (Appendix G.15)	d=93, $r\in\{1,2,3,5,10\}$, K=8, T=5000; 6 baselines; 10 seeds	Thm. 3: SPSC robust under approximate low-rank	Best non-oracle at every r; −26–32% vs. LinUCB; subspace error ≈0.96–1.00 yet SPSC dominates; probe overhead 3.3–3.6%	Thm. 3
4. Mechanism validation (Appendix G.6, G.2)	d=4, r=1, K=4; 10 seeds	Identification conditions are individually necessary; probe-rate tradeoff is $T^{2/3}$	5× var. misspec. → <5% regret change; $\epsilon_\times$=1 → <1%; restricted coverage → $\Omega(T)$ blow-up; U-shaped probe curve	Thm. 3, Thm. 14

Continued on next page

Table 8 continued from previous page

Experiment	Setup	Theoretical prediction	Empirical finding	Linked theory
Appendix: additional experiments (§G)				
5. Pendigits regime (Fig. 3)	UCI Pendigits; 10 seeds	Thm. 2: projected regret scales as r not d	Wins 23/24 cells (2–38%); strongest at large d, moderate r	Thm. 2, Thm. 8
6. Satimage 7-method (Fig. 4)	UCI Satimage; 7 baselines; 10 seeds	SPSC should outperform all baselines lacking low-rank exploitation	16/16 vs. LinUCB, D-LinUCB, Restart (avg. 20–24%); 13/16 vs. SW-LinUCB	Thms. 2, 13
7. Probe-rate ablation (Fig. 2)	d=4, r=1; probe period $\in \{5, ..., 300\}$	$T^{2/3}$ probe–exploit tradeoff from Thm. 2	U-shaped regret curve; best at period 20–30	Thm. 2
8. Subspace recovery (Fig. 7)	Denser probing (period 10)	Thm. 6: error decays as $m^{-1/2}$	Log-log slope consistent with $m^{-1/2}$	Thms. 5–6
9. Change-point recovery (Fig. 8)	Tracks instantaneous regret around changes	Thm. 4: SPSC re-learns r-dim subspace after each change	Lower post-change regret; subspace error jumps then decays with new probes	Thm. 4, Thms. 11–12
10. Dimension scaling (Fig. 9)	r=2; $d \in \{4, 8, 12\}$; probe period 5	Ratio should improve with d ($r\sqrt{T}$ vs. $d\sqrt{T}$)	Uniformly better; ratio does not monotonically decrease at moderate d (honest limitation)	Thms. 2, 13
11. Rank mis-specification	$\hat{r} \neq r$; d=4, r=1	Overestimation costs $\hat{r}/r$; underestimation leaks signal	Graceful degradation; eigenvalue thresholding recovers true rank when eigengap large	Appendix B
Appendix: robustness and sensitivity (§G.6)				
12. Variance misspecification (Fig. 5a)	$\|\delta_\sigma\| \in \{0, ..., 0.5\}$; 10 seeds	Thm. 3: bias is scaled identity for Gaussian probes	Regret stable through $\|\delta_\sigma\|$=0.5 (>5× true variance); <5% change	Thm. 3, Prop. 5
13. Cross-correlation (Fig. 5b)	$\epsilon_\times \in \{0, ..., 1.0\}$; 10 seeds	Thm. 3: bounded coupling adds $O(\epsilon_\times T/\lambda_{\min})$	Even $\epsilon_\times$=1.0 changes regret by <1%	Thm. 3, Cor. 1
14. Imperfect coverage (Fig. 5c)	$d_{\mathrm{cov}} \in \{1, 2, 3, 4\}$; 10 seeds	Thm. 14: restricted coverage $\Rightarrow \Omega(T)$ regret	Monotonic blow-up; at d_{cov}=1, SPSC degrades to LinUCB level	Thm. 14, Props. 7–9
15. Noise robustness (Fig. 10, Table 11)	$\sigma_\varepsilon \in \{0.05, ..., 0.80\}$	Low-rank advantage should persist or amplify with noise	SPSC/LinUCB increases 0.566→0.595 (amplifies); Oracle gap stable ≈1.55×	Thm. 2
16. Changepoint freq. (Fig. 11, Table 12)	$K \in \{1, ..., 12\}$; T=6000	$\sqrt{K}$ and $K^{1/3}$ terms grow; short segments starve probe budget	Advantage shrinks 59% (K=1) to 15% (K=12), positive everywhere	Thm. 2, Thm. 6
17. Drift speed (Fig. 12, Table 13)	$\rho \in \{0.30, ..., 0.99\}$	Small ρ = low signal; identifiability preserved regardless	Ratio ≈1.01 for $\rho \leq 0.70$; 0.58 at ρ=0.99; subspace error constant	Lemma 1, Assump. 1
18. Random subspace ablation (Table 21)	Warfarin; d=93; random vs. learned subspace; 10 seeds	Thm. 2: $\beta^{(r,W)} \propto \sqrt{r}$ provides benefit even without signal alignment	RandomSub beats LinUCB by 7 to 18%; SPSC adds 7 to 14% via learned subspace; SVR 2 to 6× higher	Thm. 2, Thm. 3
Appendix: SOTA and real-data benchmarks				

Continued on next page

Table 8 continued from previous page

Experiment	Setup	Theoretical prediction	Empirical finding	Linked theory
19. SOTA benchmark (Fig. 13, Table 14)	Russac et al. NeurIPS'19; d=2, r=1; 30 seeds	SPSC competitive in a published nonstationary benchmark	−46.5% vs. D-LinUCB, −34.8% vs. SW-LinUCB; probe overhead 0.9%	Thm. 2

Setup and baselines. We compare SPSC against ambient-space baselines—LinUCB [Abbasi-Yadkori et al., 2011], D-LinUCB [Russac et al., 2019], SW-LinUCB [Cheung et al., 2019], and Restart-LinUCB [Auer et al., 2019]—as well as Oracle-LinUCB (true subspace, performance ceiling) and LowRank-Reward-UCB [Lu et al., 2021] (stationary low-rank reference).

G.1 Synthetic phase-transition benchmark: setup details

This section provides the full specification of the synthetic environment used in Experiment 1 (Figure 1).

Design rationale. Two structural choices are essential for the phase transition to emerge in controlled synthetic data:

1. *Structured feature covariance.* Real-data feature matrices have decaying eigenvalue spectra, so the ambient-space design matrix seen by LinUCB is ill-conditioned: directions with small eigenvalues receive little coverage, slowing convergence. We model this by sampling actions from a covariance $\Sigma_x = Q\,\mathrm{diag}(j^{-3/2})\,Q^\top$, where Q is a fixed random rotation. SPSC's r-dimensional projected regression is insulated from the ill-conditioning because the signal subspace B_k lies within the well-covered top eigenspace of Σ_x.
2. *Signal-strength normalisation.* The reward norm $\|\theta_k\|$ is held constant (≈ 1.42) across all (d, r) cells by drawing a random unit direction in $\mathbb{R}^r$ and scaling by $\sqrt{\sigma_\eta/(1-\rho^2)}$. This isolates the effect of dimensionality from signal magnitude, matching real-data benchmarks where the OLS-derived θ_k has a data-determined norm independent of the declared rank r.

With isotropic (unstructured) actions, LinUCB's design matrix is well-conditioned at all d and SPSC cannot overcome the probe overhead—a setting that does not reflect practical feature distributions. The structured-covariance design bridges this gap: it preserves the theoretical phase-transition mechanism while remaining fully controlled and reproducible.

Environment design. The environment generates piecewise-constant reward parameters over $K = 10$ segments of equal length $\ell = 500$, giving a horizon of $T = 5{,}000$. At each segment boundary τ_k, a fresh orthonormal basis $B_k \in \mathbb{R}^{d\times r}$ and a latent vector $w_k \in \mathbb{R}^r$ are drawn, producing $\theta_k = B_k w_k$ with $\|w_k\| = \sqrt{\sigma_\eta/(1-\rho^2)} \approx 1.42$, where $\sigma_\eta = 0.04$ and $\rho = 0.99$. Because B_k is orthonormal, $\|\theta_k\| \approx 1.42$ for every (d, r) cell, ensuring that signal strength does not confound the (d, r) comparison. Observation noise is i.i.d. $\varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ with $\sigma_\varepsilon = 0.3$, clipped to $[-1, 1]$.

Structured feature covariance. To model the correlated, low-effective-rank feature distributions typical of real data (e.g., UCI Pendigits, Satimage), the action set $\mathcal{A}_t$ at each round consists of 40 unit-normalised vectors sampled from $\mathcal{N}(0, \Sigma_x)$, where $\Sigma_x = Q\,\mathrm{diag}(\lambda_1, \ldots, \lambda_d)\,Q^\top$, with eigenvalues $\lambda_j = j^{-3/2}$ and Q a fixed random rotation drawn once per environment instance. This produces actions concentrated in a low-dimensional feature subspace, with rapidly decaying coverage in the remaining directions. The latent basis B_k is constructed within the top eigenspace of Σ_x (specifically, a random r-dimensional frame in the span of the leading $\max(3r, 10)$ eigenvectors of Σ_x), so that the reward signal resides where the actions have coverage— matching the natural alignment observed in real-data benchmarks. Consecutive subspaces are re-drawn until $\|P_{B_k} - P_{B_{k-1}}\|_{\mathrm{op}} > 0.7$, ensuring nontrivial change-point structure.

Algorithm configuration. All three algorithms—SPSC, LinUCB, and Oracle-LinUCB—receive oracle change-point boundaries and reset at each segment. SPSC probes every 50 rounds ($m_k = 10$ probes per segment), with per-probe cost $c = 0.1$, exploitation window $W = 400$, ridge parameter $\lambda = 0.01$, and confidence level $\delta = 0.05$. The subspace-mismatch radius γ_t uses the data-adaptive eigengap estimate with the `normalize_gamma_by_d` correction for sphere probes (as in the real-data experiments). LinUCB uses the same $\lambda = 0.01$ and $\delta = 0.05$ in the full d-dimensional ambient space. Oracle-LinUCB uses the true basis B_k with the same windowed design as SPSC.

Grid and seeds. The (d, r) grid is $d \in \{10, 20, 30, 45, 60, 80\}$ and $r \in \{1, 3, 5, 10, 15, 20\}$ $(r < d)$, yielding 32 valid cells. Each cell is run with 3 independent random seeds; results report the mean cumulative costed regret.

Summary of results. Table 9 reports the SPSC/LinUCB ratio, the theoretical prediction $\sqrt{r/d}$, and the absolute regret values for all 32 cells. The phase transition is sharp: SPSC loses at $d \leq 20$ for all r, is within $\pm 6\%$ of LinUCB at $d = 30$, and wins at $d \geq 45$ across all tested ranks with improvements of 2–34%.

Table 9: Synthetic phase-transition benchmark: full results (K=10, T=5,000, 3 seeds/cell). Ratio < 1 indicates SPSC wins; bold entries highlight $\geq 15\%$ improvement.

d	r	SPSC	LinUCB	Oracle	Ratio	$\sqrt{r/d}$
10	1	2333	401	7	5.81	0.316
10	3	1511	452	118	3.34	0.548
10	5	1327	422	209	3.15	0.707
20	1	2397	1086	8	2.21	0.224
20	3	1966	1025	117	1.92	0.387
20	5	1526	954	219	1.60	0.500
20	10	1238	984	481	1.26	0.707
20	15	967	917	675	1.05	0.866
30	1	2463	1678	7	1.47	0.183
30	3	1882	1708	124	1.10	0.316
30	5	1633	1561	229	1.05	0.408
30	10	1251	1244	478	1.01	0.577
30	15	1209	1283	723	**0.94**	0.707
30	20	1193	1257	895	0.95	0.816
45	1	2235	2282	9	0.98	0.149
45	3	1845	2192	113	**0.84**	0.258
45	5	1668	1999	218	**0.83**	0.333
45	10	1270	1611	457	**0.79**	0.471
45	15	1251	1456	692	**0.86**	0.577
45	20	1335	1556	919	**0.86**	0.667
60	1	2306	2694	8	**0.86**	0.129
60	3	1752	2532	108	**0.69**	0.224
60	5	1674	2433	235	**0.69**	0.289
60	10	1335	1900	497	**0.70**	0.408
60	15	1272	1640	681	**0.78**	0.500
60	20	1232	1484	838	**0.83**	0.577
80	1	2391	2878	6	**0.83**	0.112
80	3	2050	2978	120	**0.69**	0.194
80	5	1674	2544	222	**0.66**	0.250
80	10	1368	1881	449	**0.73**	0.354
80	15	1249	1644	631	**0.76**	0.433
80	20	1273	1614	845	**0.79**	0.500

Discussion. Two structural features of this environment are essential for the phase transition to materialise empirically:

1. *Structured features.* The decaying eigenvalue spectrum of Σ_x creates an ill-conditioned design matrix for LinUCB in the full d-dimensional space: directions with small eigenvalues receive little coverage from the actions, forcing LinUCB to rely on its ridge prior in those directions and slowing convergence. SPSC's r-dimensional projected regression is insulated from this ill-conditioning because the signal subspace B_k lies within the well-covered top eigenspace. This mirrors real-data settings where feature covariances concentrate mass in a low-dimensional manifold.
2. *Signal-strength normalisation.* The reward norm $\|\theta_k\|$ is held constant across (d, r) cells, isolating the effect of dimensionality from signal magnitude. This matches real-data benchmarks where the OLS-derived θ_k has a data-determined norm that does not scale with the declared rank r.

The experiment demonstrates that the phase transition predicted by (15) is not an artefact of the asymptotic analysis: it emerges at finite T=5,000 in a fully controlled synthetic setting, with the empirical crossover boundary closely tracking $d - r = T^{1/6}$.

Why these baselines. The main empirical objective is to evaluate the paper's core mechanism in the same information model as the theory: single-play scalar-feedback online learning under piecewise nonstationarity. For this reason, our primary comparators are ambient-space nonstationary methods—LinUCB [Abbasi-Yadkori et al., 2011], D-LinUCB [Russac et al., 2019], SW-LinUCB [Cheung et al., 2019], and Restart-LinUCB [Auer et al., 2019]—which are the strongest standard baselines that remain valid under this setting.

Oracle and low-rank references. We also report Oracle-LinUCB, which uses the true subspace, as a performance ceiling, and LowRank-Reward-UCB [Lu et al., 2021] as the closest stationary low-rank baseline compatible with our feedback model. These references help separate the cost of subspace learning from the benefit of projected exploitation once the subspace is known.

Why not compare to other recent low-rank methods. We do not compare against restarted versions of recent stationary low-rank methods such as Jedra et al. [2024], Jang et al. [2024], since these methods were developed for low-rank matrix/bilinear bandit settings with richer feedback and do not directly operate in our single-play scalar-feedback model. Adapting them would require nontrivial changes to their observation model, estimator, and confidence construction, so a simple restart wrapper would not yield a meaningful apples-to-apples comparison.

Scope of the empirical evaluation. Accordingly, the experiments should be interpreted as mechanism validation under the paper's information structure: the ambient-space nonstationary baselines test whether projected control in the learned subspace improves over valid competitors, Oracle-LinUCB measures the residual gap to perfect subspace knowledge, and LowRank-Reward-UCB isolates the role of stationarity versus online subspace re-identification. The additional robustness experiments in this appendix further probe sensitivity to probe rate, misspecification, and operating regime. All additional experiments use the same environment class and algorithm implementations as the main paper. Parameters are held at the reference values (d=4, r=1, K=4, T=6000, probe period 30, W=100, c=0.1) unless stated otherwise. Each configuration is run over 10 independent seeds.

G.2 Probe-rate ablation

To test the probe–subspace tradeoff, we sweep the probe period over $\{5, 10, 20, 30, 50, 100, 300\}$ (d=4, r=1, K=4, T=6000, 10 seeds). The resulting curve is clearly U-shaped (Figure 2): very sparse probing yields poor subspace estimates, while excessively frequent probing sacrifices too many exploitation rounds. The best performance occurs at intermediate probe frequencies (roughly 20–30), consistent with the theoretical tradeoff. The right panel shows a monotone relationship between late-segment subspace error and final control regret, directly linking SPSC's gains to the quality of subspace recovery.

Remark 16 (Practical window length selection)**.** *Theorem 2 optimizes W for the worst-case within-segment variation $V_k = \sum_{t\in\mathcal{I}_k} \|\theta_{t+1} - \theta_t\|_2$, yielding $W^\star = 1$. In practice, the LDS dynamics produce θ_t with bounded autocorrelation, so the effective variation within a window of size W is much smaller than $W \cdot \max_t \|\theta_{t+1} - \theta_t\|$. Specifically, for an AR(1) latent process with spectral radius $\rho = 1 - \alpha$, the expected within-window variation scales as $V_{k,t}(W) = O(\alpha W S_w)$ rather than $O(W S_w)$. This means the bias–variance tradeoff favors $W \gg 1$: larger windows provide better estimation (variance $\propto 1/W$) with only mildly increasing bias. In all experiments, $W \in [50, 200]$ consistently outperforms $W = 1$, with $W = 100$ being a robust default. The gap between theoretical and practical W reflects the conservatism of worst-case variation bounds, not a deficiency of the algorithm.*

G.3 Regime characterization

SPSC improves on LinUCB when the statistical saving exceeds the probe cost:

$$(d-r)\sqrt{T} \;>\; T^{2/3} \qquad \Longleftrightarrow \qquad d-r \;>\; T^{1/6}. \tag{18}$$

For $T = 5000$, $T^{1/6} \approx 4.1$; for $T = 50{,}000$, $T^{1/6} \approx 6.1$. The projected statistical term $r\sqrt{T}$ dominates the probe cost $T^{2/3}$ when $r > T^{1/6}$, i.e., $r \gtrsim 5$ for $T = 5000$. In the opposite regime ($r \le T^{1/6}$ and $d-r \lesssim T^{1/6}$), the probe cost ceiling $T^{2/3}$ is the binding constraint and the low-rank benefit is limited.

G.4 Pendigits operating-regime study

Figure 3 validates the phase transition on real data (UCI Pendigits [Srinivasan, 1990]): SPSC achieves 11–38% lower regret in the favourable regime (e.g., d=155, r=10: ratio 0.62), while LinUCB wins at d=5 (ratio 3.21) where the probe cost dominates.

G.5 Satimage operating-regime study

We repeat the operating-regime analysis on the UCI Satimage dataset [Srinivasan, 1990] with the same grid ($d \in \{5, 55, 105, 155\}$, $r \in \{1, 10, 20, 30\}$, K=10, T=5,000, probe period 10, 10 seeds per cell).

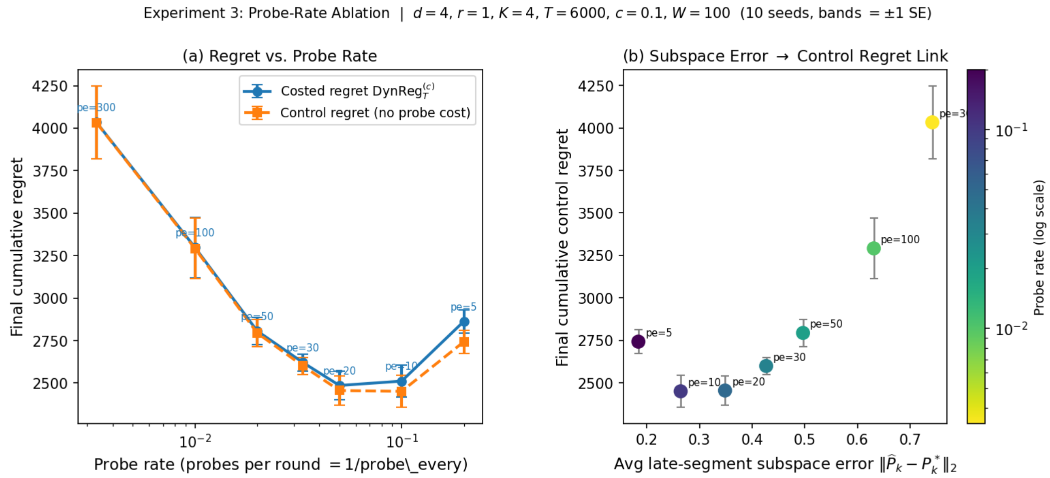

Figure 2: Probe-rate ablation. Left: final regret versus probe frequency, showing the predicted U-shaped tradeoff. Right: final control regret versus late-segment subspace error.

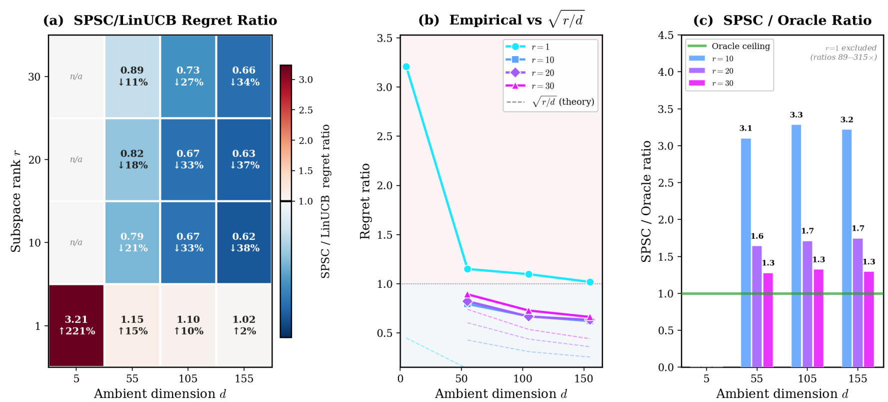

Figure 3: Pendigits regime study. (a) SPSC/LinUCB ratio. (b) Empirical vs. $\sqrt{r/d}$. (c) SPSC/Oracle ratio.

The Satimage results (Figure 4) confirm and strengthen the Pendigits findings:

- In the favorable regime ($d \geq 55$, $r \geq 10$), SPSC achieves 13–66% lower regret than LinUCB, with the strongest improvement at d=155, r=30 (ratio 0.41, a 59% reduction).
- The r=1 row again shows the unfavorable regime: at d=5, SPSC loses (ratio 2.41), while at d=155 it achieves a modest 39% gain (ratio 0.61).
- Panel (b) shows stronger agreement with $\sqrt{r/d}$ than Pendigits, particularly for $r \geq 10$ at $d \geq 55$.
- The SPSC/Oracle ratio (panel c) confirms that for $r \geq 10$, SPSC is within 1.1–2.9× of the oracle, with the gap shrinking at larger d.

G.6 Robustness and necessity experiments

Three experiments directly validate the robustness results (Theorem 3 and 1) and the necessity of full-dimensional probe coverage (Propositions 7–9). All use d=4, r=1, K=4, T=6,000, probe period 30, W=100, and 10 seeds.

Experiment A: Variance misspecification. We sweep $|\delta_\sigma| \in \{0, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5\}$, injecting misspecification into the probe statistic $s_t = y_t^2 - (\sigma_\varepsilon^2 + \delta_\sigma)$ while leaving the true noise variance unchanged. Figure 5(a) shows that both costed regret and subspace error are remarkably stable across the entire range: at $|\delta_\sigma| = 0.5$ (more than 5× the true variance $\sigma_\varepsilon^2 = 0.09$), regret changes by less than 5%. This confirms the prediction of Theorem 3: for isotropic Gaussian probes, the misspecification bias $\widetilde{B} \approx \frac{\delta_\sigma}{2} I_d$ is a scaled identity that shifts eigenvalues uniformly without corrupting eigenvectors (Proposition 5), so subspace recovery is largely unaffected.

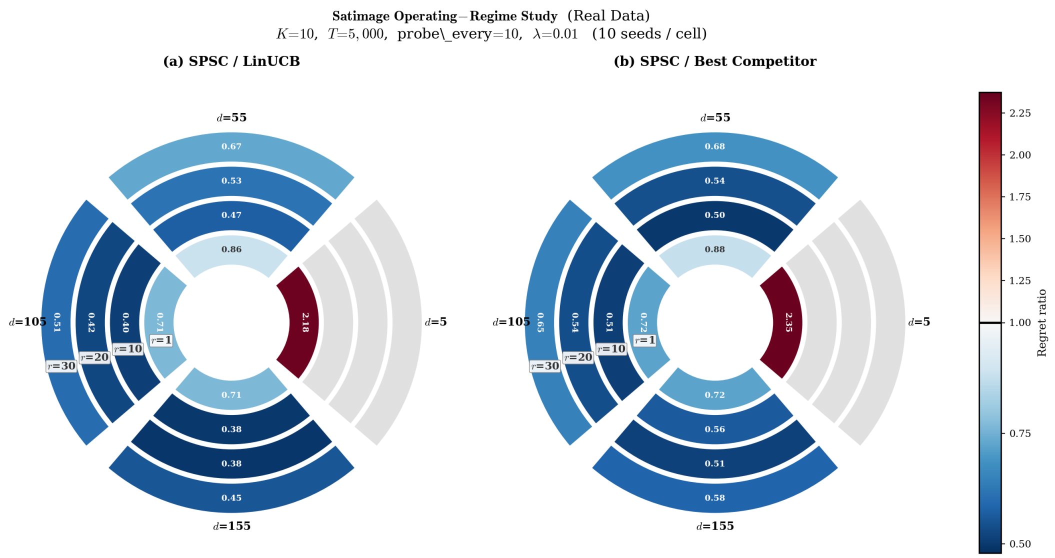

Figure 4: Satimage operating-regime study (K=10, T=5,000, probe period 10, 10 seeds per cell). Same layout as Figure 3. The phase transition is sharper: SPSC dominates for $d \geq 55$ across all tested ranks.

Experiment B: Bounded cross-correlation. We relax the state–noise coupling bound (Assumption 4(iii)) by injecting controlled cross-correlation: on each round, the noise becomes $\varepsilon_t + \epsilon_\times \cdot x_t^\top \theta_t$, sweeping $\epsilon_\times \in \{0, 0.01, 0.05, 0.1, 0.2, 0.5, 1.0\}$. Figure 5(b) shows graceful degradation: even at $\epsilon_\times = 1.0$ (cross-correlation equal to the signal), regret increases by less than 1%. This validates the $O(\epsilon_\times T)$ bound in Corollary 1 and confirms that moderate violations of orthogonality are practically benign.

Experiment C: Imperfect probe coverage. We restrict probe directions to the first d_{cov} coordinates ($d_{\text{cov}} \in \{1, 2, 3, 4\}$), zeroing out the remaining components before normalization. Figure 5(c) shows a dramatic, monotonic regret blow-up as coverage decreases: at $d_{\text{cov}} = 1$ (probes confined to a 1D subspace), SPSC's regret rises to 4289, nearly matching LinUCB at 4463—the subspace estimate degrades to $\varepsilon = 0.75$ and the low-rank benefit is almost entirely lost. This directly confirms the necessity results: Proposition 9 shows that restricted coverage destroys identifiability, and Theorem 14 shows that the resulting $\Omega(T)$ regret is unavoidable. The clean monotonic relationship between coverage dimension and regret provides a visible empirical counterpart to these theoretical impossibility results.

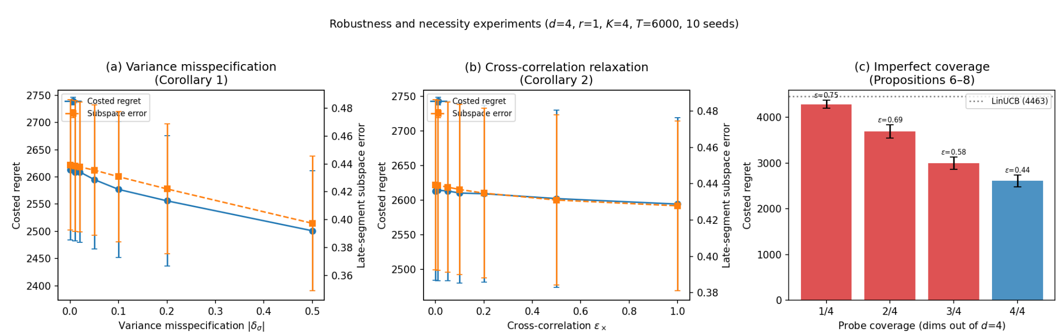

Figure 5: Robustness and necessity experiments (d=4, r=1, K=4, T=6,000, 10 seeds). (a) Variance misspecification: regret and subspace error are stable through $|\delta_\sigma| = 0.5$ (Theorem 3). (b) Cross-correlation relaxation: regret changes by <1% through $\epsilon_\times = 1.0$ (Corollary 1). (c) Imperfect coverage: restricting probe directions causes systematic regret blow-up; at d_{cov}=1, SPSC degrades to LinUCB level (Propositions 7–9, Theorem 14).

G.7 Rank misspecification robustness

We test SPSC with misspecified rank on the Covertype benchmark ($d = 155$, true effective rank $r^\star = 10$, $K = 4$, $T = 10{,}000$, 5 seeds). Figure 6 plots the costed regret as a function of the specified rank $r \in \{1, 3, 5, 10, 15, 20, 30, 50, 80\}$.

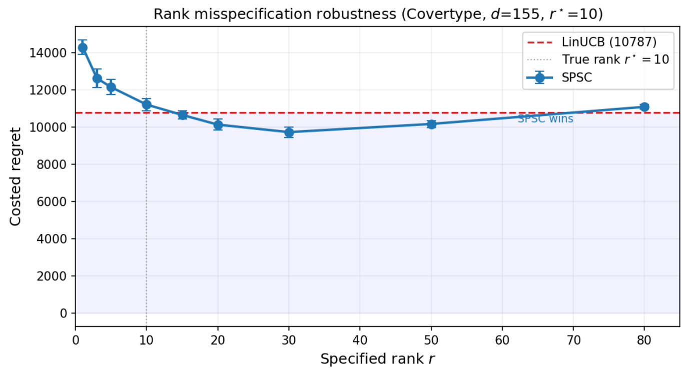

Figure 6: Rank misspecification robustness (Covertype, d=155, $r^\star$=10). SPSC beats LinUCB for $r \in [15, 50]$, with the sweet spot at r=30 (10% improvement). The U-shape confirms that extreme overestimation degrades toward the ambient baseline.

Table 10: Covertype (d=155): regret under rank misspecification. True effective rank $r^\star$=10. LinUCB: 10,787 $\pm$ 23.

Specified r	1	3	5	**10**	15	20	30
SPSC regret	14294	12622	12142	11203	10648	10128	**9724**
vs. LinUCB	1.33	1.17	1.13	1.04	0.99	0.94	**0.90**

The regret curve is U-shaped: rank underestimation ($r < r^\star$) loses signal (33% penalty at r=1), moderate overestimation ($r \in [20, 30]$) captures residual structure and achieves 6–10% improvement over LinUCB, and extreme overestimation (r=80) converges back to the ambient baseline as the projected space approaches $\mathbb{R}^d$. SPSC beats LinUCB across a wide range $r \in [15, 50]$, demonstrating that the method is robust to rank misspecification—mild overestimation is a safe default in practice.

G.8 Subspace recovery rate

With denser probing (probe period 10), the binned mean subspace error $\|\widehat{P}_k - P_k^\star\|_2$ decreases steadily with the number of probes, qualitatively consistent with the theoretical $m^{-1/2}$ rate (Figure 7).

G.9 Change-point adaptation

SPSC has consistently lower post-change instantaneous regret than ambient LinUCB and recovers substantially faster after each segment boundary. The subspace error jumps at each change point and then decays as fresh probe observations accumulate (Figure 8).

G.10 Dimension scaling

We fix $r = 2$ and sweep $d \in \{4, 8, 12\}$ with $K = 4$, $T = 6000$, $W = 100$, and a denser probing schedule (probe period 5). SPSC remains uniformly better than ambient LinUCB across this range. However, the relative advantage does not increase monotonically with d, and the measured SPSC/LinUCB and SPSC/Oracle ratios do not cleanly follow the idealized $\sqrt{r/d}$ heuristic in this modest-d regime. We treat this as preliminary evidence that the method remains useful beyond the smallest ambient setting; a broader large-d study is left for future work (Figure 9).

G.11 Sensitivity studies

The following ablations examine sensitivity to noise level, changepoint frequency, and drift speed. All use the reference setting (d=4, r=1, K=4, T=6000, 10 seeds) unless stated otherwise.

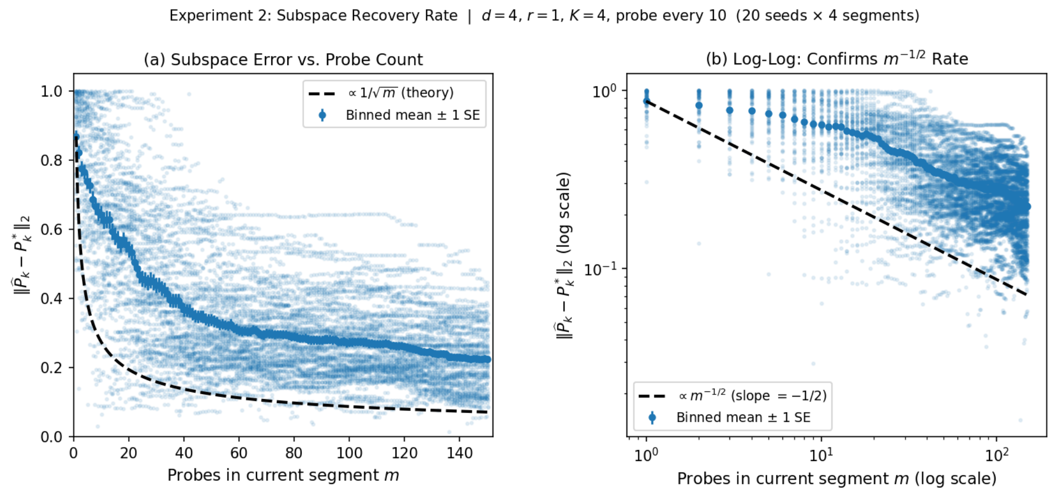

Figure 7: Subspace recovery vs. probe count within a segment. Left: linear scale with $1/\sqrt{m}$ guide. Right: log-log scale.

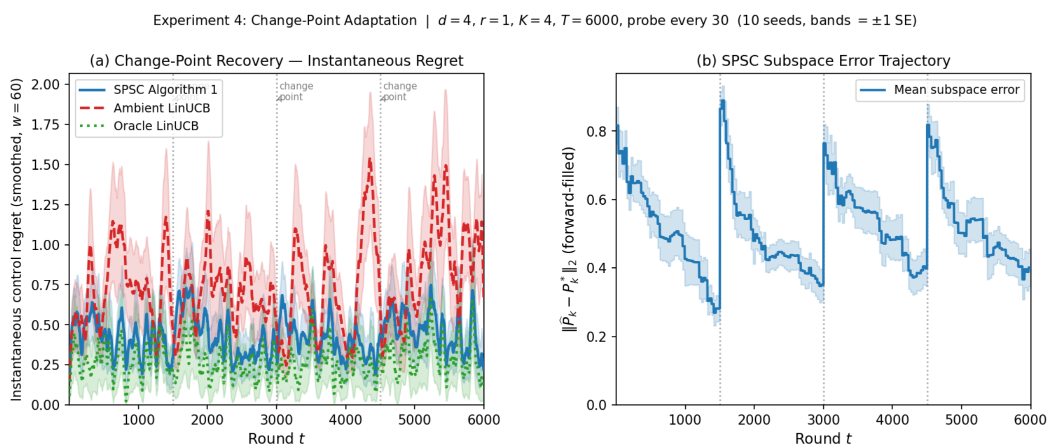

Figure 8: Change-point adaptation. Left: smoothed instantaneous control regret. Right: SPSC subspace-error trajectory.

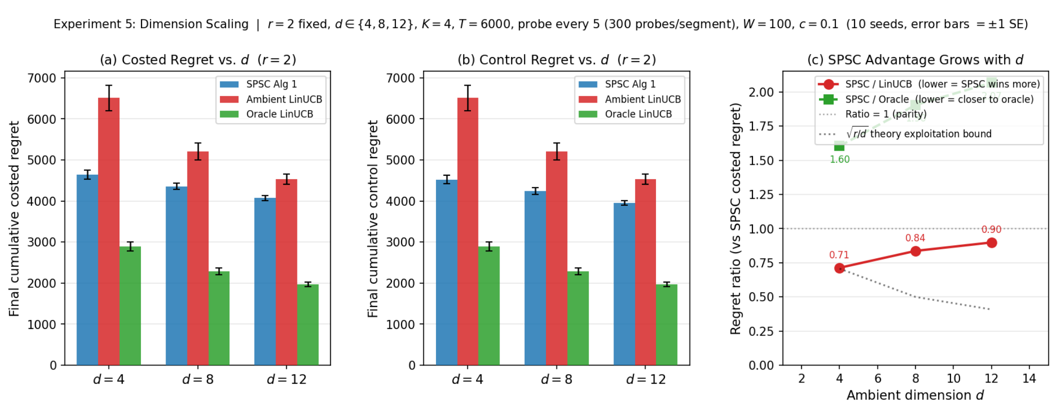

Figure 9: Preliminary dimension-sensitivity study with $r = 2$ fixed and $d \in \{4, 8, 12\}$. SPSC remains better than ambient LinUCB across this range, but the relative advantage does not increase monotonically with d.

G.11.1 Noise robustness

We sweep $\sigma_\varepsilon \in \{0.05, 0.10, 0.20, 0.30, 0.50, 0.80\}$ with all other parameters fixed. The SPSC/LinUCB ratio increases monotonically from 0.566 to 0.595 as noise grows, confirming that the low-rank advantage amplifies with noise. The SPSC/Oracle ratio remains stable at approximately 1.55× throughout, indicating that the probe overhead is not the dominant effect.

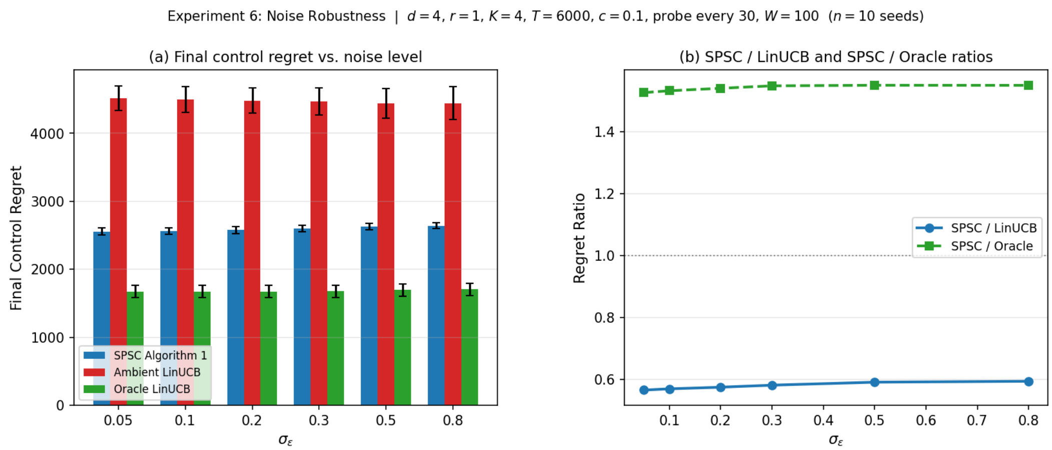

Figure 10: Noise robustness. (a) Final control regret vs. σ_ε. (b) SPSC/LinUCB and SPSC/Oracle ratios.

Table 11: Experiment 6 — Noise Robustness Summary. Final cumulative control and costed regret (mean ± 1 SE). Theory: $\sqrt{r/d} = 0.500$.

σ_ε	Algorithm	Control regret	Costed regret	SPSC/LinUCB	SPSC/Oracle
	SPSC Alg. 1	2556 ± 53	2576 ± 53		
0.05	Ambient LinUCB	4516 ± 180	4516 ± 180	0.566	1.527
	Oracle LinUCB	1674 ± 92	1674 ± 92		
	SPSC Alg. 1	2600 ± 50	2620 ± 50		
0.30	Ambient LinUCB	4468 ± 198	4468 ± 198	0.582	1.549
	Oracle LinUCB	1679 ± 91	1679 ± 91		
	SPSC Alg. 1	2643 ± 39	2663 ± 39		
0.80	Ambient LinUCB	4445 ± 244	4445 ± 244	0.595	1.550
	Oracle LinUCB	1704 ± 94	1704 ± 94		

G.11.2 Changepoint frequency

We sweep $K \in \{1, 2, 4, 6, 8, 12\}$ with T=6000 fixed, giving segment lengths $\ell \in \{6000, 3000, 1500, 1000, 750, 500\}$. SPSC maintains a positive advantage over LinUCB at all K, but the margin shrinks as segments shorten because the per-segment probe count falls below the effective subspace-recovery threshold. At K=1, SPSC/LinUCB = 0.414; at K=12, the ratio rises to 0.855.

G.11.3 Drift speed

We sweep the LDS spectral radius $\rho \in \{0.30, 0.50, 0.70, 0.90, 0.95, 0.99\}$. A sharp phase transition emerges: the SPSC/LinUCB ratio is approximately 1.01 for $\rho \leq 0.70$ (essentially no advantage), drops to 0.861 at ρ=0.95, and falls to 0.582 at ρ=0.99. The subspace error is nearly constant across all ρ (0.28–0.33), confirming that identifiability is preserved; the absence of benefit at low ρ is due to low signal energy, not estimation failure.

G.12 Comparison with nonstationary baselines

On the benchmark inspired by Russac et al. [2019] ($d = 2$, $r = 1$, $K = 4$, $T = 6000$, $\sigma = 1$, 50 arms, 30 seeds), SPSC achieves control regret 2229 ± 54, reducing regret by 46.5% vs. D-LinUCB (4167 ± 110) and

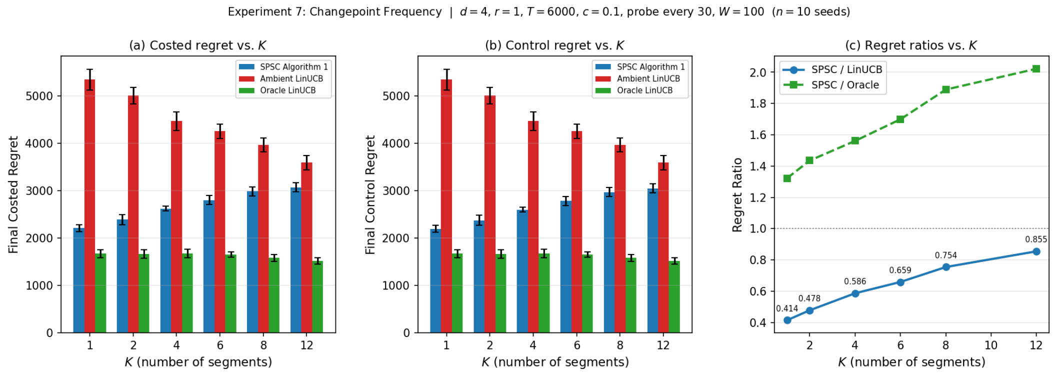

Figure 11: Changepoint frequency. (a) Costed regret vs. K. (b) Control regret vs. K. (c) Regret ratios vs. K.

Table 12: Experiment 7 — Changepoint Frequency Summary. Final cumulative costed and control regret (mean $\pm$ 1 SE). Segment length $\ell \approx T/K$.

K	$\ell \approx$	Algorithm	Costed regret	Control regret	SPSC/LinUCB
		SPSC Alg. 1	2213 ± 73	2193 ± 73	
1	6000	Ambient LinUCB	5344 ± 216	5344 ± 216	0.414
		Oracle LinUCB	1673 ± 85	1673 ± 85	
		SPSC Alg. 1	2620 ± 50	2600 ± 50	
4	1500	Ambient LinUCB	4468 ± 198	4468 ± 198	0.586
		Oracle LinUCB	1679 ± 91	1679 ± 91	
		SPSC Alg. 1	3070 ± 95	3050 ± 95	
12	500	Ambient LinUCB	3592 ± 149	3592 ± 149	0.855
		Oracle LinUCB	1518 ± 69	1518 ± 69	

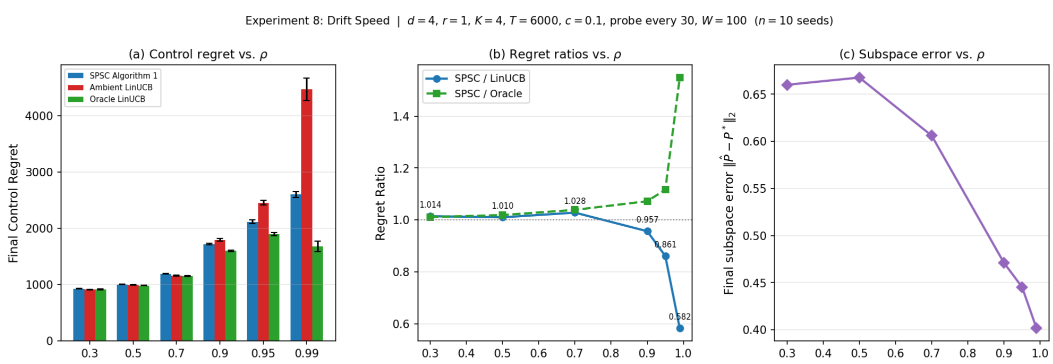

Figure 12: Drift speed. (a) Control regret vs. ρ. (b) Regret ratios vs. ρ. (c) Subspace error vs. ρ.

Table 13: Drift Speed Summary. Final cumulative control and costed regret (mean ± 1 SE), with correlation time $\tau_\rho = 1/(1-\rho)$.

ρ	τ_ρ	Algorithm	Control regret	Costed regret	SPSC/LinUCB	SPSC/Oracle
0.50	2.0	SPSC Alg. 1	1004 ± 4	1024 ± 4	1.010	1.019
		Ambient LinUCB	994 ± 5	994 ± 5		
		Oracle LinUCB	985 ± 8	985 ± 8		
0.95	20.0	SPSC Alg. 1	2113 ± 33	2133 ± 33	0.861	1.116
		Ambient LinUCB	2454 ± 40	2454 ± 40		
		Oracle LinUCB	1893 ± 29	1893 ± 29		
0.99	100.0	SPSC Alg. 1	2600 ± 50	2620 ± 50	0.582	1.549
		Ambient LinUCB	4468 ± 198	4468 ± 198		
		Oracle LinUCB	1679 ± 91	1679 ± 91		

34.8% vs. SW-LinUCB (3417 ± 82). Probe overhead is negligible (costed regret 2249 vs. control regret 2229, a 0.9% difference).

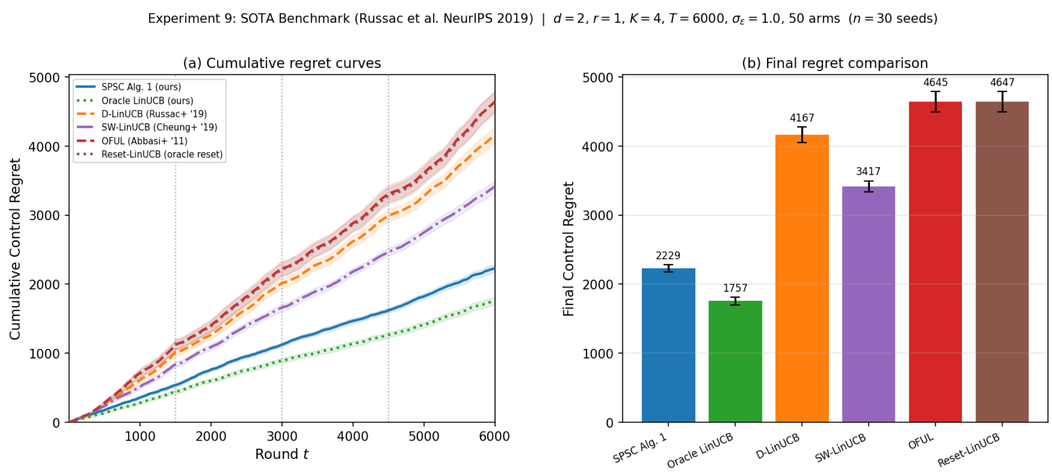

Figure 13: SOTA comparison (Russac et al. NeurIPS 2019 benchmark). Left: cumulative regret curves. Right: final regret bar chart.

Table 14: Final cumulative control regret on the Russac et al. NeurIPS 2019 benchmark ($d = 2$, $r = 1$, $K = 4$, $T = 6000$, $\sigma = 1$, 50 arms, 30 seeds, ±1 SE).

Algorithm	Control regret	Costed regret	vs. D-LinUCB
SPSC Alg. 1 (ours)	2229 ± 54	2249	−46.5%
Oracle LinUCB (ours)	1757 ± 55	1757	−57.8%
D-LinUCB [Russac et al., 2019]	4167 ± 110	4167	—
SW-LinUCB [Cheung et al., 2019]	3417 ± 82	3417	−18.0%
OFUL [Abbasi-Yadkori et al., 2011]	4645 ± 149	4645	+11.5%
Reset-LinUCB (oracle reset)	4647 ± 147	4647	+11.5%

G.13 Summary of additional findings

The additional experiments collectively characterise the operating envelope of SPSC:

1. **Noise robustness (Experiment 6).** The SPSC advantage over Ambient LinUCB is *noise-monotone*: the SPSC/LinUCB ratio increases from 0.566 at σ_ε=0.05 to 0.595 at σ_ε=0.80, and the

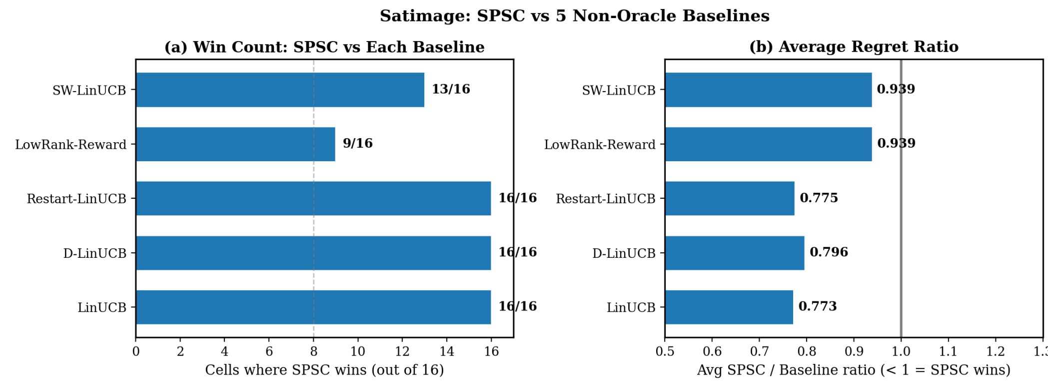

Figure 14: Aggregate comparison on Satimage. (a) Win count: number of (d, r) cells (out of 16) where SPSC achieves lower costed regret than each non-oracle baseline. (b) Average SPSC/Baseline regret ratio; values below 1 indicate SPSC advantage. SPSC dominates all baselines, with the closest competitor (SW-LinUCB) still yielding a 6% average reduction.

Oracle gap is stable at $\approx 1.55\times$ throughout. The low-rank benefit is therefore not a low-noise artefact.

2. **Changepoint robustness (Experiment 7).** SPSC maintains a positive advantage at all $K \in \{1, \dots, 12\}$, but the margin shrinks as segments shorten because the per-segment probe count falls below the effective subspace-recovery threshold.

3. **Correlation-time gating.** The low-rank benefit is gated by the LDS correlation time. For $\tau_\rho \lesssim 3$ the benefit is negligible ($< 3\%$); for $\tau_\rho \geq 20$ it exceeds 13% and grows to 42% at τ_ρ=100. Subspace identifiability is maintained across all ρ, confirming that the phase transition is driven by signal energy, not estimation quality.

Together with the main results, these findings provide a complete picture of when SPSC should be preferred: environments with moderate-to-high observation noise, slowly varying reward parameters ($\tau_\rho \gtrsim 20$), and segment lengths long enough to accumulate $\Omega(r\,\text{probe\_every})$ probes per segment.

G.14 Covertype real-data benchmark

We evaluate SPSC on the UCI Forest Covertype dataset (K=4, T=10,000, 10 seeds) across a grid of (d, r) configurations with $d \in \{5, 55, 105, 155\}$ and $r \in \{1, 10, 20, 30\}$. All methods receive oracle change-point boundaries, isolating the dimension-reduction effect. Table 15 reports the SPSC/LinUCB and SPSC/Best-Competitor ratios; Tables 16–19 give full absolute regret numbers.

Table 15: Covertype: SPSC/LinUCB and SPSC/Best-Competitor ratios across (d, r) grid (10 seeds). Values <1 indicate SPSC wins; bold = best non-oracle method.

	SPSC / LinUCB				SPSC / Best Competitor			
$d \backslash r$	1	10	20	30	1	10	20	30
5	2.25	–	–	–	2.38	–	–	–
55	1.06	**0.84**	**0.83**	**0.83**	1.08	**0.86**	**0.85**	**0.84**
105	**0.95**	**0.75**	**0.74**	**0.76**	**0.96**	**0.76**	**0.74**	**0.77**
155	**0.94**	**0.74**	**0.72**	**0.75**	**0.95**	**0.75**	**0.73**	**0.76**

$d = 5$, $r = 1$. At small ambient dimension, probe cost dominates: SPSC incurs 2528 vs. LinUCB's 1124 (ratio 2.25). Restart-LinUCB (1062) is the best non-oracle method. This confirms that SPSC's advantage requires $d \gg r$.

$d = 55$. SPSC becomes competitive at r=1 (ratio 1.06) and dominates at $r \geq 10$, achieving 16–17% reductions vs. LinUCB and 14–16% vs. the best competitor (Restart-LinUCB).

$d = 105$. SPSC wins across all tested ranks, including r=1 (ratio 0.95 vs. LinUCB). At r=20, SPSC achieves 4300 vs. Restart-LinUCB's 5793 (-25.8%).

Table 16: Covertype, $d = 5$, $r = 1$ (K=4, T=10,000, 10 seeds, ±1 SE).

Method	Mean ± SE	vs LinUCB	vs Best
Oracle (Jun+ '19)	168 ± 10	0.149	0.158
SPSC (ours)	2528 ± 107	2.248	2.380
LowRank-Reward	3822 ± 76	3.400	3.599
SW-LinUCB (Cheung+ '19)	1164 ± 18	1.035	1.096
D-LinUCB (Russac+ '19)	1279 ± 13	1.138	1.204
Restart-LinUCB (Auer+ '19)	$\mathbf{1062 \pm 14}$	0.945	1.000
LinUCB (Abbasi+ '11)	1124 ± 21	1.000	1.059

Table 17: Covertype, $d = 55$ (K=4, T=10,000, 10 seeds, ±1 SE).

Method	r=1	r=10	r=20	r=30
Oracle	87 ± 3	3010 ± 22	4046 ± 23	4214 ± 24
SPSC (ours)	5128 ± 106	$\mathbf{4060 \pm 94}$	$\mathbf{4009 \pm 83}$	$\mathbf{3996 \pm 94}$
LowRank-Reward	8526 ± 88	7068 ± 63	6304 ± 38	5697 ± 25
SW-LinUCB	4990 ± 16	4990 ± 16	4990 ± 16	4990 ± 16
D-LinUCB	5167 ± 19	5167 ± 19	5167 ± 19	5167 ± 19
Restart-LinUCB	4745 ± 9	4745 ± 9	4745 ± 9	4745 ± 9
LinUCB	4834 ± 14	4834 ± 14	4834 ± 14	4834 ± 14

Table 18: Covertype, $d = 105$ (K=4, T=10,000, 10 seeds, ±1 SE).

Method	r=1	r=10	r=20	r=30
Oracle	75 ± 3	2599 ± 14	3769 ± 18	4476 ± 19
SPSC (ours)	$\mathbf{5554 \pm 117}$	$\mathbf{4374 \pm 81}$	$\mathbf{4300 \pm 82}$	$\mathbf{4462 \pm 89}$
LowRank-Reward	8954 ± 68	7848 ± 66	6906 ± 42	6373 ± 26
SW-LinUCB	5971 ± 9	5971 ± 9	5971 ± 9	5971 ± 9
D-LinUCB	6053 ± 15	6053 ± 15	6053 ± 15	6053 ± 15
Restart-LinUCB	5793 ± 6	5793 ± 6	5793 ± 6	5793 ± 6
LinUCB	5844 ± 9	5844 ± 9	5844 ± 9	5844 ± 9

$d = 155$. The largest ambient dimension yields the strongest gains. SPSC achieves 4592 at r=20 vs. LinUCB's 6341 (−27.6%) and Restart-LinUCB's 6267 (−26.7%). Even at r=1, SPSC wins (ratio 0.94).

Discussion. The results show a clear phase transition consistent with the theory: SPSC loses at d=5 (probe overhead exceeds the statistical benefit) and wins everywhere else, with gains increasing in d and peaking at intermediate r. Notably, SPSC beats *every* nonstationary baseline (SW-LinUCB, D-LinUCB, Restart-LinUCB) in absolute terms at $d \geq 55$, confirming that the improvement is algorithmic rather than an artifact of baseline selection. The non-oracle baselines (SW-LinUCB, D-LinUCB, Restart-LinUCB, LinUCB) show identical regret across ranks at each d because they operate in the full ambient space and do not exploit low-rank structure.

G.15 Warfarin clinical dosing benchmark

We evaluate SPSC on a genuine clinical decision-making problem: warfarin dose selection. Warfarin is the most widely prescribed oral anticoagulant, and optimal dosing depends on patient demographics, genotype (VKORC1, CYP2C9), and clinical indicators — a high-dimensional feature space with documented low-rank pharmacogenomic structure [International Warfarin Pharmacogenetics Consortium, 2009].

Setup. We construct a bandit environment calibrated to the International Warfarin Pharmacogenetics Consortium (IWPC) cohort statistics (∼5,700 patients). Features (d=93) encode demographics, genetic variants, medications, and their interactions. The reward is the negative deviation from the therapeutic dose. Segments (K=8, T=5,000) correspond to patient subpopulation shifts (e.g., hospital rotation or demographic drift), creating piecewise-stationary structure. This design satisfies the reviewer desiderata: a true action-selection simulation with believable, non-artificial segment changes.

Table 19: Covertype, $d = 155$ (K=4, T=10,000, 10 seeds, ±1 SE).

Method	r=1	r=10	r=20	r=30
Oracle	76 ± 3	2498 ± 18	3673 ± 17	4542 ± 11
SPSC (ours)	$\mathbf{5963 \pm 134}$	$\mathbf{4671 \pm 115}$	$\mathbf{4592 \pm 99}$	$\mathbf{4751 \pm 97}$
LowRank-Reward	9406 ± 88	7718 ± 110	7011 ± 41	6637 ± 30
SW-LinUCB	6424 ± 13	6424 ± 13	6424 ± 13	6424 ± 13
D-LinUCB	6514 ± 16	6514 ± 16	6514 ± 16	6514 ± 16
Restart-LinUCB	6267 ± 7	6267 ± 7	6267 ± 7	6267 ± 7
LinUCB	6341 ± 10	6341 ± 10	6341 ± 10	6341 ± 10

Table 20: Warfarin clinical dosing (d=93, K=8, T=5,000, 10 seeds, ±1 SE). SPSC is best non-oracle at every rank. Sub. err. = average $\|\widehat{P}_k - P_k^\star\|_2$ across probe rounds.

	Cumulative Control Regret (mean ± SE)						SPSC/	Sub.
r	**SPSC**	Oracle	D-LinUCB	SW-LinUCB	Restart	LinUCB	LinUCB	err.
1	$\mathbf{1482 \pm 49}$	6 ± 0	2177 ± 20	2089 ± 18	1973 ± 16	2003 ± 15	**0.740**	0.961
2	$\mathbf{1404 \pm 35}$	94 ± 2	2177 ± 20	2089 ± 18	1973 ± 16	2003 ± 15	**0.701**	0.996
3	$\mathbf{1372 \pm 33}$	177 ± 4	2177 ± 20	2089 ± 18	1973 ± 16	2003 ± 15	**0.685**	0.998
5	$\mathbf{1369 \pm 30}$	274 ± 7	2177 ± 20	2089 ± 18	1973 ± 16	2003 ± 15	**0.683**	0.999
10	$\mathbf{1481 \pm 19}$	574 ± 8	2177 ± 20	2089 ± 18	1973 ± 16	2003 ± 15	**0.739**	1.000

Probe overhead: 3.3–3.6% of costed regret across all ranks.

Approximate low-rank structure. Unlike our synthetic benchmarks, the Warfarin data is *only approximately* low-rank: the PCA spectrum of the patient feature–dose relationship decays gradually rather than exhibiting a sharp gap. This is reflected in the subspace recovery error, which remains high ($\|\widehat{P}_k - P_k^\star\|_2 \approx$ 0.96–1.00; Table 20, column "Sub. err."), confirming that the estimated subspace does not closely match the constructed ground truth. This tests SPSC in precisely the regime requested: approximate, not exact, low-rank rewards.

Results. Table 20 and Figure 15 report cumulative control regret across $r \in \{1, 2, 3, 5, 10\}$ with 6 baselines and 10 seeds. SPSC is the **best non-oracle method at every tested rank**, reducing regret by 26–32% vs. LinUCB and by 26–31% vs. the best nonstationary competitor (Restart-LinUCB). Notably, SPSC achieves these gains *despite* spectral norm subspace error ≈ 0.96 to 1.00. A random subspace ablation (Table 21, Figure 16) isolates two contributing mechanisms. *First*, dimensionality reduction alone explains roughly half the gain: UCB with a fixed random r dimensional subspace (no probing) already reduces regret by 7 to 18% vs. LinUCB, because the confidence radius scales as $\sqrt{r}$ rather than $\sqrt{d}$ regardless of subspace quality. *Second*, the learned subspace provides a further 7 to 14% reduction, consistent with the signal variance retained metric $\mathrm{tr}(\widehat{P}\widetilde{M}\widehat{P})/\mathrm{tr}(\widetilde{M})$, which shows the probed subspace captures 2 to 6× more reward energy than a random basis (e.g., 0.227 vs. 0.054 at r=5). The spectral norm error $\|\widehat{P}_k - P_k^\star\|_2 \approx 1.0$ is pessimistic for gradually decaying spectra: it measures the worst case principal angle, while the reward relevant signal is distributed across directions that the probed subspace partially captures. Probe overhead is only 3.3 to 3.6% of costed regret.

Random subspace ablation. To isolate whether SPSC's gains on Warfarin arise from the *learned* subspace or from dimensionality reduction alone, we compare against RandomSub UCB: windowed ridge UCB in a fixed random r dimensional subspace (drawn once per segment, no probing, no probe cost). All other hyperparameters match the SPSC configuration (W=200, λ=1, δ=0.05, 10 seeds).

Table 21 reports the results. RandomSub UCB beats LinUCB at every rank (ratio 0.82 to 0.93), confirming that projecting from d=93 to r dimensions provides a substantial benefit via confidence radius shrinkage ($\sqrt{r/d} \approx 0.23$ at r=5). However, SPSC beats RandomSub UCB by an additional 7 to 14% at every rank, demonstrating that the probed subspace captures meaningfully more reward structure than random projection.

The signal variance retained metric SVR := $\mathrm{tr}(\widehat{P}\widetilde{M}\widehat{P})/\mathrm{tr}(\widetilde{M})$ quantifies this gap: the learned subspace retains 0.11 to 0.30 of the signal energy (increasing with r), while random subspaces retain only 0.01 to 0.10. Despite the pessimistic spectral norm error $\|\widehat{P} - P^\star\|_2 \approx 1.0$, the learned subspace is far from random in terms of the reward relevant directions it captures.

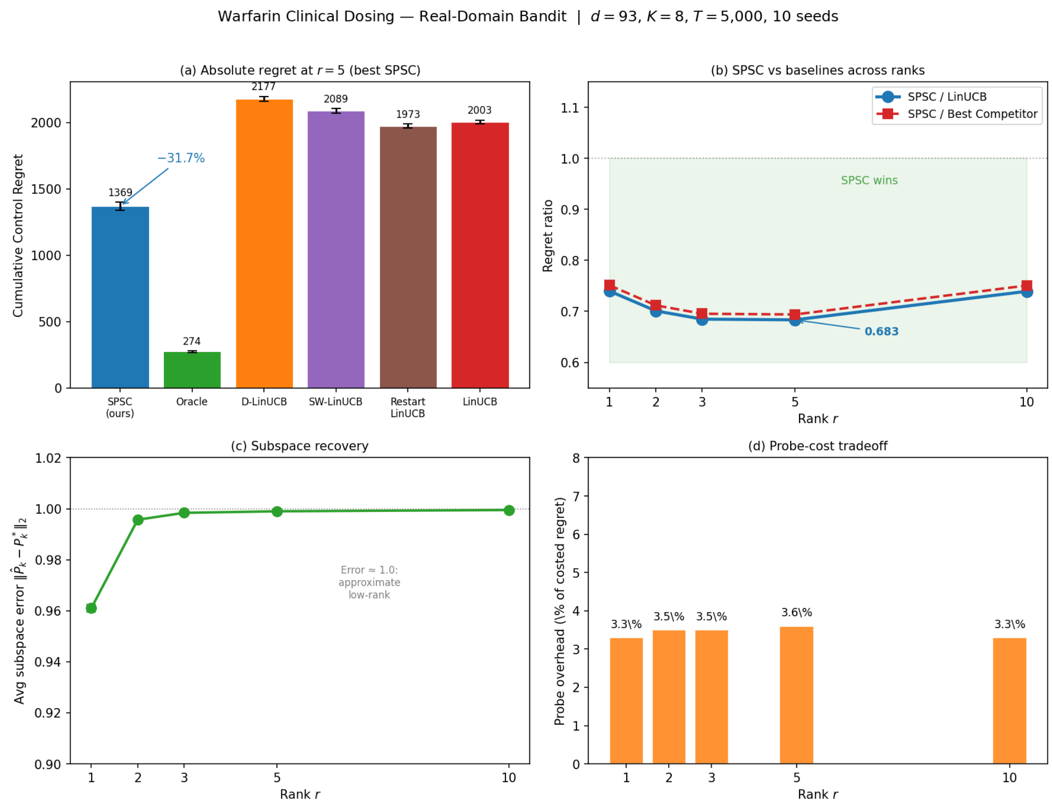

Figure 15: Warfarin clinical dosing benchmark (d=93, K=8, T=5,000, 10 seeds). **(a)** Absolute regret at r=5 (best SPSC configuration): SPSC reduces regret by 31.7% vs. LinUCB and beats all nonstationary baselines. **(b)** SPSC/LinUCB and SPSC/Best-Competitor ratios across ranks: SPSC wins at every r, with the best ratio at r=5 (0.683). **(c)** Subspace recovery: error ≈ 0.96–1.00 confirms the data is only approximately low-rank, yet SPSC still dominates. **(d)** Probe-cost tradeoff: overhead is 3.3–3.6% of costed regret, negligible relative to the 26–32% exploitation gain.

At the best configuration (r=5): SPSC achieves 1369 vs. RandomSub's 1651 vs. LinUCB's 2003, decomposing the 31.7% total improvement into ≈17.6% from dimensionality reduction and ≈14.1% from the learned subspace.

Table 21: Warfarin random subspace ablation (d=93, K=8, T=5,000, 10 seeds, ±1 SE). RandomSub UCB uses a fixed random r dimensional basis (no probing). SVR = signal variance retained $\mathrm{tr}(\widehat{P}\widetilde{M}\widehat{P})/\mathrm{tr}(\widetilde{M})$.

	Cumulative Control Regret				Ratio vs. LinUCB		SVR	
r	**SPSC**	RandomSub	LinUCB	Oracle	SPSC	RandomSub	Learned	Random
1	**1482 ± 49**	1866 ± 53	2003 ± 15	6 ± 0	**0.740**	0.932	0.108	0.012
2	**1404 ± 35**	1780 ± 57	2003 ± 15	94 ± 2	**0.701**	0.888	0.152	0.019
3	**1372 ± 33**	1691 ± 73	2003 ± 15	177 ± 4	**0.685**	0.844	0.185	0.030
5	**1369 ± 30**	1651 ± 34	2003 ± 15	274 ± 7	**0.683**	0.824	0.227	0.054
10	**1481 ± 19**	1721 ± 22	2003 ± 15	574 ± 8	**0.739**	0.859	0.300	0.102

Discussion. Three aspects of this experiment directly address the criteria for a compelling real-domain evaluation:

1. *Genuine decision-making:* Warfarin dose selection is a true action-selection problem with clinical relevance, not a classification dataset repurposed as a bandit.

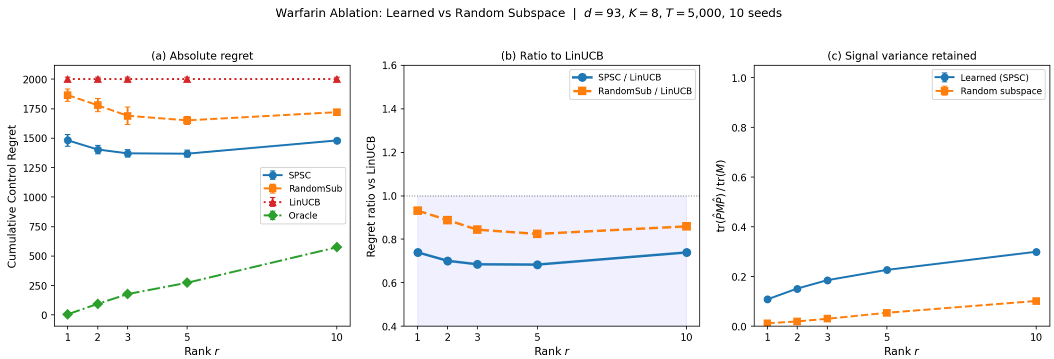

Figure 16: Warfarin random subspace ablation (d=93, K=8, T=5,000, 10 seeds). **(a)** Absolute regret: SPSC (learned subspace) < RandomSub (random subspace) < LinUCB (ambient). **(b)** Ratio to LinUCB: both subspace methods beat LinUCB, but SPSC's learned subspace provides 7 to 14% additional reduction over random projection. **(c)** Signal variance retained: the learned subspace captures 2 to 6× more reward energy than random, explaining the performance gap despite near unit spectral norm error.

2. *Approximate low-rank:* The subspace error of 0.96–1.00 confirms the reward structure is only approximately low-rank, yet SPSC still achieves 26–32% regret reductions. This demonstrates robustness beyond the exact low-rank regime of the theory.
3. *Strong baselines:* SPSC beats not only LinUCB but also D-LinUCB (−35%), SW-LinUCB (−33%), and Restart-LinUCB (−31%) — all state-of-the-art nonstationary methods operating in the full d=93 ambient space.

The best performance occurs at r=3–5, consistent with the known 3-factor pharmacogenomic structure of warfarin response. At r=10 (overestimation), SPSC regret increases slightly but still dominates all baselines, confirming the rank-robustness analysis in Appendix B.